Section 15 RF Coils

Contents

1. QUAD BODY COIL (QBC) CHECK/ADJUSTMENT

The Quad Body Coil is already adjusted in the factory. Therefore only a test has to be done to check the QBC and to determine the values for the MRL as a reference.
Verify that the QBC did not move during transportation by checking all excenter screws When the excenters were loose, the QBC has changed position and re-adjustment of the QBC is necessary (See appendix A)

1.1. TOOLS

The next tools are needed from the measurement accessories:

- Quadrature flux probe
- 2 Linear flux probes
- QBC flux probe holder
- Rotation knob (22.5 degr.)
- 50 ohm coaxial cable 15 m
- QBC Trimmer tool

1.2. HYBRID BOX + QBC RESONANCE FREQUENCY

1. Switch off the gradient amplifier.
2. Lower the patient table.
3. Remove the bridge section, the picu front cover and the front cone.
4. The gradient/RF coil assembly and the hybrid box in front of the QBC are visible now.
5. Setup the flux-probe holder with the rotation knob and the two (linear) flux-probes in the magnet iso-center. The flux-probes should coincide with the detune boards in the body coil. The flux probe holder must be positioned horizontal. The cables should leave the bore via the backside of the magnet.
6. Mount the probes on both ends of the holder. See Figure 1.
7. Remove the magnet left top cover to get access to the TRSW board and the Coilint board in the PFEI.
8. Disconnect cable MME X5 (QBC_receive) from the coilint.
9. Disconnect the two tune signals MMF X2 and MMF X3 from the TRSW board (PFEI). Connect one of the two flux-probes to tune A at connector MMF X2 of the TRSW board.
10. Connect the other flux probe to MME X5 of the coilint. See also Figure 2.

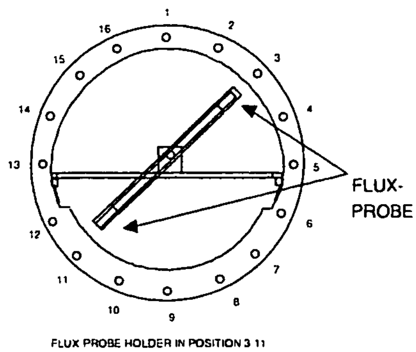


Figure 1 Flux-probe holder in QBC

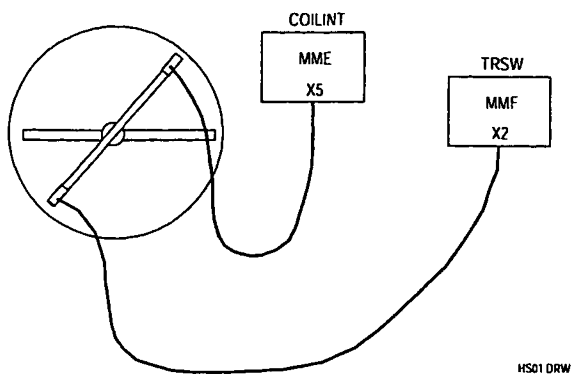


Figure 2 Flux probe connections

11. Login : GYROSCAN.
12. Select: SCAN CONTROL.
13. Select: SCAN UTILITIES.
14. Select: ENTER SERVICE MODE.
15. Select: SYSTEM TUNING
16. Select: SYSTEM TUNING TOOLS.
17. Select: INSTALLATION PROCEDURES.
18. Select: BODY COIL PROCEDURES
19. Select: ADJUST BBM RECEIVE

20. Start the measurement with <proceed> (the central frequency is set to 21.298000 MHz by the program)
21. Rotate the probe-holder between rod 3 - 11.
22. Select auto scale to make the signal visible on the screen.
23. Check the offset frequency and the Q-factor and compare them with the specification. The frequency offset can be changed with trimmer C37. Trimmer C37 can be reached via an opening in the top plate of the hybrid box. See also Figure 3.
24. Compare the measured values with the specification. Write down the found value in the MRL (section 22).

- *Specification:* *freq. offset (F1) < ± 10 kHz*
  Q > 400

If the Q value is not within specification the QBC must be adjusted with the procedure in chapter 12 "Appendix A: QBC Adjustment". If the specification has been met, write down the measured values in the MRL as a reference.

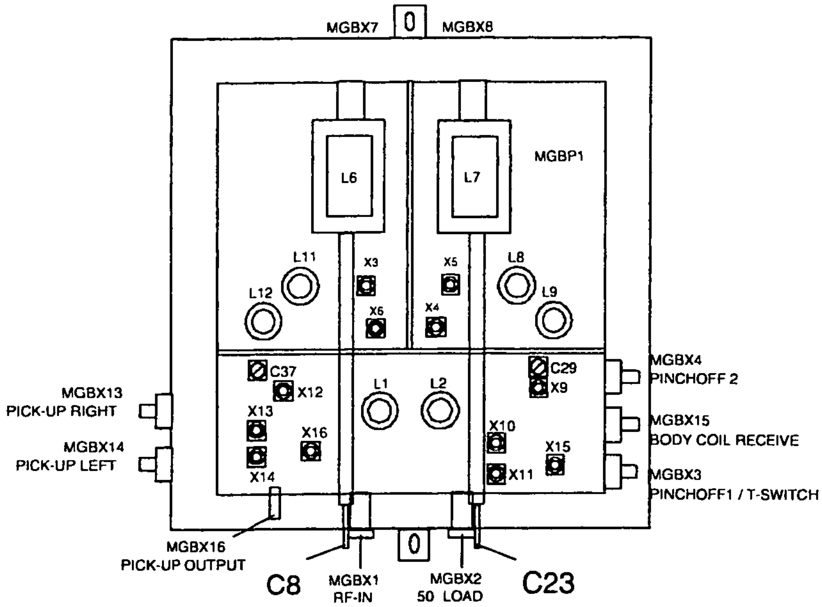


Figure 3 Hybrid Box

22. Rotate the probe-holder between rod 7 - 15.
23. Now use trimmer C29 to adjust the frequency offset within specification.
24. Compare the measured values with the specification. Write down the found value in the MRL (section 22).

Specification: *freq. Offset (F2) < ± 10 kHz and (F2) = F(1) ± 2.5 kHz*
Q > 400

NOTE

The specification for both frequency offsets (F1) and (F2) is < 10 kHz but beside that, both frequencies should not differ more than 2.5 kHz.

If the Q value is not according specification you have to adjust the QBC with the procedure in "Appendix A. QBC Adjustment" at the end of this section.

If the measured values are according the specification the QBC and the hybrid box are correctly adjusted. Write down the measured values in the MRL (Section 22) and restore the original situation.

1.3. DETERMINATION OF THE BODYCOIL PARAMETERS

NOTE

The program uses 21.298 Mhz as reference frequency.

To perform this measurement the cone-cover and the bridge section must be mounted.

1. Mount the **Quadrature** flux probe in the QBC flux probe holder and position the holder in the center of the body coil **with the connector of the Q-flux probe pointed to the backside of the magnet.** Also the cable must be lead via the backside of the magnet.
2. Connect a cable between tune A at connector MMF X2 and the Quadrature flux probe. See Figure 4.

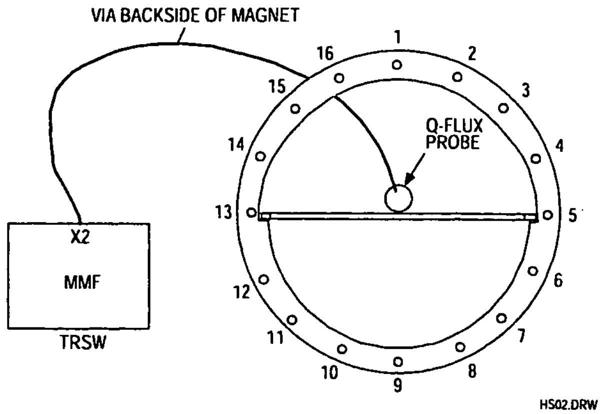


Figure 4

3. Login : GYROSCAN.
4. Select: SCAN CONTROL.
5. Select: SCAN UTILITIES.
6. Select: ENTER SERVICE MODE.
7. Select: SYSTEM TUNING
8. Select: SYSTEM TUNING TOOLS.
9. Select: INSTALLATION PROCEDURES.
10. Select: BODY COIL PROCEDURES
11. Select: DETERMINE BODY COIL PARAMETERS

12. Start the measurement with <Proceed>.
    No errors are allowed.
13. Write the actual body coil frequency (Act. coil freq.) and Q-factor (Measured Q) in the MRL (Section 22) as a reference.

2. DETERMINATION OF THE HEAD COIL PARAMETERS

2.1. INTRODUCTION

The procedure written in this chapter is to check and adjust the QHC (Quad Head Coil) parameters.
It can be necessary to adapt the cable to the other side of the coil. See appendix B at the end of this section (section 15) for procedure.
Adjustment is needed in case the QHC does not meet the specifications or you have re-installed the cable with Quad Driver board to the other side of the coil

The QHC is checked on frequency, Q-factor and orthogonality. The operating frequency for the QHC is 21.295 Mhz Frequency-offset with respect to this frequency and orthogonality can be adjusted if not in specification.

2.2. QUAD HEAD COIL FREQUENCY AND ORTHOGONALITY ADJUSTMENT

The quadrature head coil can be seen as two coils working in different modes: there is a phase shift of 90° between both coils. Each coil has a pre-amplifier. By disconnecting one coil (removal of a jumper on the Quad Pre-amp board), the resonance frequency of the other coil can be adjusted.
The adjustment for orthogonality must be done to ensure an exact 90° phase difference between the two coils.

Preparation:

1. Remove the magnet left top cover.
2. Remove both tune coil cables MMFX2 and MMFX3 from the TRSW board (PFEI).
3. Move the table out of the magnet and position the head coil in such a way that the rear of it can be reached. Connect the head coil to the surface coil connector 1 of the PICU.
4. Move the slideable part of the head coil to the front (scanning) position. This is necessary to reduce the influence of the quad driver board on the coil.
5. Unscrew the rear cover of the coil and mount it back to the head coil using tape in stead of screws. For the adjustment you need to take the cover off but the measurements are only valid when the rear cover is closed, because the values are influenced by this cover.
6. Place the head coil on the table in such a way that the rear of the head coil matches the table end at magnet side (reference point).
7. Make the middle (marker on coil) of the head coil move to the iso-center of the magnet via the "travel-to-scan-plane".

8. Position the linear flux probe (do not use the quadrature flux probe) in position L8 - L12 (see Figure 5). The flux probe must be positioned against the front ring of the head coil. The flux probe holder must be used without the 360° rotation knob.

9. Connect the **linear** flux probe cable to tune A at connector MMFX2. Connector MMFX3 must be disconnected.

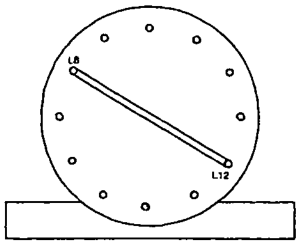


Figure 5 - Flux probe in position L8 - L12, seen from patient support end

NOTE

On the front ring of the head coil, marker lines indicate rod L5, L8, L11 and L12. Make sure the flux probe cable is apart from the head coil cable and that it leaves the bore as straight as possible.

Start the adjustment loop as follows:

10. Login : GYROSCAN.
11. Select: SCAN CONTROL.
12. Select: SCAN UTILITIES.
13. Select: ENTER SERVICE MODE.
14. Select: SYSTEM TUNING
15. Select: SYSTEM TUNING TOOLS.
16. Select: INSTALLATION PROCEDURES.
17. Select: HEAD COIL PROCEDURE

18. Enter: <PROCEED> to start the measurement process

NOTE

The central frequency used in the program is: 21.295000 Mhz. All measured values are offset frequencies with respect to the central frequency. The measured values are only valid in case the rear cover of the head coil is closed during the measurement.

STEP 1 : Frequency check/adjustment first mode

1 Place the flux probe in position L8-L12
2. Remove rear cover.
Jumper W1: Removed
Jumper W2: Position 2
See drawing Z3-7.20 for jumper location (See Figure 6).

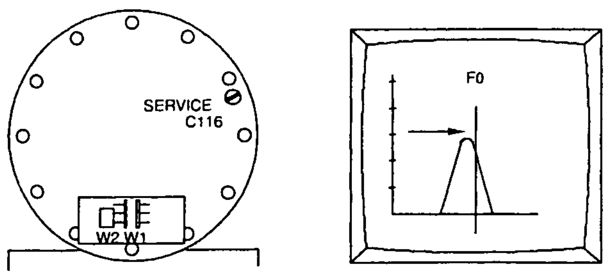


Figure 6 Frequency adjustment mode L8-L12

3. Put back the rear cover and read:

Q(1)	= Coil Q-factor.
U_1	= Voltage at Tune Top
F(1)	= Offset Frequency

Specifications:

Q(1):	> 350
F(1).	< ± 10 kHz

4. If F(1) does not meet the specifications, adjust the frequency with trimmer C116 (See Figure 6)
5. If the frequency is in specification, write down the Q-value (Q(1)), Voltage at Tune Top U1 and Offset frequency (F(1)).

STEP 2 : Frequency check/adjustment second mode

1. Install the flux probe in position L5-L11.
2. Remove rear cover. Remove jumper W2 and install jumper W1 in position 2. See Figure 8 for jumper location.

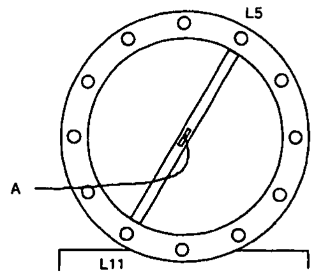


Figure 7 Probe in position L5 - L11

3. Replace rear cover read and write down:

Q(2)	= Coil Q-factor.
U_2	= Voltage at Tune Top (maximum voltage)
F(2)	= offset frequency
Specifications:	
Q(2)	> 350
F(2)	≤ ± 10 kHz

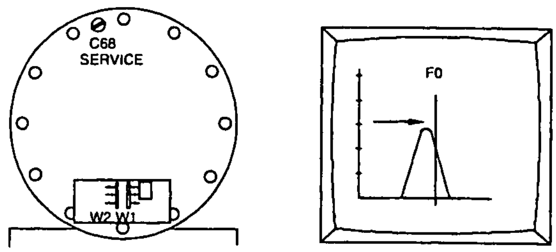
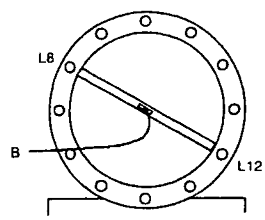


Figure 8 Frequency adjustment mode L5-L11

4. If F(2) does not meet the specifications, adjust the frequency with trimmer C68 (see Figure 8).
5. If the frequency is in specification, write down the Q value (Q(2)), Voltage at Tune Top (U_2) and Offset frequency (F(2)).

STEP 3 : Orthogonality check/adjustment (1)

1. Install flux probe to position L8 - L12. The flux probe must be positioned against the front ring of the head coil.



2. Read and write down:
   - U_3 = **Voltage at Center Frequency**

   Calculate the attenuation with next formula:

$$Att(1) = 20\log(\frac{U_2}{U_3})$$

The typical value for the orthogonality is: Att(1) > 15 dB

If this value is reached, continue with **STEP 4**.

If this value is not reached, adjust the Tune Voltage at the marker position to a value as low as possible using the next procedure:

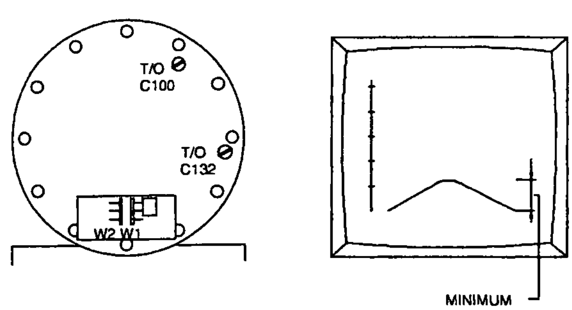


Figure 10 Adjustment of the orthogonality

3. Probe position between L8/L12 (As the previous adjustment). Put the jumpers in the following position:
   - W1 -> position 2
   - W2 -> removed

4. Turn C100 a little clockwise, see if response decreases. If not turn C100 counter clockwise. Turn C132 the same amount in the other direction(*). (Refer to the drawings manual Z3-7.18 for trimmer location)

   (*) Other direction means: If you have set for more capacity with C100 turn C132 in the other direction until you have the same amount capacity less. Minimize the response by turning both trimmers successively, as described above, until a minimum is reached.
   Write down the measured voltage U_3.

NOTE

Sometimes, depending on their initial position, both trimmers have to be turned into the same direction, sometimes in the opposite direction.

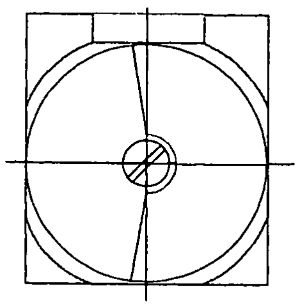


Figure 11 Trimmer in Middle position

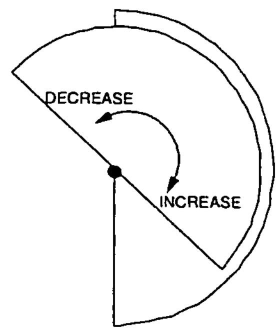


Figure 12 Increasing / decreasing capacity

NOTE

The orthogonality measurement is not accurate due to the mechanical construction of the coil. Adjust the orthogonality as good as possible. It is not necessary to reach the typical value. The correct functioning of the coil is checked during the Image Quality performance measurements.

STEP 4: Orthogonality check/adjustment (2)

1. Install flux probe to position L5 - L11. The flux probe must be positioned against the front ring of the head coil and put the jumpers in the next position:
   W1 removed
   W2 position 2

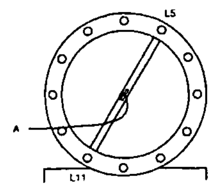


Figure 13 Flux probe position mode L5-L11

2. Read and write down:
   - U_4 = **Voltage at Center Frequency**

   Calculate the attenuation with next formula:

$$Att(2) = 20\log(\frac{U_1}{U_4})$$

The typical value for the orthogonality is: Att(1) > 15 dB

If this value is reached, continue with paragraph "Final actions"

If this value is not reached, adjust the Tune Voltage at the marker position to a value as low as possible using the next procedure:

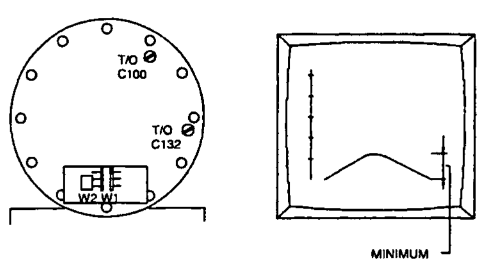


Figure 14 Adjustment of the orthogonality

Probe position between L5/L11 (As the previous adjustment). Put the jumpers in the following position:
W1 -> removed
W2 -> position 2

4. Turn C100 a little clockwise, see if response decreases. If not turn C100 counter clockwise **and** turn C132 the same amount in the other direction(*).

   (*) Other direction means: If you have set for more capacity with C100 turn C132 in the other direction until you have the same amount capacity less. Minimize the response by turning both trimmers successively, as described above, until a minimum is reached.
   Write down the measured voltage U_4.

NOTE

Sometimes, depending on their initial position, both trimmers have to be turned into the same direction, sometimes in the opposite direction.

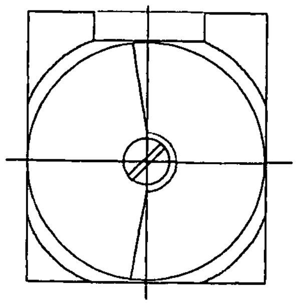


Figure 15 Trimmer in Middle position

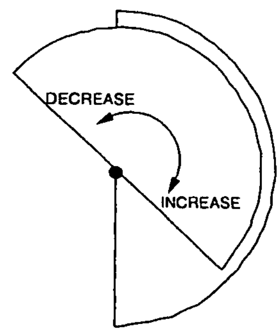


Figure 16 Increasing / decreasing capacity

NOTE

The orthogonality measurement is not accurate due to the mechanical construction of the coil. Adjust the orthogonality as good as possible. It is not necessary to reach the typical value. The correct functioning of the coil is checked during the Image Quality performance measurements.

2.3. FINAL ACTIONS

1. Repeat step 1 - 4 until no adjustments are necessary. Every adjustment can influence other parameters.
2 If you have followed the procedure and no adjustments were necessary, fill in the measured values, in Table 1 Install both jumpers in position 2 and mount the rear cover.
3. Remove the flux probe-holder out of the head coil and reconnect tune coil cables MMF X2 and MMF X3 Read and record as a reference in MRL (no specifications).
   F(head) = Offset frequency
   Q(head) = Coil Q-factor

10. Restore normal situation.
11. Write down the final values in the table in the MRL (section 22) as a reference.

Probe position	Parameters with specification		Measured values
Step 1) Jumper position: W1: Removed W2: Position 2 L8 - L12	Q(1)	> 350	
	Voltage U_1		
	F(1)	< ± 10 kHz	
Step 2) Jumper position: W1: Position 2 W2: Removed L5 - L11	Q(2)	> 350	
	Voltage U_2		
	F(2)	< ± 10 kHz	
Step 3) Jumper position. W1. Position 2 W2: Removed L8 - L12	Voltage U_3		
	Att(1)	typical value > 15 dB	
Step 4) Jumper position: W1: Removed W2: Position 2 L5 - L11	Voltage U_4		
	Att(2)	typical value > 15 dB	
Both jumpers positioned	F(head)	n.a.	
	Q(head)	n.a.	

3. APPENDIX A: QBC ADJUSTMENT

This procedure must be followed in case the QBC is mis-adjusted or not within specification.

3.1. PRINCIPLE

The adjustments can be divided in three parts.

First the QBC will be set in a defined start position.
After that the next steps are followed:

1. Hybrid box transmit curve

The transmit curve of the hybrid box is checked/adjusted. The coil is designed to have a flat transmit curve to improve the slice selection definition of the excitated volume.

2. Hybrid box resonance frequency

The f_0 of the two pre-amps in the hybrid-box are adjusted to the correct frequency and the Q-factor is checked.

3. Orthogonality adjustment

When the Q-factor with the previous measurement was not in specification, a readjustment of the orthogonality is necessary.

When the adjustments are done, the empty body coil frequency must be determined.

3.2. TOOLS

1. 2 flux-probes + cables
2. flux-probe holder
3. rotation knob for flux-probe holder
4. quadrature flux probe 21 MHz 4522 117 70251
5. trimmer/lock tool for coil trimmers

3.3. DEFINING START POSITION

1. Switch off the gradient amplifier.
2. Lower the patient table.
3. Remove the bridge section, the picu front cover and the front cone if not already done.
4. Remove the left magnet top cover.
5. The gradient/RF coil assembly and the hybrid box in front of the QBC are visible now
6. Setup the flux-probe holder with the rotation knob and the two (linear) flux-probes in the magnet iso-center The flux-probes should coincide with the detune boards in the body coil. The flux probe holder must be positioned horizontal. The cables should leave the bore via the backside of the magnet.

The gradient/RF coil assembly and the Hybrid box in front of the QBC are visible now.

7. Disconnect MGBX1 and MGBX3 and MGBX4 from the hybrid box. These two cables can carry 300 V, so for security reasons they have to be disconnected before the hybrid box may be opened.

CAUTION

When the connectors MGBX1 and MGBX3 and MGBX4 are connected to the hybrid box, 300 V can be present inside the hybrid box.

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8. Open the hybrid-box by taking off the top-plate.
9. Mark the two coils L6 and L7 to be able to put them back in the same position after the adjustment, and take the two coils out of the hybrid box. See also Figure 19

NOTE

It is important to mark the two coils L6 and L7 and remount them in the exact same way after the adjustment as they were before. It prevents that you have to do extra adjustments at the end of the adjustment procedure.

In this situation the body coil is electrically disconnected from the rest of the system, so the resonance frequency of the coil can be adjusted.

10. Close the hybrid box again.
11. Mark the position of the pick-up coils and remove them from the QBC.
12. Reconnect MGBX1, MGBX3 and MGBX4 to the hybrid box.
13. Setup the flux-probe holder with the two (linear) flux-probes in the magnet iso-center. The flux-probes should coincide with the detune boards in the body coil.
14. Mount the probes on both ends of the holder.
15. Disconnect the two tune signals MMFX2 and MMFX3 from the TRSW board (PFEI).
16. Connect one of the two flux-probes to tune A at connector MMFX2.

17. Disconnect connector MME X5 (QBC_receive) from the coilint.
    Connect the other flux-probe to MME X5 (coilint).
    See also Figure 17 and 18.

18. Use the special tool for the trimmers, and release the locking nut to make it possible to adjust the trimmers and turn all rod-trimmers carefully completely clockwise (CW). You have to stop when you feel resistance (end of trimmer).
    The resonance frequency of the QBC will be very low now.

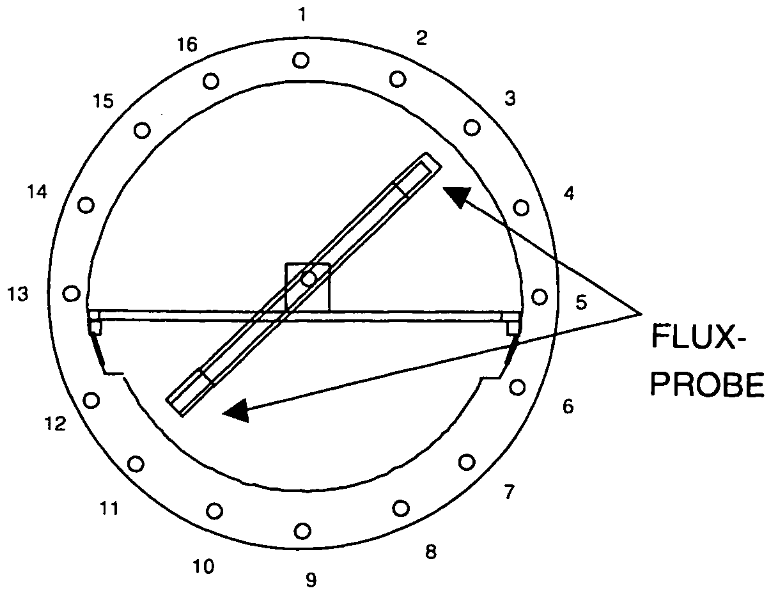


Figure 17 Flux-probe holder in QBC

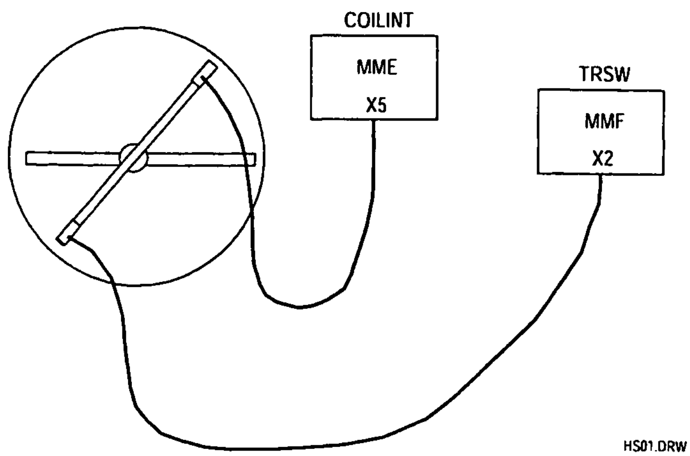


Figure 18 Flux probe connections

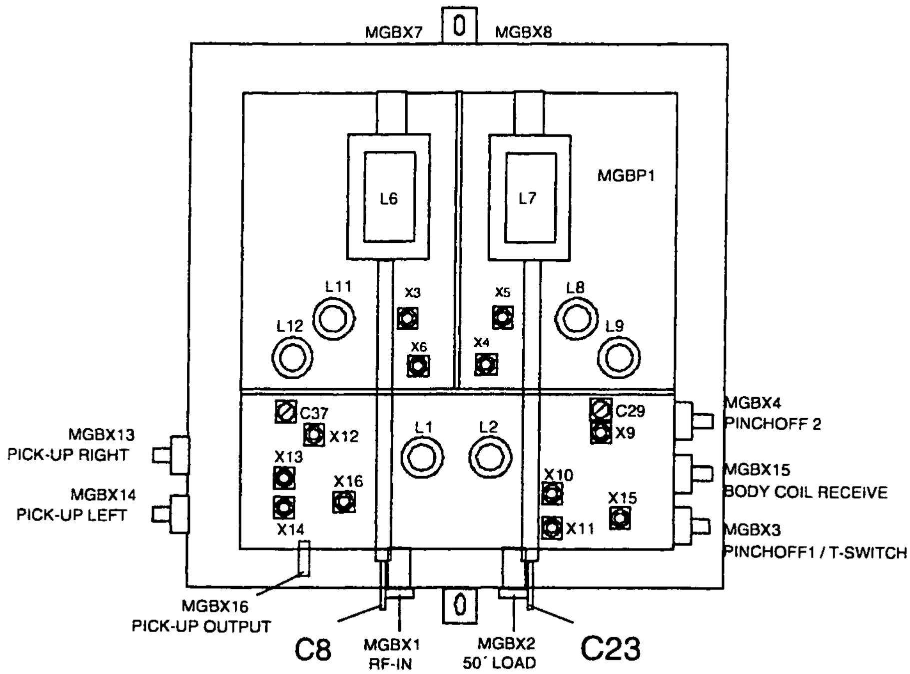


Figure 19 Hybrid Box

19. Start the bodycoil adjustment:
    Select: SCAN CONTROL.
    Select: SCAN UTILITIES.
    Select: ENTER SERVICE MODE.
    Select: SYSTEM TUNING
    Select: SYSTEM TUNING TOOLS.
    Select: INSTALLATION PROCEDURES.
    Select: BODY COIL PROCEDURES
20. Select: Adjust frequency course. Start the measurement with proceed
21. Now the trimmers of rods nr. 3, 7, 11, 15 have to be turned 15 full turns (360 degrees) counter clockwise (CCW).
22. Rotate the flux probe holder to the horizontal position, i.e. between rod 5 - 13.
23. Check the frequency of the Tune top (Offset_horizontal). (Measurement only valid if the top of the tune curve is visible on the screen)

24. Rotate the flux probe holder to the vertical position, i.e. between rod 1 - 9.
25. Check the frequency of the Tune top (Offset_vertical). (Measurement only valid if the top of the tune curve is visible on the screen)

26. If the tune top voltages can not be determined, the trimmers of rod, 3, 7, 11, 15, have to be turned 5 full turns CCW between measuring the tune-curves, until the tune curve gets visible on the screen and the offsets for horizontal and vertical mode are about -60 kHz.

If offset_horizontal > offset_vertical goto STEP A first and then to STEP B.
If offset_vertical > offset_horizontal goto STEP B first and then to STEP A.

If both offsets are about the same, then start with A first.

3.4. STEP A

1. Rotate the flux probe holder in horizontal position.
2. Adjust the offset to a minimum value (< 10 kHz), using the trimmers at rod 4, 5, 6, 12, 13 and 14. Always turn these trimmers in the same direction, with the same amount of turns. Turning CCW of a trimmer will INCREASE the resonance frequency.
   1 turn will change the resonance frequency with approx. 20 kHz.

3.5. STEP B

1. Rotate the flux-probe in vertical position.
2. Adjust the offset to a minimum value (< 10 kHz), using the trimmers at rod 16, 1, 2, 8, 9 and 10. Always turn these trimmers in the samme direction wioth the same amount of turns.
   Turning CCW of a trimmer will INCREASE the resonance frequency.
   1 turn will change the resonance frequency with approx. 20 kHz.
3. Do STEP A in case not already done.

3.6. STEP C

1. Rotate the flux-probe to position 3-11. Adjust the frequency offset to a minimum using the trimmers of rod 3 and 11.
2. Rotate the flux-probe to position 7-15. Adjust the frequency offset to a minimum using the trimmers of rod 7 and 15.

3.7. STEP D

1. Select: Adjust frequency fine in Bodycoil procedures.

Always adjust the two trimmers belonging to one pair equally. This means turn each trimmer in the same direction with the same rotation.

specification for each probe position:

Offset < ± 5 kHz
Q > 400

2. Rotate the flux-probe holder in position 1 - 9.
   Use the trimmers 1 and 9 to adjust the offset to a minimum value (specification < ± 5 kHz)
3 Rotate the flux-probe holder in position 8 - 16.
4. Use the trimmers 8 and 16 to adjust the offset to a minimum value (specification < ± 5 kHz)
5. Rotate the flux-probe holder in position 2 - 10.
6. Use the trimmers 2 and 10 to adjust the offset to a minimum value. (specification < ± 5 kHz)
7. Rotate the flux-probe holder in position 5 - 13.
   Use the trimmers 5 and 13 to adjust the offset to a minimum value (specification < ± 5 kHz)
8. Rotate the flux-probe holder in position 6 - 14.
   Use the trimmers 6 and 14 to adjust the offset to a minimum value. (specification < ± 5 kHz)
9 Rotate the flux-probe holder in position 4 - 12.
   Use the trimmers 4 and 12 to adjust the offset to a minimum value. (specification < ± 5 kHz)
10. Rotate the flux-probe holder in position 3 - 11.
    Use the trimmers 3 and 11 to adjust the offset to a minimum value. (specification < ± 5 kHz)
11. Rotate the flux-probe holder in position 7 - 15.
    Use the trimmers 7 and 15 to adjust the offset to a minimum value. (specification < ± 5 kHz)

Repeat step D until no adjustments are needed any more.
If all the values are in specification, restore the normal situation by following the next procedure:

12. Stop the tuning loop.
13. Disconnect MGBX1, MGBX3 and MGBX4. Open the Hybrid box.
14. Remount the two 560nH coils (L6 and L7) in the hybrid box in the same position as they were before (check with the marks) and close the Hybrid box.
15. Remove the flux-probe holder from the magnet bore. Dismount the two flux probes from the holder.
16. Remount the pick-up coils in the magnet bore. The distance between the edge of the pick-up coil and the body coil must be 14 mm (see Z3-6.14). Use spacers or shim plates to get the correct distance. Be sure to connect the pick-up coils to the correct connector (PU-right to MGBX13 and PU left to MGBX14).
17. Reconnect the body coil receive signal MMEX5 to the coilint.
18. Reconnect both tune cables MMFX2 and MMFX3.

3.8. HYBRID BOX TRANSMIT CURVE :

1. Put one of the linear flux probes in the rotation knob, and put the rotation knob in the holder. Place the holder in the magnet iso-center.
2. Rotate the holder to get the flux probe in a plane between rod 3 - 11.
3. Connect this flux probe to the pick-up coil input on the coilint board MMEX8 see also Figure 20.

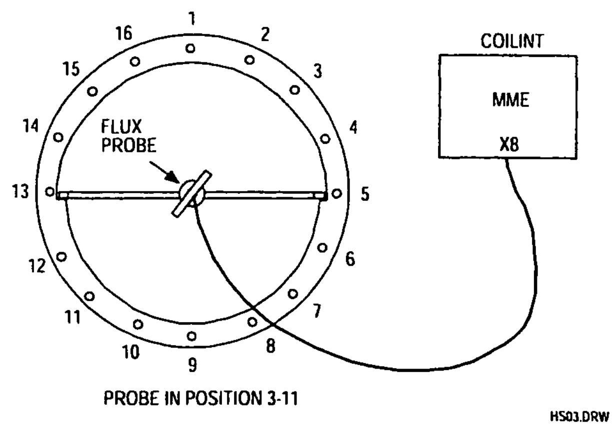


Figure 20 flux-probe positioning

4. Switch on the gradients
5. Select: *Adjust BBM Xmit*
6. Start the measurement with 4 times <proceed>.(First a reference measurement is done)
7. Select Display grid
8. Select Autoscale
9. On the image monitor a response is displayed with two 'humps'. Use trimmer C8 on the front of the hybrid box to get a symmetrical response, i.e. the amplitude of the two 'humps' must be as identical as possible. See Figure 22.

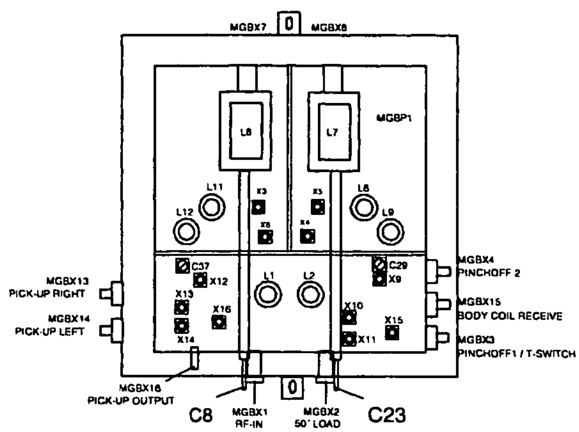


Figure 21 Hybrid Box

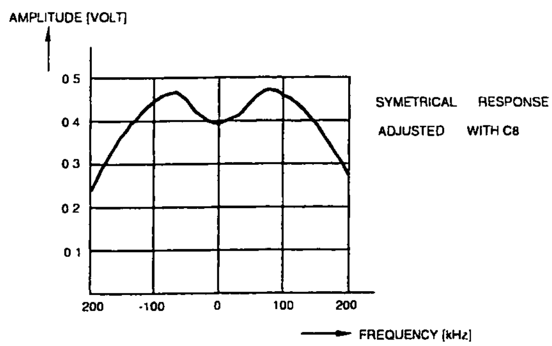


Figure 22 "Two humps"

10. Rotate the flux-probe 90° in the plane between the rods 7 - 15.

11. The two humps will be gone now. Use trimmer C23 to adjust the response signal.
    The response signal must be as flat as possible and symetrical.
12. Select. Display Min/Max

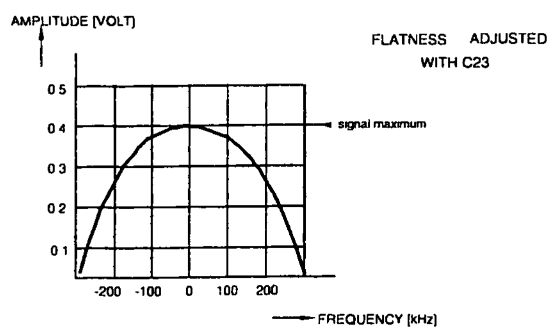


Figure 23 Signal adjustment

Repeat the adjustment for the Hybrid box transmit curve (the adjustments of trimmers C8 and C23) if the result is not satisfactory.

13. Replace the rotation knob and the flux probe with the quad flux-probe. The quad flux-probe can be mounted directly in the holder (without the 360° rotation knob). Place the holder in the magnet iso-center. The Quad-flux probe must be placed in the bore via the patient support side; **the cable connector pointing to the patient support** but the cable should leave the bore via the backside. The holder must be placed horizontal. Connect the quad flux-probe to MME X8.
14. Use C23 to adjust the flatness of the tune curve within specification. The response signal must be as flat and symetrical as possible.

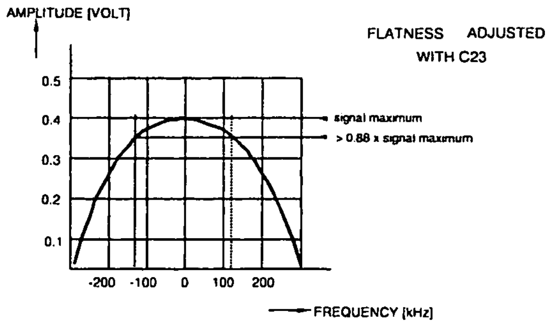


Figure 24 Adjustment bandwidth

Check that the signal is in specification with the next procedure:

15. Measure the signal maximum and log this value (Umax).
16. Select <Mod Display> and modify the value for Vertical Min to the value 0.88 x Umax.
17. The lowest values on the screen are the -1dB points of the curve. The center frequency is set to 21.285 kHz. For the transmit curve we are interested in the behaviour around 21.285 Mhz. To do this we have to measure the following: See also Figure 24.

18. Determine the frequency offset of the left -1dB point by playing with the zoom function. (Offset 1)
19. Determine the frequency offset of the right -1dB point by playing with the zoom function. (Offset 2)
20. Pick the lowest of the two values and multiply with 2.

21. The found value is the -1dB Bandwith. (B_{-1dB})

    Specification: $B_{-1dB} \geq 200$ *kHz*

22. When the specification has been met, stop the adjustment loop by pressing <STOP SCAN> on the console.
23. Switch off the gradients.
24. Remove the flux-probe holder from the magnet bore, and disconnect the quad flux-probe. Reconnect the pick-up coil signal to the pick-up coil input on the coilint.

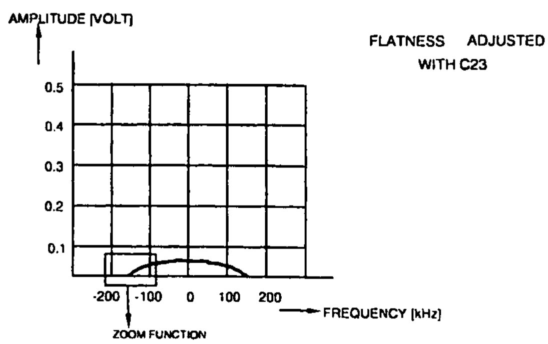


Figure 25 Bandwith check

3.9. HYBRID BOX RESONANCE FREQUENCY

1. Setup the flux-probe holder as shown in Figure 26. Connect one flux-probe to MMF X2 of the TRSW board (PFEI) and the other to MME X5 on the coilint (PFEI)
2. Select: *Adjust BBM receive* in Bodycoil procedures
3. Start the measurement with <proceed>
4. Rotate the probe-holder between rod 3 - 11.

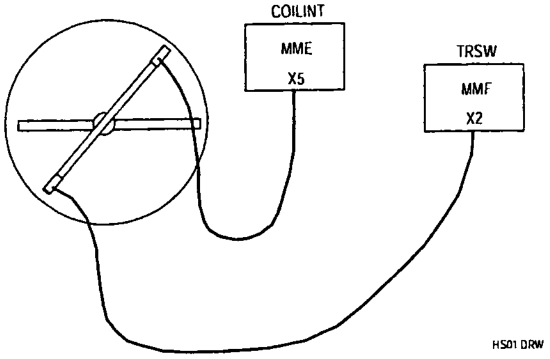


Figure 26 Measurement set-up

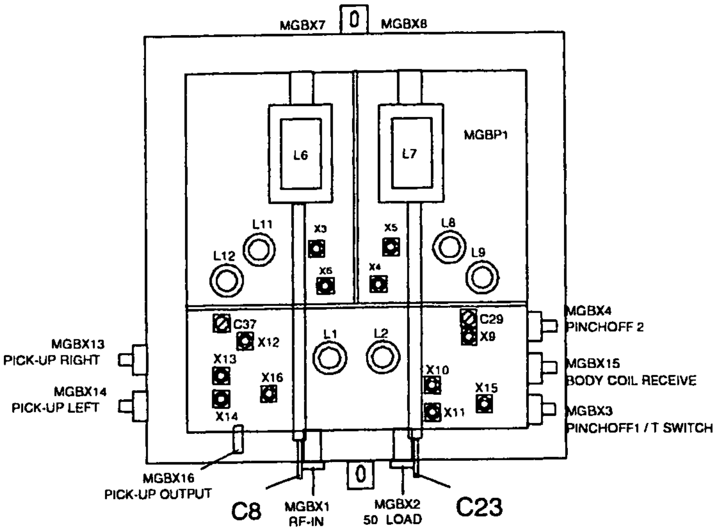


Figure 27 Hybrid Box

5. Adjust the offset frequency to a minimum value. The offset can be changed with trimmer C37. Trimmer C37 can be reached via an opening in the top plate of the hybrid box box.
   See also Figure 27.

   Specification: *freq. offset F(1): < ± 10 kHz*
   Q > 400

6. Rotate the probe-holder between rod 7 - 15.
7. Now use trimmer C29 to adjust the frequency offset to a minimum value.

   Specification: *freq. offset F(2): < ± 10 kHz and F(2) = F(1) ± 2.5 kHz*
   Q >400

3.10. ORTHOGONALITY QBC

1. Connect all cables for the QBC according the labels.
2. Mount the quadrature flux-probe in the flux-probe holder in the center of the QBC. The connector of the Quadrature flux probe must point to the patient support side of the magnet.
3. Disconnect Tune A and Tune B signals MMF X2 and MMF X3.
4. Connect a cable between Tune A MMF X2 and the Quadrature flux-probe. The cable must leave the bore via the front side of the magnet as straight as possible to avoid cable resonances.
5. Select *Adjust BBM* receive in body coil procedures.
6. **Minimize** this signal by turning the trimmers for rod 5 and 13 following the next procedure:
   - Measure the signal level at the resonance frequency.(use a marker)
   - Turn the trimmers for rod 5 and 13 each a half a turn clockwise.
   - Measure the signal level again. (at the resonance frequency)

   If the signal level is lower, continue by turning the trimmers for rod 5 and 13 clockwise and measure the signal level after each half a turn until you have found a minimum.
   If the signal level goes up, continue by turning the trimmers for rod 5 and 13 each a half a turn counter clockwise and measure the signal after each half a turn until you have found a minimum.

3.11. HYBRID BOX RESONANCE FREQUENCY

After the adjustment of the orthogonality, the hybrid box resonance frequency has to be repeated.

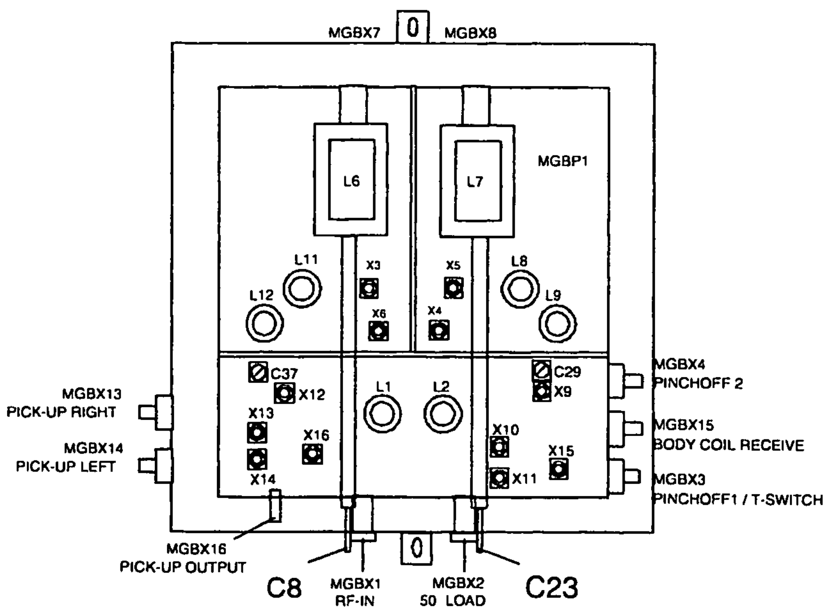


Figure 28 Hybrid Box

1. Setup the flux-probe holder as shown in Figure 26. Connect one flux-probe to tune A at connector MMFX2 of the TRSW board in the PFEI and the other to MME X5 on the coilint.
2. Select *Adjust BBM* receive in Bodycoil procedures
3. Start the measurement with <proceed>
4. Rotate the probe-holder between rod 3 - 11.

5. Adjust the offset frequency to a minimum value. The offset can be changed with trimmer C37. Trimmer C37 can be reached via an opening in the top plate of the Hybrid box box.

   Specification: *freq. offset F(1): < ± 10 kHz*
   Q > 400

6. Rotate the probe-holder between rod 7 - 15.
7. Now use trimmer C29 to adjust the frequency offset to a minimum value.

   Specification: *freq. offset F(2): < ± 10 kHz and F(2) = F(1) ± 2.5 kHz*
   Q >400

NOTE

The specification for both frequency offsets (F1) and (F2) is < 10 kHz but beside that, both frequencies should not differ more than 2.5 kHz.

3.12. EMPTY BODY COIL FREQUENCY DETERMINATION

From the 0.5T Body Coil the resonance frequency needs to be determined. This value will be called the empty_body_coil_frequency, and is used as a reference value. No tools are required to do this measurement.

1. To perform this measurement the bridge section and both cone-covers must be mounted, the bodycoil must be empty, no tools are required.
   The program uses 21.298 Mhz as reference frequency.

2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES.
4. Select. ENTER SERVICE MODE.
5. Select: SYSTEM TUNING
6. Select: SYSTEM TUNING TOOLS.
7. Select: INSTALLATION PROCEDURES.
8. Select: BODY COIL PROCEDURES
9. Select: DETERMINE BODY COIL PARAMETERS
10. Press <PROCEED> to start the measurement.
11. Read: - actual coil frequency
    - measured Q

12. Write down the empty body coil frequency and the Q in the MRL as a reference.

If you were doing an installation, continue with "Determination of the head coil parameters".

4. APPENDIX B: ADAPTION QHC CABLE TO RIGHT SIDE

4.1. INTRODUCTION

The quad head coil is standard delivered and adjusted with the cable on the left side of the coil.
This procedure can be used for changing the cable to the right side.

4.2. PROCEDURE DESCRIPTION

1. Put the head coil on a table and remove the cover on the bottom side that covers the driver PCB.
2. Disconnect the cables to the PCB and dismount the driver. On the other side of the coil, the bottom cover is already prepared to mount the Driver board. The PCB must be rotated over 180 degrees to mount the board on the right side.
3. Remove the bracket from the right side that covers the hole for the cable feed through.
4. Remove the cable from the left side. The hole can be covered with the bracket.
5. Dismount the rear cover of the head coil and dismount the cable bracket from the cover.
6. Turn it over 180 degrees so that the cable points to the right side of the coil.
7. Mount the bracket again and mount the cover back to the coil.
8. Mount the cables on the right side of the coil and connect them to the driver PCB.
   The connection cables on each side of the driver PCB should be separated from each other as good as possible to reduce the coupling. Use ty-wraps to secure the cables to the bottom cover if necessary.
9. Mount the cover for the PCB.

4.3. ADJUSTMENT QHC

After the cable has been moved to the right side the coil has to be checked/readjusted.
For procedure, see the relevant paragraphs in this section (15).

Section 15 RF Coils

Contents

1. QUAD BODY COIL (QBC) CHECK/ADJUSTMENT

The Quad Body Coil (QBC) is already adjusted in the factory. Therefore only a test has to be done to check the QBC and to determine the values for the MRL as a reference.
Verify that the QBC did not move during transportation by checking all excenter screws. When the excenters were loose, the QBC has changed position and re-adjustment of the QBC is necessary (See appendix A).

1.1. TOOLS

- Quadrature flux probe (4522 117 70151)
- Linear flux probe
- QBC flux probe holder
- Rotation knob
- 15 meter 50 ohm coaxial cable

1.2. HYBRID BOX + QBC RESONANCE FREQUENCY

1. Switch-off the gradient system.
2. Lower the patient table.
3. Remove the bridge section, the PICU front cover and the front cone.
   The QBC and the hybrid box in front of the QBC are visible now.
4. Remove the left magnet top cover
5. Mount the quadrature flux probe in the QBC flux probe holder from the backside of the magnet and position the holder in the center of the body coil with the connector of the Q-flux probe pointed to the backside of the magnet. Also the cable must be routed via the backside of the magnet
   The holder must be positioned horizontal.
6. Remove both tune coil cables MMFX2 and MMFX3 from the TRSW board (PFEI).
7. Connect a 50 ohm coaxial cable between tune A at connector MMFX2 and the quadrature flux probe. See Figure 1.

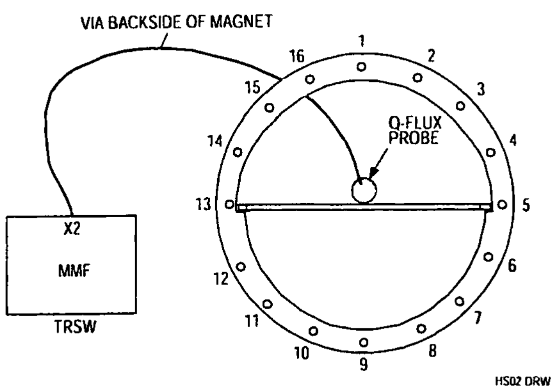


Figure 1 Measurement set-up for Hybrid + QBC resonance frequency (Front View)

8. Login : GYROSCAN.
9. Select: SCAN CONTROL.
10. Select: SCAN UTILITIES.
10. Select: ENTER SERVICE MODE.
11. Select: SYSTEM TUNING
12. Select: SYSTEM TUNING TOOLS.
13. Select: INSTALLATION PROCEDURES.
14. Select: BODY COIL PROCEDURES
15. Select: ADJUST QBC RECEIVE
16. Enter: Proceed to start the measurement
17. Select: AUTOSCALE

18. Determine the values for the frequency offset and the Q-factor.

Offset $\leq \pm$ 10 kHz
Q: $\geq$ 260 (Typical value Q: $\geq$ 290)

19. If the measured values are within specification, restore the original situation and continue with paragraph "Determination of Body coil parameters".
If the offset or Q-value are not within specification, adjustment of this mode must be done, according next paragraph "Hybrid box + QBC resonance frequency adjustment.

1.3. HYBRID BOX + QBC RESONANCE FREQUENCY ADJUSTMENT

If one of the values in previous paragraph were not within specification adjustment of the hybrid box must be done as follows:

Trimmers C31 and C32 in the hybrid box are used to adjust the two modes of the QBC and hybrid box. To reach both trimmers, the top plate of the hybrid box must be removed. See Figure 2.

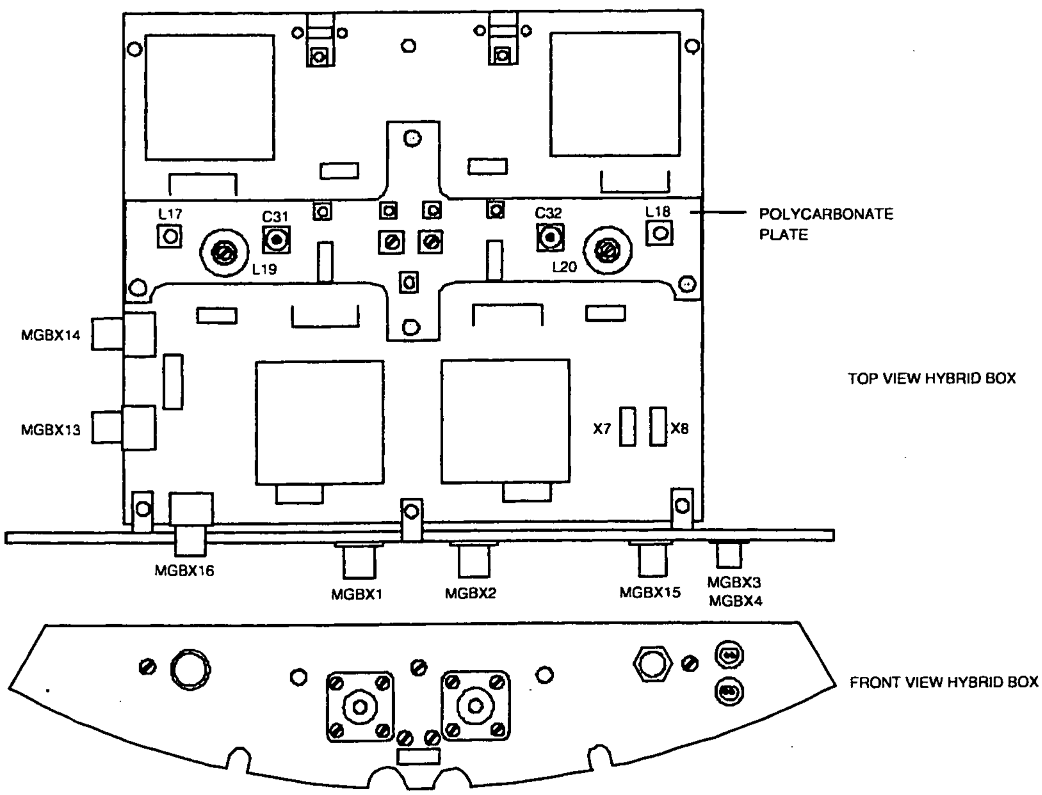


Figure 2 Hybrid box 1.0 T (15 kW)

1. Position the **linear flux probe** with use of the rotation knob in the QBC flux probe holder in the center of the QBC. Connect the cable of the flux probe to tune A signal at MMFX2. The cable should leave the magnet via the backside to prevent coupling with the hybrid box. See Figure 3.

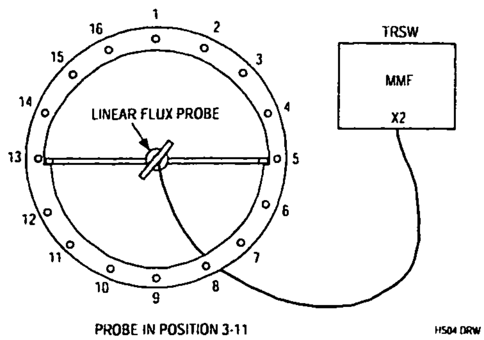


Figure 3 Frequency adjustment mode 3 - 11 (front view)

Procedure:

2. Position the flux probe in position 3 - 11. (3)

3. Login : GYROSCAN.
4. Select: SCAN CONTROL.
5. Select: SCAN UTILITIES.
6. Select: ENTER SERVICE MODE.
7. Select: SYSTEM TUNING
8. Select: SYSTEM TUNING TOOLS.
9. Select: INSTALLATION PROCEDURES.
10. Select: BODY COIL PROCEDURES
11. Select: ADJUST QBC RECEIVE

12. Adjust the frequency with trimmer C31 as good as possible to the resonance frequency. The offset can be checked on the monitor.
    Specification: *F(1) < 10 kHz*

13. Position the flux probe in position 7-15 (4).

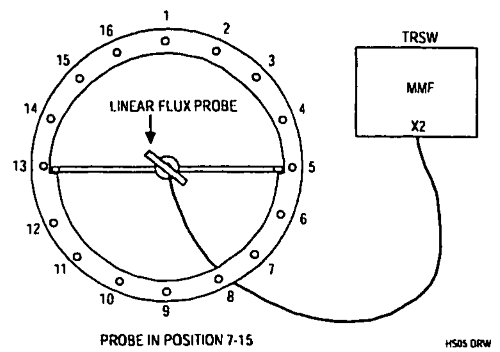


Figure 4 Frequency adjustment mode 7 - 15 (front view)

14. Adjust the frequency with trimmer C32 as good as possible to the resonance frequency. The offset can be checked on the monitor.
    Specification: *F(2) < 10 kHz*

15 Remove the linear flux probe and mount the quadrature flux probe in the QBC flux probe holder and position the holder in the center of the body coil with the connector of the Q-flux probe pointed to the backside of the magnet. Also the cable must be lead via the backside of the magnet. See Figure 5.

NOTE

The quadrature flux probe must be positioned in this way because it makes a quadrature RF-field. If the Q-flux probe is positioned via the front side, the orthogonal field would be picked up by the QBC which results in no signal on the monitor.

16. Connect a 50 ohm coaxial cable between tune A at connector MMFX2 and the quadrature flux probe. See Figure 5.

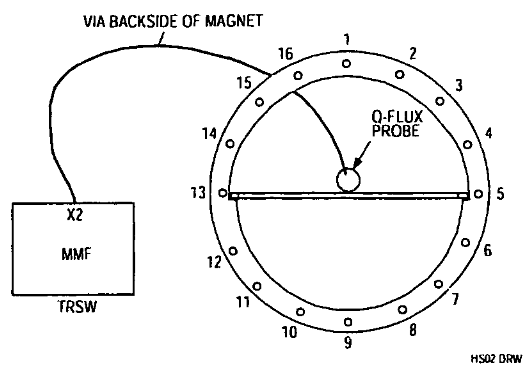


Figure 5 Connections of the Quadrature flux probe (Front View)

17. Select. ADJUST QBC RECEIVE
18. Determine the values for the frequency offset and the Q-factor.

Offset $\leq \pm$ 10 kHz
Q: $\geq$ 260 (Typical value Q: $\geq$ 290)

19. If the specification can not be reached, adjustment of the body coil is necessary. See appendix A at the end of this section.

1.4. DETERMINATION OF BODY COIL PARAMETERS

To perform this measurement the bridge section and both cone-covers must be mounted.

1. Mount the **quadrature** flux probe in the QBC flux probe holder and position the holder in the center of the body coil **with the connector of the Q-flux probe pointed to the backside of the magnet**. Also the cable must be lead via the backside of the magnet.
2. Connect a cable between tune at connector MMFX2 and the quadrature flux probe.
   See Figure 5.
   The program uses 42.576 MHz as reference frequency.

3. Login : GYROSCAN.
4. Select: SCAN CONTROL.
5. Select: SCAN UTILITIES.
6. Select: ENTER SERVICE MODE.
7. Select: SYSTEM TUNING
8. Select: SYSTEM TUNING TOOLS.
9. Select: INSTALLATION PROCEDURES.
10. Select: BODY COIL PROCEDURES
11. Select: DETERMINE BODY COIL PARAMETERS
12. Now the measurement can be started with <Proceed>.
    No errors are allowed.

13. Write the actual body coil frequency and Q-factor in the MRL (Section 22) as a reference.

2. ADJUSTMENT OF THE QUAD HEAD COIL

2.1. INTRODUCTION

The Quad Head Coil (QHC) for the 1.0T system is provided with an electronic load.
The quad head coil is already adjusted in the factory Adjustment during installation is only necessary if the cable has to be installed at the other side of the coil or if the Image Quality measurements for the head coil are not within specifications.

2.2. ADAPTATION CABLE TO THE RIGHT SIDE

The procedure to adapt the cable to the other side of the coil, is written in Appendix B at the end of this section. After the adaptation of the cable, readjustment/check of the head coil is necessary.

2.3. QUAD HEAD COIL FREQUENCY AND ORTHOGONALITY ADJUSTMENT

NOTE

The following procedure is only necessary when the cable is adapted to the right side of the coil, or when the QHC adjustment is suspected e.g. when the Image Quality measurements are not within specification.

The QHC is checked on frequency, Q-factor and orthogonality. The adjustment frequency for the QHC is 42.576 Mhz. Frequency-offset with respect to this frequency and orthogonality can be adjusted if not within specification. Q-factor can not be adjusted but is optimized by adjusting frequency and orthogonality.

The quad head coil can be seen as two coils working in different modes: there is a phase shift of 90° between both coils. Each coil has a pre-amplifier. By disconnecting one coil (removal of a jumper on the quad pre-amp board), the resonance frequency of the other coil can be adjusted.
The adjustment for orthogonality must be done to ensure an exact 90° phase difference between the two coils.

Preparation:

1. Read the previous Note if not already done.
2. Remove the left magnet top cover.
3. Remove both tune coil cables MMFX2 and MMFX3
4. Connect the head coil to the Surface coil connector 1 (SC1).
5. Move the slide-able part of the head coil to the front (scanning) position. This is necessary to reduce the influence of the quad driver board on the coil.

6. Unscrew the rear cover of the coil and mount it back to the head coil using tape in stead of screws. For the adjustment you need to take the cover off but the measurements are only valid when the rear cover is closed, because the values are influenced by this cover.
7. Place the head coil on the table in such a way that the rear of the head coil aligns with the end of the patient table, closest to the magnet (reference point).
8. Use the light visor and the marker on the head coil to position the head coil in the iso-center of the magnet via the "travel-to-scan-plane".
9. Connect the **linear flux probe** cable to MMFX2. (DO NOT USE THE QUADRATURE FLUX PROBE)
10. Position the linear flux probe in position L8 - L12 (see Figure 6). The flux probe holder must be positioned against the front ring of the head coil. The flux probe holder must be used without the 360° rotation knob.

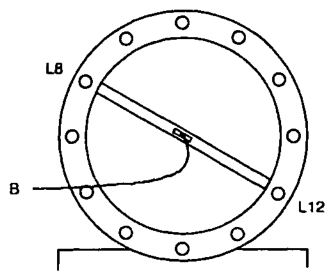


Figure 6 Flux probe in position L8 - L12, seen from patient support end

NOTE

On the front ring of the head coil, marker lines indicate rod L5, L8, L11 and L12. Make sure the flux probe cable is apart from the head coil cable and that it leaves the bore as straight as possible.

Start the adjustment loop as follows:

11. Login : GYROSCAN.
12. Select: SCAN CONTROL.
13. Select: SCAN UTILITIES.
14. Select: ENTER SERVICE MODE.
15. Select: SYSTEM TUNING
16. Select: SYSTEM TUNING TOOLS.
17. Select: INSTALLATION PROCEDURES.
18. Select: HEAD COIL PROCEDURES

19. Enter: <PROCEED> to start the measurement
20. Select: AUTOSCALE

NOTE

The measured values are only valid in case the rear cover of the head coil is closed during the measurement. Disconnect the head coil when changing the jumpers.

STEP 1 : Frequency check/adjustment first mode

1. Place the flux probe in position L8-L12
2. Remove rear cover.
   Jumper W1: Position 2
   Jumper W2: Removed
   See drawing for jumper location (See Figure 7).

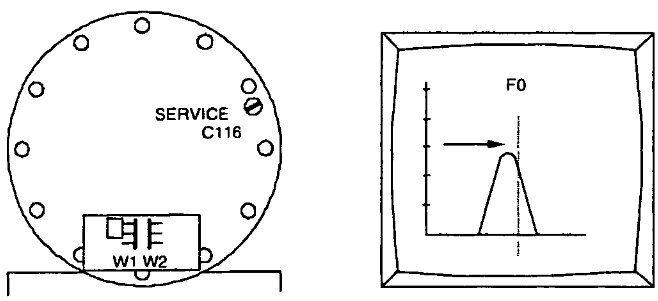
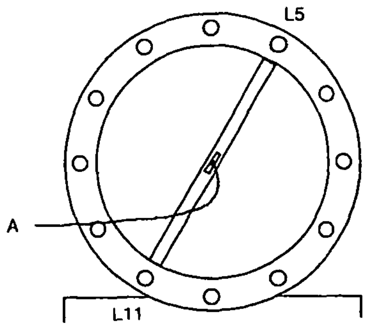


Figure 7 Frequency adjustment Mode L8 - L12

3. Put back the cover, read and write down:
   Q(1) = Coil Q-factor.
   U_1 = Voltage at Tune Top (maximum voltage)
   F(1) = offset frequency
   Specifications:
   Q(1): > 250
   F(1): < ± 25 kHz
4. If F(1) does not meet the specifications, adjust the frequency with trimmer C116 (See Figure 7)
5. If the frequency is within specification, write down the Q-value (Q(1)), Voltage at Tune Top U1 and the Offset frequency (F(1)).

STEP 2 : Frequency check/adjustment second mode

1. Install the flux probe in position L5-L11.
2. Remove rear cover. Remove jumper W1 and install jumper W2 in position 2. See Figure 9 for jumper location.



3. Replace rear cover, read and write down:

Q(2)	= Coil Q-factor.
U_2	= Voltage at Tune Top (maximum voltage)
F(2)	= offset frequency
Specifications:	
Q(2)	> 250
F(2)	≤ ± 25 kHz

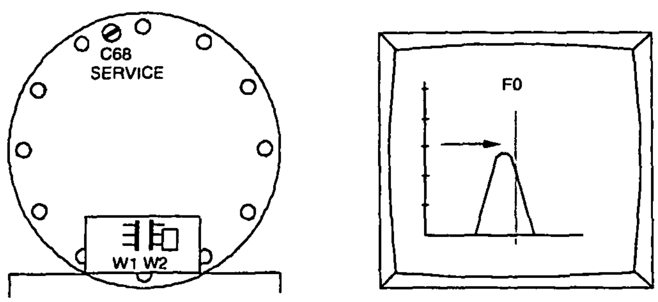


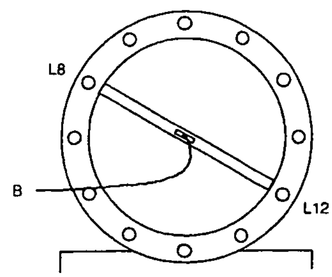
Figure 9 Frequency adjustment mode L5 - L11

4. If F(2) does not meet the specifications, adjust the frequency with trimmer C68 (see Figure 9).

5. If the frequency is within specification, write down the Q value (Q(2)), Voltage at Tune Top (U_2) and the Offset frequency (F(2)).

STEP 3: Orthogonality check/adjustment (1)

1 Install flux probe to position L8 - L12. The flux probe must be positioned against the front ring of the head coil.



2. Read and write down:
   - U_3 = **Voltage at Center Frequency**

   Calculate the attenuation with next formula

$$Att(1) = 20\log\left(\frac{U_2}{U_3}\right)$$

The typical value for the orthogonality is: Att(1) > 17 dB

If this value is reached, continue with **STEP 4**.

If this value is not reached, adjust the Tune Voltage at the marker position to a value as low as possible using the next procedure:
(See Drawings manual Z3-7.19 for exact trimmer location)

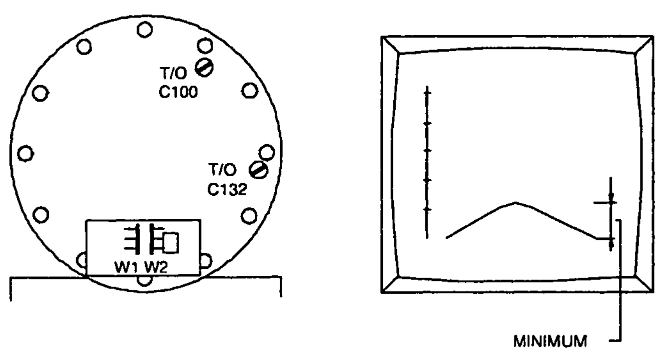


Figure 11 Adjustment of the orthogonality

3. Probe position between L8/L12 (As the previous adjustment). Put the jumpers in the following position:
   W1 -> removed
   W2 -> position 2

4. Turn C100 a little clockwise, see if response decreases. If not turn C100 counter clockwise. Turn C132 the same amount in the other direction(*).

   (*) Other direction means: If you have set for more capacitance with C100, turn C132 in the other direction until you have the same amount capacitance less. Minimize the response by turning both trimmers successively, as described above, until a minimum is reached.
   Write down the measured voltage U3.

NOTE

Sometimes, depending on their initial position, both trimmers have to be turned in the same direction, or sometimes in the opposite direction.

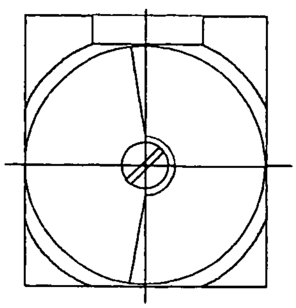


Figure 12 Trimmer in Middle position

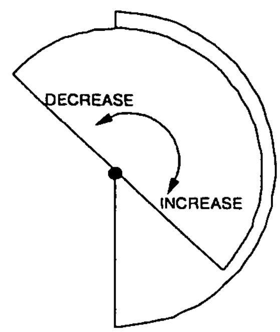


Figure 13 Increasing / decreasing capacitance

NOTE

The orthogonality measurement is not accurate due to the mechanical construction of the coil. Adjust the orthogonality as good as possible. It is not necessary to reach the typical value. The correct functioning of the coil is checked during the Image Quality performance measurements.

STEP 4 : Orthogonality check/adjustment (2)

1. Install flux probe to position L5 - L11. The flux probe must be positioned against the front ring of the head coil and put the jumpers in the next position:
   W1 position 2
   W2 removed (See Figure 15)

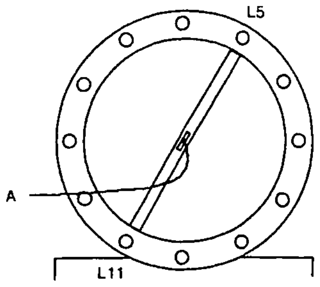


Figure 14 Flux probe position mode L5-L11

2. Read and write down:
   - U_4 = **Voltage at Center Frequency**

   Calculate the attenuation with next formula:

$$Att(2) = 20\log(\frac{U_1}{U_4})$$

The typical value for the orthogonality is: Att(1) > 17 dB

If this value is reached, continue with paragraph **"Final actions"**.

If this value is not reached, adjust the Tune Voltage at the marker position to a value as low as possible using the next procedure:
(See Drawings manual Z3-7.19 for exact trimmer location)

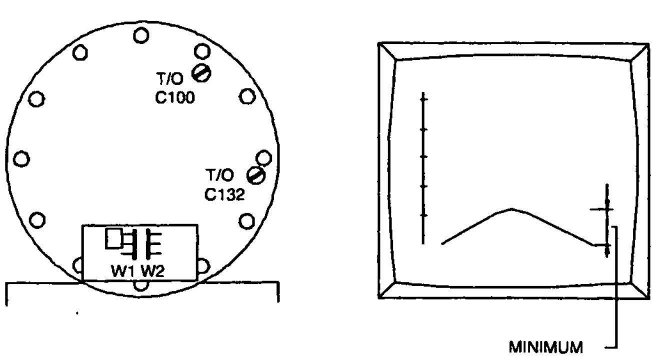


Figure 15 Adjustment orthogonality

3. Probe position between L5/L11 (As the previous adjustment). Put the jumpers in the following position:
   W1 -> position 2
   W2 -> removed (see Figure 15)
4. Turn C100 a little clockwise, see if response decreases. If not turn C100 counter clockwise **and** turn C132 the same amount in the other direction(*).

   (*) Other direction means: If you have set for more capacitance with C100 turn C132 in the other direction until you have the same amount capacitance less. Minimize the response by turning both trimmers successively, as described above, until a minimum is reached.
   Write down the measured voltage U_4.

NOTE

Sometimes, depending on their initial position, both trimmers have to be turned in the same direction, or sometimes in the opposite direction.

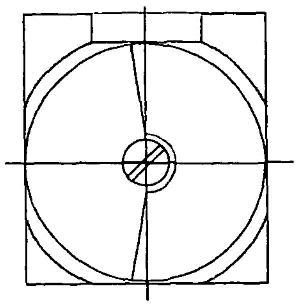


Figure 16 Trimmer in Middle position

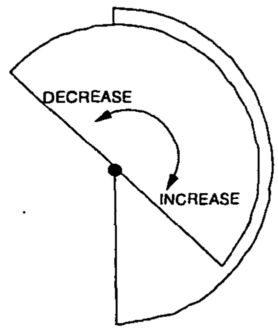


Figure 17 Increasing / decreasing capacitance

NOTE

The orthogonality measurement is not accurate due to the mechanical construction of the coil. Adjust the orthogonality as good as possible. It is not necessary to reach the typical value. The correct functioning of the coil is checked during the Image Quality performance measurements.

2.4. FINAL ACTIONS

1. Repeat step 1 - 4 until no adjustments are necessary. Every adjustment can influence other parameters.
2. If you have followed the procedure and no adjustments were necessary, fill in the measured values, in the MRL according Table 1. (next page)
3. Install both jumpers in position 2 and mount the rear cover.
4. Remove the flux-probe holder out of the head coil and reconnect tune coil MMFX2 and MMFX3 to the TRSW board.
5. Read and write down as a reference in MRL (no specification):
   F(head) = Offset frequency
   Q(head) = Coil Q-factor
6. Restore normal situation.
7. Write down the final values in the table in the MRL (Section 22) as a reference.

Probe position	Parameters with specification		Measured values
Step 1) Jumper position: W1: Position 2 W2: Removed L8 - L12	Q(1)	> 250	
	Voltage U_1		
	F(1)	< ± 25 kHz	
Step 2) Jumper position: W1. Removed W2. Position 2 L5 - L11	Q(2)	> 250	
	Voltage U_2		
	F(2)	< ± 25 kHz	
Step 3) Jumper position: W1: Removed W2: Position 2 L8 - L12	Voltage U_3		
	Att(1)	typical value > 17 dB	
Step 4) Jumper position: W1: Position 2 W2: Removed L5 - L11	Voltage U_4		
	Att(2)	typical value > 17 dB	
Both jumpers positioned (reference values)	F(head)	n.a.	
	Q(head)	n.a.	

Table 1 - Measurement results

3. APPENDIX A: READJUSTMENT PROCEDURE QUAD BODY COIL

Only if the hybrid box + quad body coil could not be adjusted within the required specifications, (re)adjustment of the separated QBC should be done. Adjustment is also necessary when the QBC is replaced.

The re-adjustment is done in three steps:

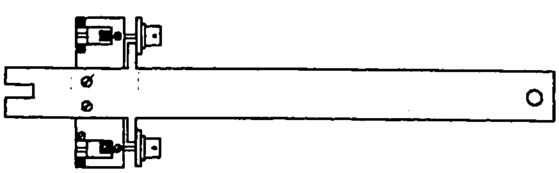
1. Horizontal and vertical mode are adjusted to the resonance frequency.
2. The diagonal modes are set to the resonance frequency
3. The complete QBC is set to the correct frequency.

3.1. TOOLS

- Flux probe holder
- Rotation knob for flux probe
- QBC adjustment tool
- Extension cables 1 m (belonging to QBC adjustment tool)
- 50 ohm coaxial cable 15 m
- Quadrature flux probe 42 MHz.
- Two special trimmer adjustment tools.

3.2. MEASUREMENT SET-UP

1. Switch off the gradient system.
2. Remove the lower system filter box cover and switch off the FEPSU.
3. Remove the bridge section, the PICU cover and the front cone cover (if not already done).The hybrid box and QBC are visible now.
4. Disconnect the cables from the hybrid box.
5. Open the hybrid box by taking off the top-plate and loosen the screws of the connectors for the two rigid coax connections of the QBC (two screws, per coax connection!) Refer to drawing Z3-7.15
6. Remove the hybrid box.
7. Remove the inner top-cover of the QBC by carefully pulling out the covers by hand: start on one side of the cover, when one side is loose, the cover can be taken out. Mark the front and rear side of the Top-cover of the QBC to be able to put it back in the same way.
8. Remove the inner lower-cover: start on one side.
9. Mount the special QBC adjustment tool at the QBC rigid coax connection. Do not use force to fasten the screws.



4522 131 3500. SERVICE TOOL QBC ADJUST T10-NT / ACS-NT

Figure 18 QBC adjusting tool

10. The QBC adjustment tool (with the two special 1 1 meter extension cables) must be connected to the QBC connections in stead of the hybrid box. (see Figure 20). The extension cables must be left open on one end. **(not terminated).**

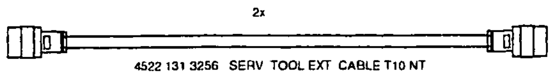


Figure 19 QBC adjustment tool extension cables

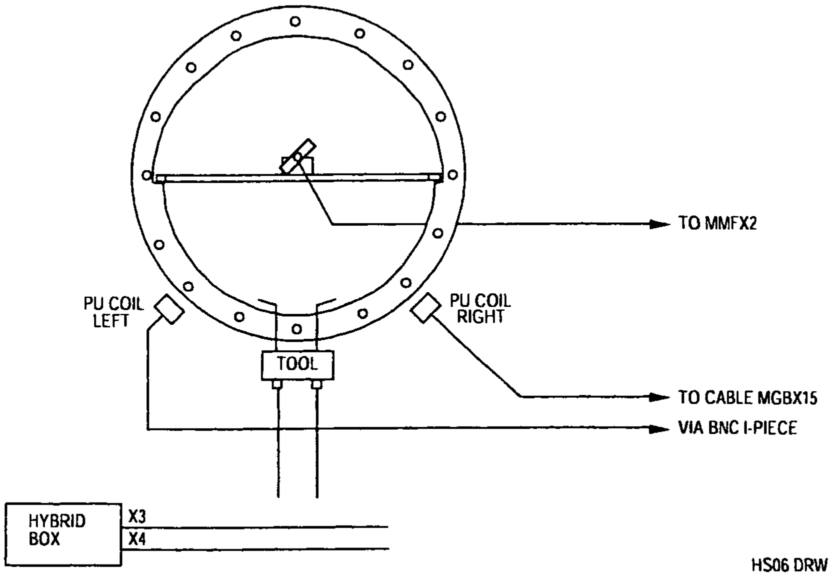


Figure 20 Measurement set-up for adjusting the QBC without hybrid box (front view)

11. Mount the top plate to the hybrid box with a few screws (Safety !).
12. Position the hybrid box just outside the magnet bore in such a way that the cables MGBX3 and MGBX4 can be connected to the hybrid box. This will prevent that you will get error messages during the Manual coil tuning program. Make sure that the bottom side of the hybrid box does not make contact with the gradient coil. The top plate of the hybrid box must be present to avoid touching the 300 Volts.
13. Connect the left pick-up coil (looking from the front of the magnet) to the MR response cable MGBX15 by using a BNC I-piece. The pick-up coils are used to receive.
14. Mark or measure the present position of the pick-up coils and shift both pick-up coils as close as possible to the QBC to get the best signal.

15. On the front side and the back side of the QBC you can find the adjustment screws for the trimmers. Loosen the screws that lock the trimmers for all QBC rods. (see Figure 21 and Figure 22)

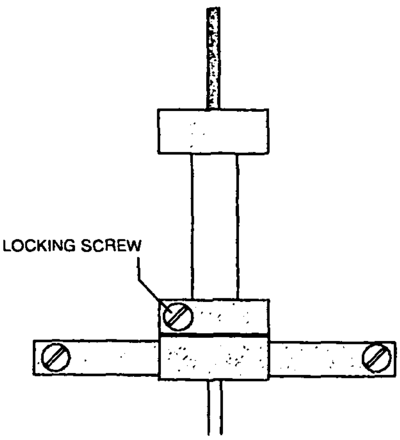


Figure 21 Trimmer in detail

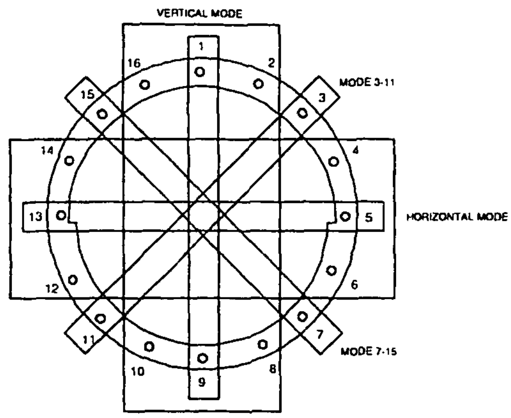


Figure 22 QBC frequency mode

You can see that two types of rods are used: thick rods (with thick trimmers) and thin rods (with thin trimmers). For each type of trimmer a separate trimmer tool (supplied with the system) must be used.

16. Mount the top QBC inner cover (check for correct orientation). Position one side of the cover in its place and use your hands to push the other side in position. Use your fist if necessary to make the cover "click" in position.
    During the measurement the top-inner cover must be mounted because it introduces a frequency shift. The bottom cover can be left out.
17. Remove the left magnet top cover and disconnect both tune cables MMFX2 and MMFX3.
18. Position the QBC flux probe holder horizontal in the center of the QBC and mount the linear flux probe via the rotation knob in the flux probe holder. Connect the flux probe to tune A at MMFX2. The flux probe is used to transmit. It is advised to route the cable via the backside of the magnet to the tune box.
19. Switch on the FEPSU.

3.3. ADJUSTMENT QBC

1. Login : GYROSCAN.
   Select: SCAN CONTROL.
   Select: SCAN UTILITIES.
   Select: ENTER SERVICE MODE.
   Select: SYSTEM TUNING
   Select: SYSTEM TUNING TOOLS.
   Select: INSTALLATION PROCEDURES.
   Select: BODY COIL PROCEDURES

2. Select ADJUST FREQUENCY QBC and start the measurement.

3.4. FINE ADJUSTMENT SEPARATE MODES

1. Position the flux probe to position 7 - 15 (see Figure 23).

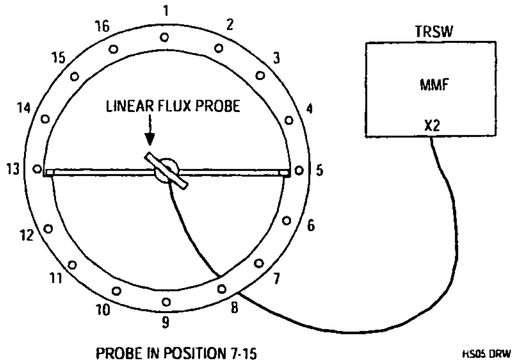


Figure 23 Flux probe position 7 - 15

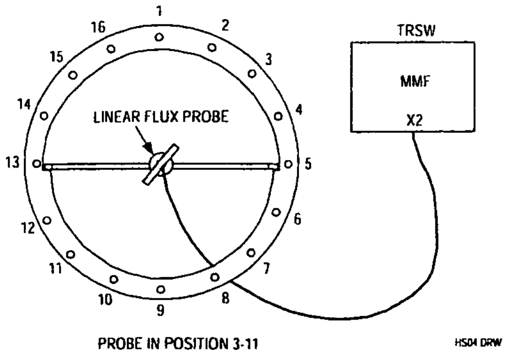


Figure 24 Flux probe position 3 - 11

2. Measure the frequency offset and write down this value in the table below.
3. Position the flux probe in position 3 - 11 (see Figure 22) and connect the right pick-up coil in stead of the left pick-up coil to the receive cable MGBX15.
4. Measure the offset and write down the value in the table below.
5. Connect the left pick-up coil and position the flux probe in horizontal position and measure the offset Fill in the table.
6. Put the flux probe in vertical position and measure the offset. Fill in the table.

Linear Flux probe	Specification	Measured
Mode 7 – 15 Mode 3 – 11	F < ± 10 kHz F < ± 10 kHz	
Mode horizontal	F < ± 10 kHz	
Mode vertical	F < ± 10 kHz	

To adjust the QBC, use the following trimmers according the measured offset: (See Figure 22)

Mode 7 – 15: trimmers 7 and 15

Mode 3 – 11: trimmers 3 and 11

Horizontal mode: For offset > 20 kHz. use trimmers 4, 5, 6 and 12, 13, 14
For offset < 20 kHz. use trimmers 5 and 13

Vertical mode: For offset > 20 kHz. use trimmers 16, 1, 2 and 8, 9, 10
For offset < 20 kHz. use trimmers 1 and 9

NOTE

Turn all trimmers in the same direction (inwards (clockwise) for lower frequency or outwards (counter-clock wise) for higher frequency). For each rod the trimmers on the front and backside have to be turned with the same amount in the same direction.

Specification: *Frequency offset for each mode ≤ 10 kHz / frequency difference between modes ≤ 5 kHz.*

7. Start with the mode with the largest offset and adjust all modes to the centre frequency (± 10 kHz.)
8. Use the left pickup coil for modes 7 - 15, horizontal mode and vertical mode, and the right pickup coil for the mode 3 - 11.
9. Use small frequency steps (e.g. 1/4 turn at a time).
10. Check the modes until no adjustments are necessary any more.
11. Dismount the QBC inner covers and lock all trimmers.
12. Re-install the QBC inner covers.
13. Measure all modes, fill in the next table and stop the measurement.

Linear Flux probe	Specification	Measured
Mode horizontal Mode vertical	F < ± 10 kHz F < ± 10 kHz	
Mode 3 - 11	F < ± 10 kHz	
Mode 7 - 15	F < ± 10 kHz	

NOTE

If you loose count during adjustment, you should start with the initial position. The procedure to adjust the QBC from the start position is described in DSS.

3.5. PREPARATION FOR MEASURING QBC RECEIVE MODE

Switch off the FEPSU.

1. Remove the flux probe and flux probe holder from the QBC.
2. Disconnect cables MGBX3 and MGBX4 from the hybrid box.
3. Remove the QBC adjustment tool from the QBC. Be careful with the coaxial connections of the QBC when disconnecting.
4. Mount the hybrid box in front of the QBC and connect all the cables to it. Position the pick-up coils back to their original position. (Check the marks or position to 11 mm from the QBC front ring)
   Make sure that the QBC detune cable is lead in the middle underneath the hybrid box.
5. Switch on the FEPSU

3.6. HYBRID BOX + QBC RESONANCE FREQUENCY ADJUSTMENT

Trimmers C31 and C32 in the hybrid box are used to adjust the two modes of the QBC and hybrid box. Both trimmers can be reached by removing the top plate of the hybrid box.

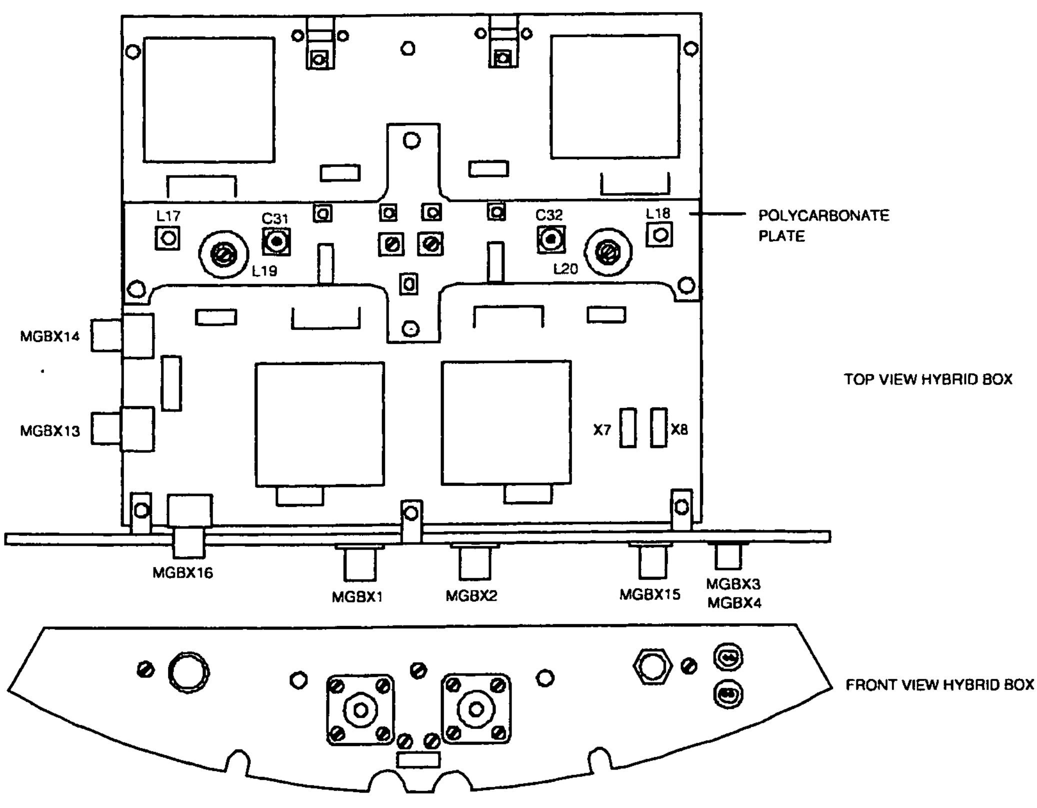
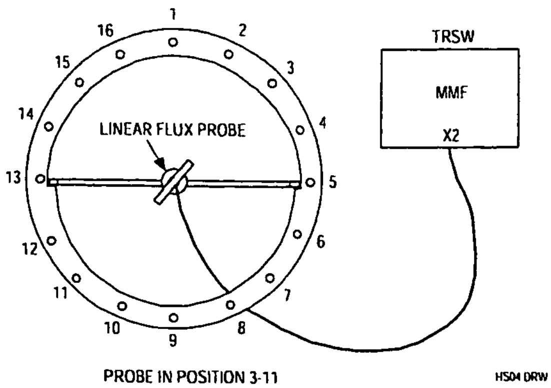


Figure 25 Hybrid box 1.0T (15 kW) with trimmer C31 and C32

6. Position the linear flux probe with use of the rotation knob in the QBC flux probe holder in the center of the QBC. Connect the cable of the flux probe to tune A signal at connector MMFX2. See Figure 26.



To prevent coupling problems with the Hybrid box, the cable must leave the bore via the backside of the magnet.

Figure 26 Frequency adjustment mode 3 - 11

Procedure:
Position the flux probe in position 3-11 (Figure 26).

7. Login : GYROSCAN.
8. Select: SCAN CONTROL.
9. Select: SCAN UTILITIES.
10. Select: ENTER SERVICE MODE.
11. Select: SYSTEM TUNING
12. Select: SYSTEM TUNING TOOLS.
13. Select: INSTALLATION PROCEDURES.
14. Select: BODY COIL PROCEDURES
15. Select: ADJUST QBC RECEIVE

16. Adjust the frequency with trimmer C31 to the resonance frequency. The offset can be checked on the monitor.

Specification: $F(1) < 10$ kHz

17. Position the flux probe in position 7-15 (Figure 27).

Remind to lead the flux probe cable via the backside of the magnet.

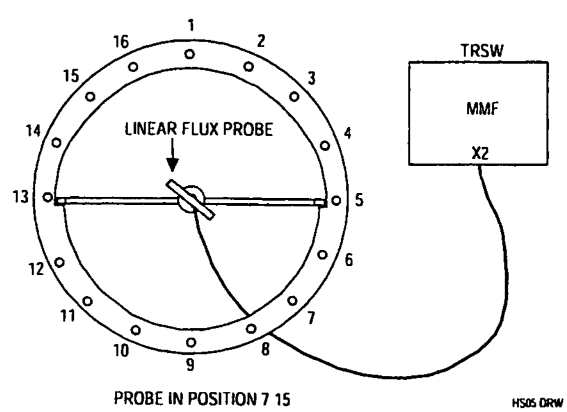


Figure 27 Frequency adjustment mode 7 - 15

Adjust the frequency with trimmer C32 to the resonance frequency. The offset can be checked on the monitor.

Specification: *F(2) < 10 kHz*

18 Mount the quadrature flux probe in the QBC flux probe holder and position the holder in the center of the body coil with the connector of the Q-flux probe pointed to the backside of the magnet. Also the cable must be lead via the backside of the magnet.

19. Connect a cable between tune A at connector MMFX2 and the quadrature flux probe. See Figure28.

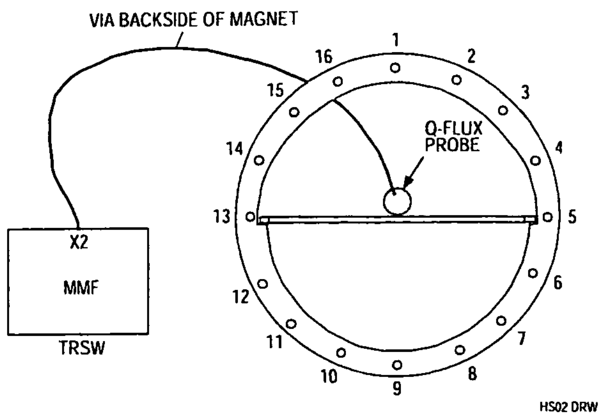


Figure 28 Connection of the Quadrature flux probe (Front View)

20. The same program is used : ADJUST QBC RECEIVE

21. Determine the values for the frequency offset and the Q-factor.

Offset ≤ ± 10 kHz
Q ≥ 260 (Typical value ≥ 290)

3.7. RESTORING NORMAL CONDITION

1. Switch off the FEPSU.

2. Connect all cables to their original position.

3. Mount the cone cover, the PICU front cover and the bridge section.

4. Switch on the FEPSU.

3.8. DETERMINATION OF BODY COIL PARAMETERS

To perform this measurement the bridge section and both cone-covers must be mounted.

1. Mount the **quadrature** flux probe in the QBC flux probe holder and position the holder in the center of the body coil **with the connector of the Q-flux probe pointed to the backside of the magnet**. Also the cable must be lead via the backside of the magnet.
2. Disconnect both tune cable MMFX2 and MMFX3 and connect a cable between tune A at connector MMFX2 and the quadrature flux probe.
   See Figure 28.
   The program uses 42.576 Mhz as reference frequency.

3. Login : GYROSCAN.
4. Select: SCAN CONTROL.
5. Select: SCAN UTILITIES.
6. Select: ENTER SERVICE MODE.
7. Select: SYSTEM TUNING
8. Select: SYSTEM TUNING TOOLS.
9. Select: INSTALLATION PROCEDURES.
10. Select: BODY COIL PROCEDURES
11. Select: DETERMINE BODY COIL PARAMETERS
12. Now the measurement can be started with <Proceed>.
    No errors are allowed.
13. Write the actual body coil frequency and Q-factor in the MRL as a reference.

4. APPENDIX B: ADAPTION QHC CABLE TO RIGHT SIDE

4.1. INTRODUCTION

The quad head coil is standard delivered with the cable on the left side of the coil.
This procedure can be used for changing the cable to the right side

4.2. PROCEDURE DESCRIPTION

1. Put the head coil on a table and remove the cover on the bottom side that covers the driver PCB
2. Disconnect the cables to the PCB and dismount the driver. On the other side of the coil, the bottom cover is already prepared to mount the driver board. In the PCB two extra holes are already present to mount the board on the right side.
3. Remove the bracket from the right side that covers the hole for the cable feed through.
4. Remove the cable from the left side. The hole can be covered with the bracket.
5. Dismount the rear cover of the head coil and dismount the cable bracket from the cover.
6. Turn it over 180 degrees so that the cable points to the right side of the coil.
7. Mount the bracket again and mount the cover back to the coil.
8. Mount the cables on the right side of the coil and connect them to the driver PCB.
   The connection cables on each side of the driver PCB should be separated from each other as good as possible to reduce the coupling. Use cable binders to secure the cables to the bottom cover.
9. Mount the cover for the PCB.

4.3. ADJUSTMENT QHC

After the cable has been moved to the right side the coil has to be checked/readjusted
For procedure, see the relevant paragraphs in this section (15).

Section 15 RF Coils

Contents

1. QUAD BODY COIL (QBC) CHECK/ADJUSTMENT

The Quad Body Coil is already adjusted in the factory. Therefore only a test has to be done to check the QBC and to determine the values for the MRL as a reference.

1.1. TOOLS

- Quadrature flux probe (4522 117 70151)
- Linear flux probe
- QBC flux probe holder
- Rotation knob
- 15 meter 50 ohm coaxial cable

1.2. HYBRID BOX + QBC RESONANCE FREQUENCY

1. Switch-off the gradient system
2. Remove the bridge section, the PICU front cover and the front cone.
   The QBC and the hybrid box in front of the QBC are visible now.
3. Remove the left magnet top cover.
4. Mount the quadrature flux probe in the QBC flux probe holder from the backside of the magnet and position the holder in the center of the body coil with the connector of the Q-flux probe pointed to the backside of the magnet. The cable must be routed via the backside of the magnet.
   The holder must be positioned horizontal.
5. Remove both tune coil cables MMFX2 and MMFX3 from the TRSW board (PFEI).
6. Connect a 50 ohm coaxial cable between tune A at connector MMFX2 and the quadrature flux probe.
   See Figure 1.

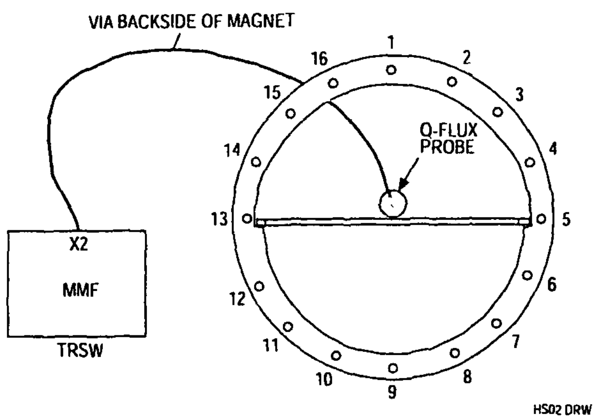


Figure 1 Measurement set-up for hybrid +QBC resonance frequency (Front View)

7. Login : GYROSCAN.
8. Select: SCAN CONTROL.
9. Select: SCAN UTILITIES.
10. Select: ENTER SERVICE MODE.
11. Select: SYSTEM TUNING
12. Select: SYSTEM TUNING TOOLS.
13. Select: INSTALLATION PROCEDURES.
14. Select: BODY COIL PROCEDURES
15. Select: ADJUST QBC RECEIVE
16. Enter: Proceed to start the measurement
17. Select: AUTOSCALE

18. Determine the values for the frequency offset and the Q-factor.

Offset ≤ ± 20 kHz
Q: ≥ 250 (typical value Q:≥ 290)

19. If the measured values are within specification, restore the original situation and continue with paragraph "Determination of body coil parameters".
    If one or more of the measured values are not within specification, adjustment of this mode has to be done, according next paragraph "Hybrid box + QBC resonance frequency adjustment.

    Verify that the QBC did not move during transportation by checking all excenter screws. When the excenters were loose, the QBC has changed position and re-adjustment of the QBC is necessary (See appendix A).

1.3. HYBRID BOX + QBC RESONANCE FREQUENCY ADJUSTMENT

If one of the values is not within specification adjustment of the hybrid box can be done as follows.

Trimmers C31 and C32 in the hybrid box are used to adjust the two modes of the QBC and hybrid box To reach both trimmers, the top plate of the hybrid box must be removed. See Figure 2.

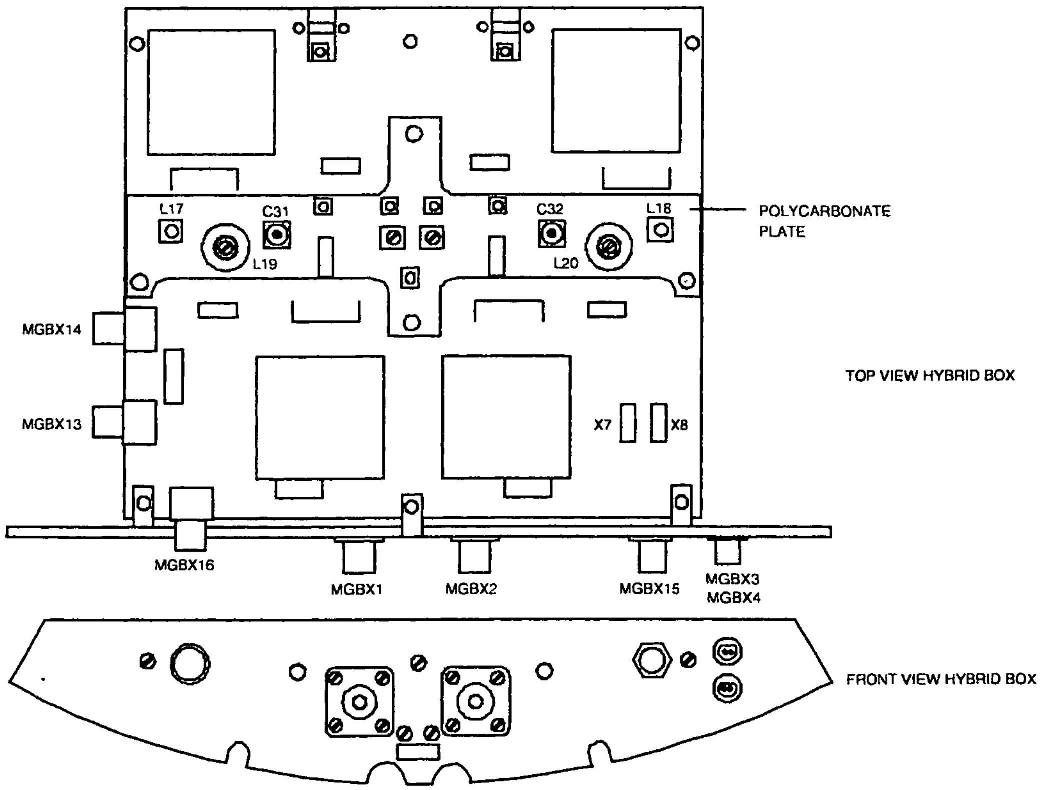


Figure 2 Hybrid box Intera 1.0T (15 kW)

1. Position the **linear flux probe** with use of the rotation knob in the QBC flux probe holder in the center of the QBC. Connect the cable of the flux probe to Tune A signal at connector MMFX2 of the TRSW board. The cable should leave the magnet via the backside to prevent coupling with the hybrid box. See Figure 3.

Procedure:

2. Position the flux probe in position 3 - 11. (Figure 3)

3. Login : GYROSCAN.
4. Select: SCAN CONTROL.
5. Select: SCAN UTILITIES
6. Select: ENTER SERVICE MODE.
7. Select: SYSTEM TUNING
8. Select: SYSTEM TUNING TOOLS.
9. Select: INSTALLATION PROCEDURES.
10. Select: BODY COIL PROCEDURES
11. Select: ADJUST QBC RECEIVE

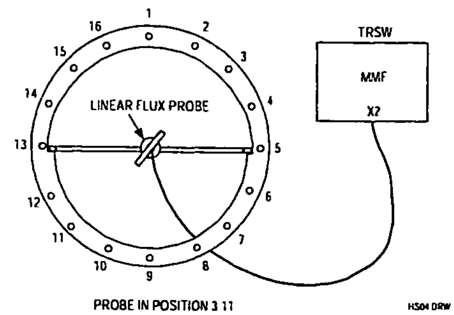


Figure 3 Freq. Adjustment mode 3 - 11

12. Adjust the frequency with trimmer C31 as good as possible to the resonance frequency. The offset can be checked on the monitor.
    Specification: *F(1) < 2 kHz*

13. Position the flux probe in position 7-15 (Figure 4).

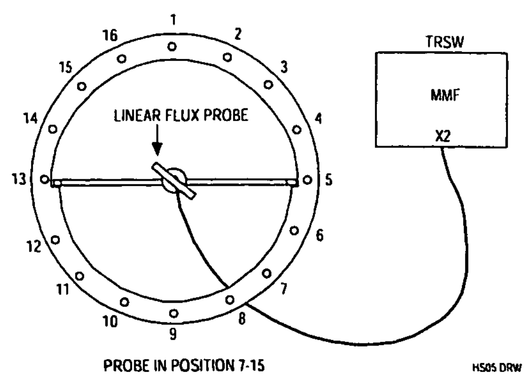


Figure 1

Figure 4 Freq. adjustment mode 7 - 15

14. Adjust the frequency with trimmer C32 as good as possible to the resonance frequency. The offset can be checked on the monitor.
    Specification: *F(2) < 2 kHz*

15. Mount the quadrature flux probe in the QBC flux probe holder and position the holder in the center of the body coil with the connector of the Q-flux probe pointed to the backside of the magnet. Also the cable must be lead via the backside of the magnet. See Figure 5.

NOTE

The quadrature flux probe must be positioned in this way because it makes a quadrature RF-field. If the Q-flux probe is positioned via the front side, the orthogonal field would be picked up by the QBC which results in no signal on the monitor. (orthogonality measurement).

16. Connect a cable between tune A at connector MMFX2 and the quadrature flux probe. See Figure 5.

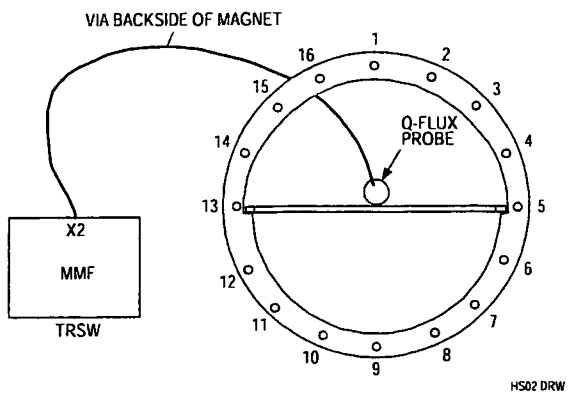


Figure 5 Connections of the quadrature flux probe (Front View)

17. The same program is used : ADJUST QBC RECEIVE
18. Determine the values for the frequency offset and the Q-factor.

Offset $\leq \pm$ 20 kHz
Q: $\geq$ 250 (typical value Q:$\geq$ 290)

19. If the specification can not be reached, adjustment of the complete body coil is necessary. See appendix A at the end of this section.

1.4. DETERMINATION OF BODY COIL PARAMETERS

To perform this measurement both cone-covers and the bridge section must be mounted.

1. Mount the **quadrature** flux probe in the QBC flux probe holder and position the holder in the center of the body coil **with the connector of the Q-flux probe pointed to the backside of the magnet**. Also the cable must be lead via the backside of the magnet.
2. Connect a cable between tune A at connector MMFX2 and the quadrature flux probe. See Figure 5.

The program uses 42.576 MHz as reference frequency.

3. Login : GYROSCAN.
4. Select: SCAN CONTROL.
5. Select: SCAN UTILITIES.
6. Select: ENTER SERVICE MODE.
7. Select: SYSTEM TUNING
8. Select: SYSTEM TUNING TOOLS.
9. Select: INSTALLATION PROCEDURES.
10. Select: BODY COIL PROCEDURES
11. Select: DETERMINE BODY COIL PARAMETERS
12. Now the measurement can be started with <Proceed>.
    No errors are allowed.
13. Write the actual body coil frequency and Q-factor in the MRL (Section 22) as a reference.

2. ADJUSTMENT OF THE QUAD HEAD COIL

2.1. INTRODUCTION

The Quad Head Coil (QHC) for the Intera 1.0T system is provided with an electronic load.
The quad head coil is already adjusted in the factory. Adjustment during installation is only necessary if the cable has to be installed at the other side of the coil or if the Image Quality measurements for the head coil are not within specifications.

2.2. ADAPTATION CABLE TO THE RIGHT SIDE

The procedure to adapt the cable to the other side of the coil, is written in Appendix B at the end of this section. After the adaptation of the cable, readjustment/check of the head coil is necessary.

2.3. QUAD HEAD COIL FREQUENCY AND ORTHOGONALITY ADJUSTMENT

NOTE

The following procedure is only necessary when the cable is adapted to the right side of the coil, or when the QHC adjustment is suspected e.g. when the Image Quality measurements are not within specification.

The QHC is checked on frequency, Q-factor and orthogonality. The adjustment frequency for the QHC is 42.576 Mhz. Frequency-offset with respect to this frequency and orthogonality can be adjusted if not within specification. Q-factor can not be adjusted but is optimized by adjusting frequency and orthogonality.

The quad head coil can be seen as two coils working in different modes: there is a phase shift of 90° between both coils. Each coil has a pre-amplifier. By disconnecting one coil (removal of a jumper on the quad pre-amp board), the resonance frequency of the other coil can be adjusted.
The adjustment for orthogonality must be done to ensure an exact 90° phase difference between the two coils.

Preparation:

1. Read the previous Note if not already done.
2. Remove the left magnet top cover.
3. Remove both tune coil cables MMFX2 and MMFX3 from the TRSW board (PFEI).
4. Connect the head coil to the surface coil connector 1 (SC1).
5. Move the slide-able part of the head coil to the front (scanning) position. This is necessary to reduce the influence of the quad driver board on the coil.

6. Unscrew the rear cover of the coil and mount it back to the head coil using tape in stead of screws. For the adjustment you need to take the cover off but the measurements are only valid when the rear cover is closed, because the values are influenced by this cover.
7. Place the head coil on the table in such a way that the rear of the head coil aligns with the end of the patient table, closest to the magnet (reference point).
8. Use the light visor and the marker on the head coil to position the head coil in the iso-center of the magnet via the "travel-to-scan-plane".
9. Connect the linear flux probe cable to MMFX2 (DO NOT USE THE QUADRATURE FLUX PROBE)
10. Position the **linear flux probe** in position L8 - L12 (see Figure 6). The flux probe holder must be positioned against the front ring of the head coil. The flux probe holder must be used without the 360° rotation knob.

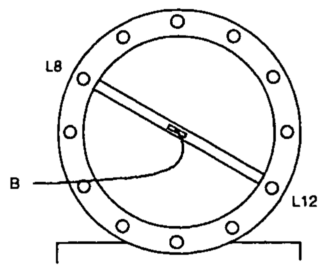


Figure 6 Flux probe in position L8 - L12, seen from patient support end

NOTE

On the front ring of the head coil, marker lines indicate rod L5, L8, L11 and L12.
Make sure the flux probe cable is apart from the head coil cable and that it leaves the bore as straight as possible.

Start the adjustment loop as follows:

1. Login : GYROSCAN.
2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES.
4. Select: ENTER SERVICE MODE.
5. Select: SYSTEM TUNING
6. Select: SYSTEM TUNING TOOLS.
7. Select: INSTALLATION PROCEDURES.
8. Select: HEAD COIL PROCEDURES

9. Enter : <PROCEED> to start the measurement process
10. Select: AUTOSCALE

NOTE

The measured values are only valid in case the rear cover of the head coil is closed during the measurement. Disconnect the head coil when changing the jumpers.

STEP 1 : Frequency check/adjustment first mode

1. Place the flux probe in position L8-L12
2. Remove rear cover.
   Jumper W1. Position 2
   Jumper W2· Removed
   See drawing Z3-7 20 for jumper location (See Figure 7).

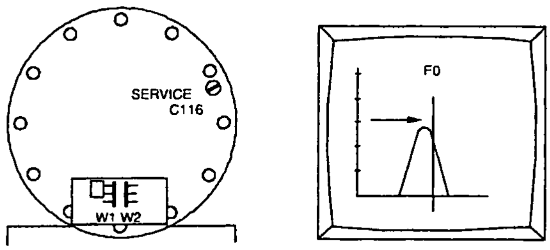
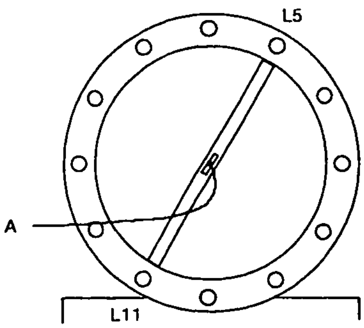


Figure 7 Frequency adjustment mode L8 - L12

3. Put back the cover, read and write down:
   Q(1) = Coil Q-factor.
   U_1 = Voltage at Tune Top (maximum voltage)
   F(1) = offset frequency
   Specifications:
   Q(1): > 250
   F(1): < ± 25 kHz
4. If F(1) does not meet the specifications, adjust the frequency with trimmer C116 (See Figure 7)
5. If the frequency is within specification, write down the Q-value (Q(1)), Voltage at Tune Top U1 and the Offset frequency (F(1)).

STEP 2: Frequency check/adjustment second mode

1. Install the flux probe in position L5-L11.
2. Remove rear cover. Remove jumper W1 and install jumper W2 in position 2. See Figure 9 for jumper location.



3. Replace rear cover, read and write down:
   Q(2) = Coil Q-factor.
   U_2 = Voltage at Tune Top (maximum voltage)
   F(2) = offset frequency
   Specifications:
   Q(2) > 250
   F(2) ≤ ± 25 kHz

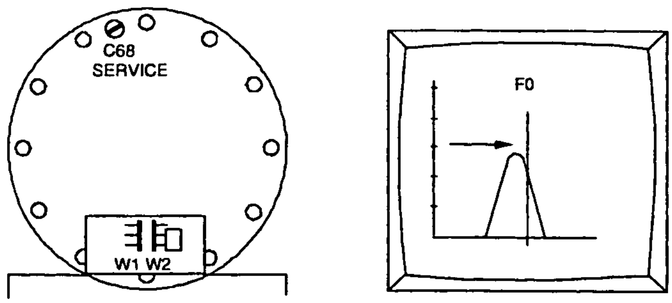
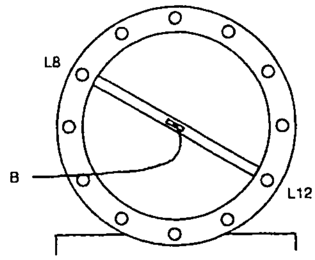


Figure 9 Frequency adjustment mode L5 - L11

4. If F(2) does not meet the specifications, adjust the frequency with trimmer C68 (see Figure 9).
5. If the frequency is within specification, write down the Q value (Q(2)), Voltage at Tune Top (U_2) and the Offset frequency (F(2)).

STEP 3 : Orthogonality check/adjustment (1)

1. Install flux probe to position L8 - L12. The flux probe must be positioned against the front ring of the head coil.



2. Read and write down:
   - U_3 = **Voltage at Center Frequency**

   Calculate the attenuation with next formula:

$$Att(1) = 20\log(\frac{U_2}{U_3})$$

The typical value for the orthogonality is: Att(1) > 17 dB

If this value is reached, continue with **STEP 4**.

If this value is not reached, adjust the Tune Voltage at center frequency to a value as low as possible using the next procedure:
(See Drawings manual Z3-7.19 for exact trimmer location)

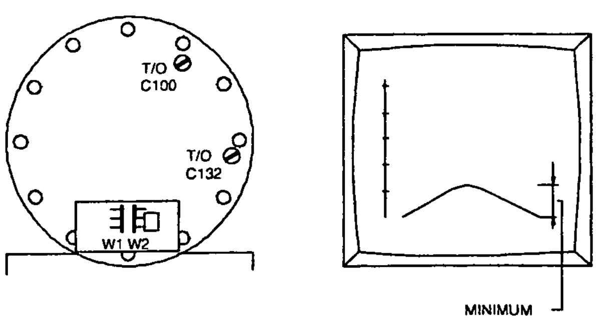


Figure 11 Adjustment of the orthogonality

3. Probe position between L8/L12 (As the previous adjustment). Put the jumpers in the following position:
   W1 -> removed
   W2 -> position 2

4. Turn C100 a little clockwise, see if response decreases If not turn C100 counter clockwise. Turn C132 the same amount in the other direction(*).

   (*) Other direction means: If you have set for more capacitance with C100, turn C132 in the other direction until you have the same amount capacitance less. Minimize the response by turning both trimmers successively, as described above, until a minimum is reached.
   Write down the measured voltage U_3.

NOTE

Sometimes, depending on their initial position, both trimmers have to be turned in the same direction, or sometimes in the opposite direction.

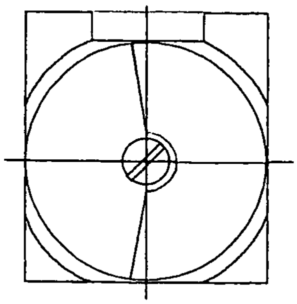


Figure 12 Trimmer in Middle position

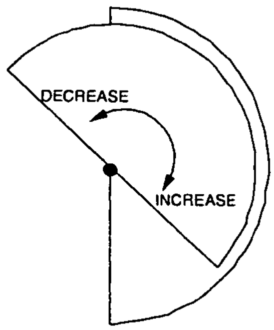


Figure 13 Increasing / decreasing capacitance

NOTE

The orthogonality measurement is not accurate due to the mechanical construction of the coil. Adjust the orthogonality as good as possible. It is not necessary to reach the typical value. The correct functioning of the coil is checked during the Image Quality performance measurements.

STEP 4 : Orthogonality check/adjustment (2)

1. Install flux probe to position L5 - L11. The flux probe must be positioned against the front ring of the head coil and put the jumpers in the next position:
   W1 position 2
   W2 removed (See Figure 14 and 15)

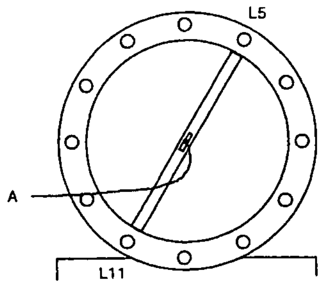


Figure 14 Flux probe position mode L5-L11

2. Read and write down:
   - U_4 = **Voltage at Center Frequency**

   Calculate the attenuation with next formula.

$$Att(2) = 20\log\left(\frac{U_1}{U_4}\right)$$

The typical value for the orthogonality is: Att(1) > 17 dB

If this value is reached, continue with. paragraph **"Final actions"**

If this value is not reached, adjust the Tune Voltage at center frequency to a value as low as possible using the next procedure:
(See Drawings manual Z3-7.19 for exact trimmer location)

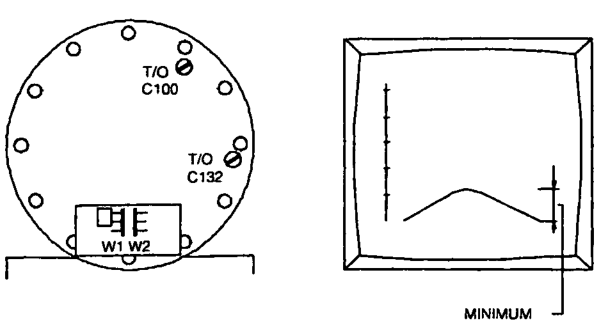


Figure 15 Adjustment orthogonality

3. Probe position between L5/L11 (As the previous adjustment). Put the jumpers in the following position:
   W1 -> position 2
   W2 -> removed (see Figure 14 and15)
4. Turn C100 a little clockwise, see if response decreases. If not turn C100 counter clockwise **and** turn C132 the same amount in the other direction(*).

   (*) Other direction means: If you have set for more capacitance with C100 turn C132 in the other direction until you have the same amount capacitance less. Minimize the response by turning both trimmers successively, as described above, until a minimum is reached.
   Write down the measured voltage U_4.

NOTE

Sometimes, depending on their initial position, both trimmers have to be turned in the same direction, or sometimes in the opposite direction.

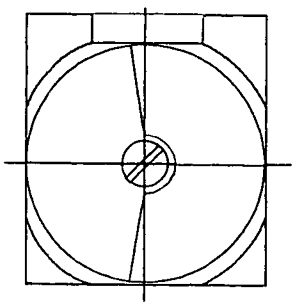


Figure 16 Trimmer in Middle position

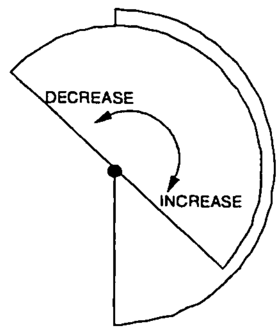


Figure 17 Increasing / decreasing capacitance

NOTE

The orthogonality measurement is not accurate due to the mechanical construction of the coil. Adjust the orthogonality as good as possible. It is not necessary to reach the typical value. The correct functioning of the coil is checked during the Image Quality performance measurements.

2.4. FINAL ACTIONS

5. Repeat step 1 - 4 until no adjustments are necessary. Every adjustment can influence other parameters.
6. If you have followed the procedure and no adjustments were necessary, fill in the measured values, in the MRL according Table 1.(next page)
7. Install both jumpers in position 2 and mount the rear cover.
8. Remove the flux-probe holder out of the head coil and reconnect tune coil cables to MMFX2 and MMFX3 of the TRSW board.
9. Read and write down as a reference in MRL (no specification):
   F(head) = Offset frequency
   Q(head) = Coil Q-factor
10. Restore normal situation.
11. Write down the final values in the table in the MRL (Section 22) as a reference.

Probe position		Parameters with specification		Measured values
Step 1) L8 - L12	Jumper position: W1: Position 2 W2: Removed	Q(1)	> 250	
		Voltage U_1		
		F(1)	< ± 25 kHz	
Step 2) L5 - L11	Jumper position: W1. Removed W2: Position 2	Q(2)	> 250	
		Voltage U_2		
		F(2)	< ± 25 kHz	
Step 3) L8 - L12	Jumper position. W1: Removed W2: Position 2	Voltage U_3		
		Att(1)	typical value > 17 dB	
Step 4) L5 - L11	Jumper position: W1: Position 2 W2: Removed	Voltage U_4		
		Att(2)	typical value > 17 dB	
Both jumpers positioned (reference values)		F(head)	n.a.	
		Q(head)	n.a	

Table 1 - Measurement results

3. APPENDIX A: READJUSTMENT PROCEDURE QUAD BODY COIL

Only if the hybrid box + quad body coil could not be adjusted within the required specifications, (re)adjustment of the separated QBC should be done. Adjustment is also necessary when the QBC is replaced.

The re-adjustment is done in three steps:

1. Horizontal and vertical mode are adjusted to the resonance frequency.
2. The diagonal modes are set to the resonance frequency
3. The complete QBC is set to the correct frequency.

3.1. TOOLS

- Flux probe holder
- Rotation knob for flux probe
- QBC adjustment tool
- Extension cables 1 m (belonging to QBC adjustment tool)
- 50 ohm coaxial cable 15 m
- Quadrature flux probe 42 MHz.
- Two special trimmer adjustment tools.

3.2. MEASUREMENT SET-UP

1. Switch off the gradient system.
2. Remove the lower system filter box cover and switch off the FEPSU.
3. Remove the bridge section, the PICU cover and the front cone cover (if not already done). The hybrid box and QBC are visible now.

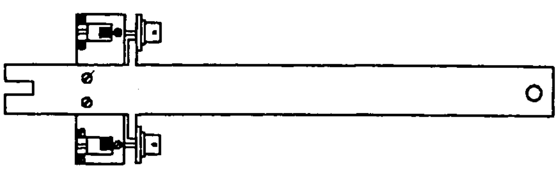


Figure 18 QBC adjusting tool

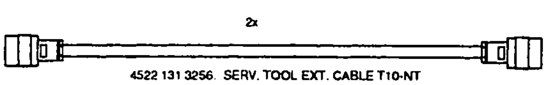


Figure 19 QBC adjustment tool extension cables

4. Disconnect the cables from the hybrid box.
5 Open the hybrid box by taking off the top-plate and loosen the screws of the connectors for the two rigid coax connections of the QBC (two screws, per coax connection!) Refer to drawing Z3-7.15
6. Remove the hybrid box.
7. Remove the inner top-cover of the QBC by carefully pulling out the covers by hand: start on one side of the cover, when one side is loose, the cover can be taken out. Mark the front and rear side of the top-cover of the QBC to be able to put it back in the same way. It could have an influence on the shimming of the magnet.
8 Remove the inner lower-cover: start on one side.
9 Mount the special QBC adjustment tool at the QBC rigid coax connection Do not use force to fasten the screws.
10. The QBC adjustment tool (with the two special 1.1 meter extension cables) must be connected to the QBC connections in stead of the hybrid box. (see Figure 20). The extension cables must be left open on one end. **(not terminated)**.

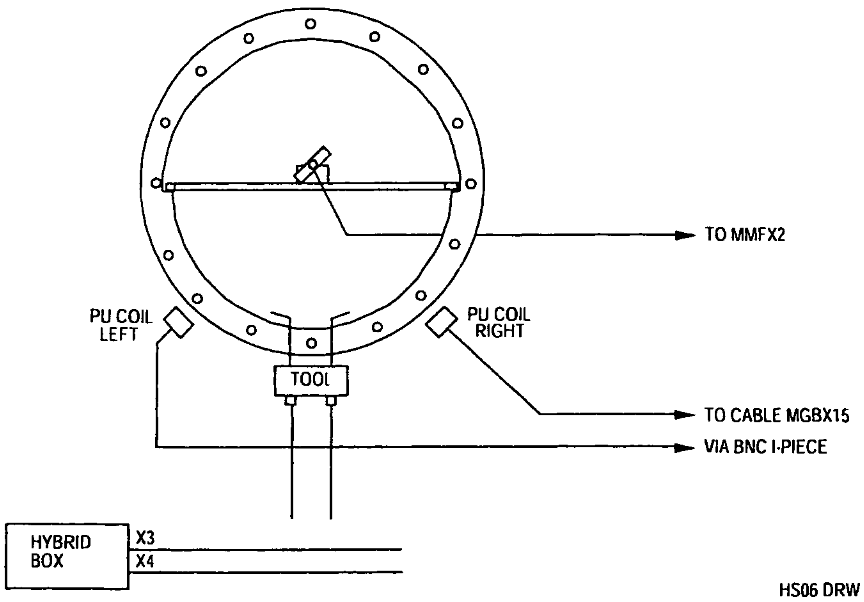


Figure 20 Measurement set-up for adjusting the QBC without hybrid box (front view)

11. Mount the top plate to the hybrid box with a few screws (Safety !).
12. Position the hybrid box just outside the magnet bore in such a way that the cables MGBX3 and MGBX4 can be connected to the hybrid box. This will prevent that you will get error messages during the Manual coil tuning program. Make sure that the bottom side of the hybrid box does not make contact with the gradient coil The top plate of the hybrid box must be present to avoid touching the 300 Volts.
13. Connect the left pick-up coil (looking from the front of the magnet) to the MR response cable MGBX15 by using a BNC I-piece. The pick-up coils are used to receive.
14. Mark or measure the present position of the pick-up coils and shift both pick-up coils as close as possible to the QBC to get the best signal.

15. On the front side and the back side of the QBC you can find the adjustment screws for the trimmers. Loosen the screws that lock the trimmers for all QBC rods (see Figure 21 and Figure 22).

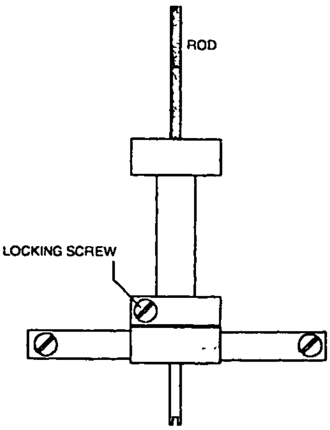


Figure 21 QBC trimmer in detail

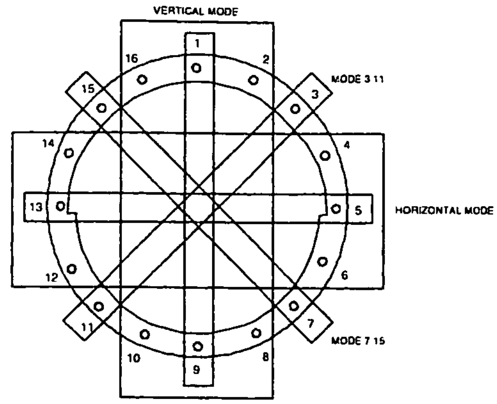


Figure 22 QBC frequency modes

You can see that two types of rods are used: thick rods (with thick trimmers) and thin rods (with thin trimmers). For each type of trimmer a separate trimmer tool must be used.

16. Mount the QBC inner covers. Start with the lower cover. Position one side of the cover in its place and use your hands to push the other side in position. Use you fist if necessary to make the cover "click" in position.(for the lower cover it can be necessary to position it first in a belly shape and push it with both hands down. To finish it you have to use your fingers underneath the rods to push the cover in position) During the measurement the inner covers must be mounted because the inner covers introduce a frequency shift which is different for the horizontal and vertical mode.

17. Remove the left magnet top cover and disconnect the tune coil cables MMFX2 and MMFX3 from the TRSW board (PFEI).

18. Position the QBC flux probe holder in the center of the QBC and mount the linear flux probe via the rotation knob in the flux probe holder. Connect the flux probe to tune A signal at MMFX2. The flux probe is used to transmit. It is advised to route the cable via the backside of the magnet.

19. Switch on the FEPSU.

3.3. ADJUSTMENT QBC

1. Login : GYROSCAN.
   Select: SCAN CONTROL.
   Select: SCAN UTILITIES.
   Select: ENTER SERVICE MODE.
   Select: SYSTEM TUNING
   Select: SYSTEM TUNING TOOLS.
   Select: INSTALLATION PROCEDURES.
   Select: BODY COIL PROCEDURES

2. Select parset ADJUST FREQUENCY QBC and start the measurement.

3.4. FINE ADJUSTMENT SEPARATE MODES

1. Position the flux probe to position 7 - 15 (see Figure 23).

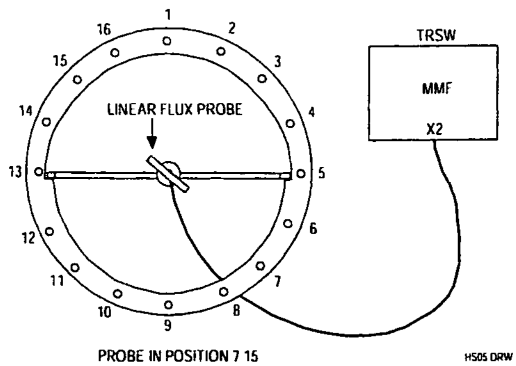


Figure 23 Flux probe position 7 - 15

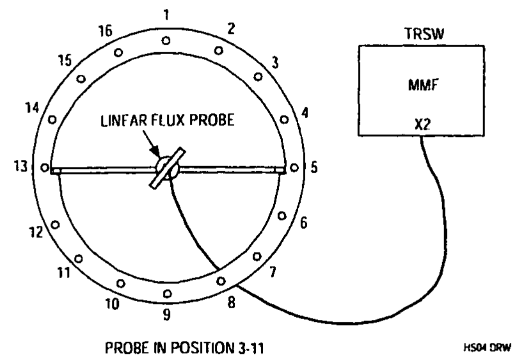


Figure 24 Flux probe position 3 - 11

2. Measure the frequency offset and write down this value in the table below.
3. Position the flux probe in position 3 - 11 (see Figure 24) and connect the right pick-up coil in stead of the left pick-up coil to the receive cable MGBX15.
4. Measure the offset and write down the value in the table below.
5. Connect the left pick-up coil and position the flux probe in horizontal position and measure the offset. Fill in the table.
6. Put the flux probe in vertical position and measure the offset. Fill in the table.

Linear Flux probe	Specification	Measured
Mode 7 - 15	F < ± 13 kHz	
Mode 3 - 11	F < ± 13 kHz	
Mode horizontal	F < ± 13 kHz	
Mode vertical	F < ± 13 kHz	

To adjust the QBC, use the following trimmers according the measured offset: (See Figure 22)

Mode 7 - 15: trimmers 7 and 15

Mode 3 - 11: trimmers 3 and 11

Horizontal mode: For offset > 20 kHz. use trimmers 4, 5, 6 and 12, 13, 14
For offset < 20 kHz. use trimmers 5 and 13

Vertical mode: For offset > 20 kHz. use trimmers 16, 1, 2 and 8, 9, 10
For offset < 20 kHz. use trimmers 1 and 9

NOTE

Turn all trimmers in the same direction (inwards (clockwise) or outwards (counter-clock wise)). For each rod the trimmers on the front and backside have to be turned with the same amount in the same direction.

7. Start with the mode with the largest offset and adjust all modes to the center frequency (± 13 kHz.)

   Specification: *Frequency offset for each mode ≤ 13 kHz*

8. Use the left pickup coil for modes 7 - 15, horizontal mode and vertical mode, and the right pickup coil for the mode 3 - 11.
9. Use small frequency steps (e.g. 1/4 turn at a time).
10. Check the modes until no adjustments are necessary any more.
11. Dismount the QBC inner covers and lock all trimmers.
12. Re-install the QBC inner covers.
13. Check all modes and fill in the following table.

Linear flux probe	Specification	Measured
Mode horizontal	F < ± 13 kHz	
Mode vertical	F < ± 13 kHz	
Mode 3 - 11	F < ± 13 kHz	
Mode 7 - 15	F < ± 13 kHz	

3.5. PREPARATION FOR MEASURING QBC RECEIVE MODE

1. Switch off the FEPSU.
2. Remove the flux probe and flux probe holder from the QBC.
3. Disconnect cables MGBX3 and MGBX4 from the hybrid box.
4. Remove the QBC adjustment tool from the QBC. Be careful with the coaxial connections of the QBC when disconnecting.
5. Reconnect the tune A and tune B signal MMFX2 an MMFX3.
6. Mount the hybrid box in front of the QBC and connect all the cables to it. Position the pick-up coils back to their original position. (Check the marks)
   Make sure that the "computer cable" is lead in the middle underneath the hybrid box.
7. Switch on the FEPSU

3.6. HYBRID BOX + QBC RESONANCE FREQUENCY ADJUSTMENT

Trimmers C31 and C32 in the hybrid box are used to adjust the two modes of the QBC and hybrid box. Both trimmers can be reached by removing the top plate of the hybrid box.

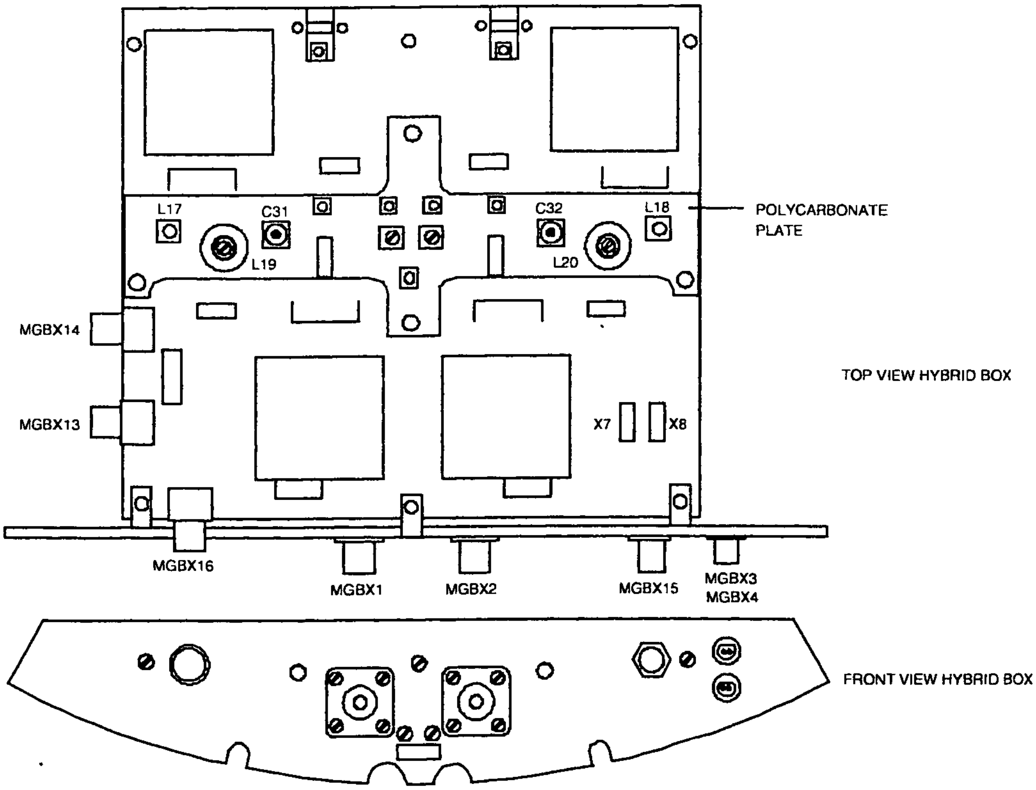
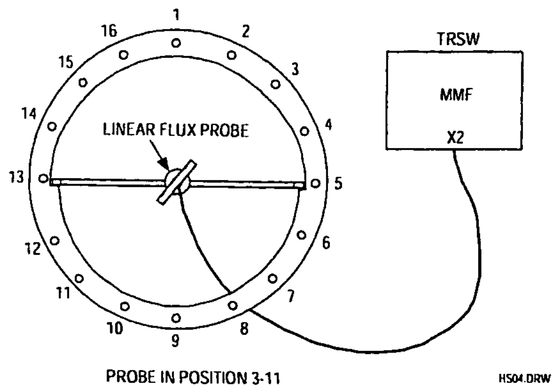


Figure 25 Hybrid box Intera 1.0T (15 kW) with trimmer C31 and C32

1. Position the linear flux probe with use of the rotation knob in the QBC flux probe holder in the center of the QBC. Connect the cable of the flux probe to tune A signal at MMFX2 of the TRSW board. See Figure 26.



To prevent coupling problems with the hybrid box, the cable must leave the bore via the backside of the magnet.

Figure 26 Frequency adjustment mode 3 – 11

Procedure:

2. Position the flux probe in position 3-11 (Figure 26).
3. Login : GYROSCAN.
4. Select: SCAN CONTROL.
5. Select: SCAN UTILITIES.
6. Select: ENTER SERVICE MODE.
7. Select: SYSTEM TUNING
8. Select: SYSTEM TUNING TOOLS.
9. Select: INSTALLATION PROCEDURES.
10. Select: BODY COIL PROCEDURES
11. Select: ADJUST QBC RECEIVE
12. Enter: Proceed to start the measurement
13. Select: AUTOSCALE

12. Adjust the frequency with trimmer C31 to the resonance frequency. The offset can be checked on the monitor.
    Specification: *F(1) < 2 kHz*

13. Position the flux probe in position 7-15 (Figure 27).

Remind to lead the flux probe cable via the backside of the magnet.

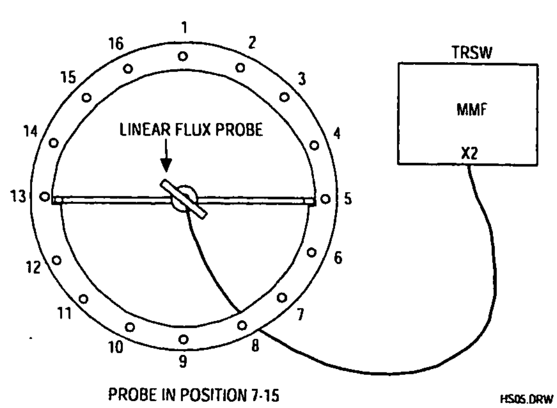


Figure 27 Frequency adjustment mode 7 - 15

14. Adjust the frequency with trimmer C32 to the resonance frequency. The offset can be checked on the monitor
   Specification: *F(2) < 2 kHz*

15 Mount the quadrature flux probe in the QBC flux probe holder and position the holder in the center of the body coil with the connector of the Q-flux probe pointed to the backside of the magnet. Also the cable must be lead via the backside of the magnet.

16. Connect a cable between tune A at connector MMFX2 and the quadrature flux probe.
   See Figure 28.

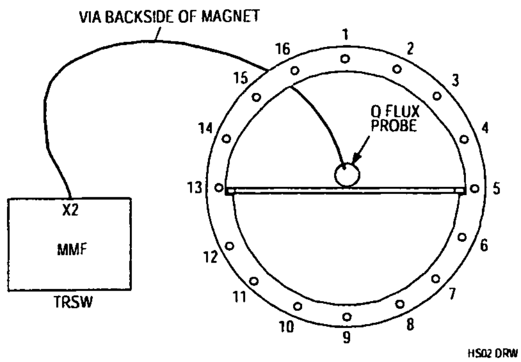


Figure 28 Connection of the Quadrature flux probe (Front View)

17. The same program is used : ADJUST QBC RECEIVE
18. Determine the values for the frequency offset and the Q-factor.

Offset ≤ ± 20 kHz
Q: ≥ 250 (Typical value ≥ 290)

3.7. RESTORING NORMAL CONDITION

1. Switch off the FEPSU.
2. Connect all cables to their original position.
3. Mount the cone covers and the bridge section.
4. Switch on the FEPSU.

3.8. DETERMINATION OF BODY COIL PARAMETERS

To perform this measurement both cone-covers and the bridge section must be mounted.

1. Mount the **quadrature** flux probe in the QBC flux probe holder and position the holder in the center of the body coil **with the connector of the Q-flux probe pointed to the backside of the magnet**. Also the cable must be lead via the backside of the magnet.
2. Connect a cable between tune A at connector MMFX2 and the quadrature flux probe.
   See Figure 28.
   - The program uses 42.576 Mhz as reference frequency.

3. Login : GYROSCAN.

4. Select: SCAN CONTROL.
5. Select: SCAN UTILITIES.
6. Select: ENTER SERVICE MODE.
7. Select: SYSTEM TUNING
8. Select: SYSTEM TUNING TOOLS.
9. Select: INSTALLATION PROCEDURES.
10. Select: BODY COIL PROCEDURES
11. Select: DETERMINE BODY COIL PARAMETERS
12. Now the measurement can be started with <Proceed>.
    No errors are allowed.
13. Write the actual body coil frequency and Q-factor in the MRL as a reference.

4. APPENDIX B: ADAPTION QHC CABLE TO RIGHT SIDE

4.1. INTRODUCTION

The quad head coil is standard delivered with the cable on the left side of the coil.
This procedure can be used for changing the cable to the right side.

4.2. PROCEDURE DESCRIPTION

1. Put the head coil on a table and remove the cover on the bottom side that covers the driver PCB.
2. Disconnect the cables to the PCB and dismount the driver. On the other side of the coil, the bottom cover is already prepared to mount the driver board. In the PCB two extra holes are already present to mount the board on the right side.
3. Remove the bracket from the right side that covers the hole for the cable feed through.
4. Remove the cable from the left side. The hole can be covered with the bracket.
5. Dismount the rear cover of the head coil and dismount the cable bracket from the cover.
6. Turn it over 180 degrees so that the cable points to the right side of the coil.
7. Mount the bracket again and mount the cover back to the coil.
8. Mount the cables on the right side of the coil and connect them to the driver PCB.
   IThe connection cables on each side of the driver PCB should be separated from each other as good as possible to reduce the coupling. Use cable binders to secure the cables to the bottom cover.
9. Mount the cover for the PCB.

4.3. ADJUSTMENT QHC

After the cable has been moved to the right side the coil has to be checked/readjusted.
For procedure, see the relevant paragraphs in this section (15).

Section 15 RF Coils

Contents

1. QUAD BODY COIL (QBC) CHECK/ADJUSTMENT

The Quad Body Coil is already adjusted in the factory. Only a test has to be done to check the QBC and to determine the values for the MRL as a reference.
Verify that the QBC did not move during transportation by checking all excenter screws. When the excenters were loose, the QBC has changed position and re-adjustment of the QBC is necessary (See appendix A).

1.1. TOOLS

- Quadrature flux probe (4522 117 70051)
- QBC flux probe holder
- 50 ohm coaxial cable 15 meter
- Linear flux probe
- Rotation knob

1.2. HYBRID BOX + QBC RESONANCE FREQUENCY

1. Switch-off the gradient system
2. Lower the patient table.
3. Remove the bridge section, the PICU cover and the front cone cover. The QBC and the hybrid box in front of the QBC are visible now.
4. Mount the **quadrature** flux probe in the QBC flux probe holder from the rear side of the magnet and position the holder horizontal in the center of the body coil with the connector of the Q-flux probe pointed to the backside of the magnet. The cable must be routed via the backside of the magnet.
5. Remove the left magnet top cover and disconnect both tune outputs MMFX2 and MMFX3 from the TRSW board (PFEI).
6. Connect a 50 ohm coaxial cable between tune A at connector MMFX2 and the quadrature flux probe. See Figure 1.

7. Login : GYROSCAN.
8. Select: SCAN CONTROL.
9. Select: SCAN UTILITIES.
10. Select: ENTER SERVICE MODE.
11. Select: SYSTEM TUNING
12. Select: SYSTEM TUNING TOOLS.
13. Select: INSTALLATION PROCEDURES.
14. Select: BODY COIL PROCEDURES
15. Select: ADJUST QBC RECEIVE
16. Enter: Proceed to start the measurement
17. Select: AUTOSCALE

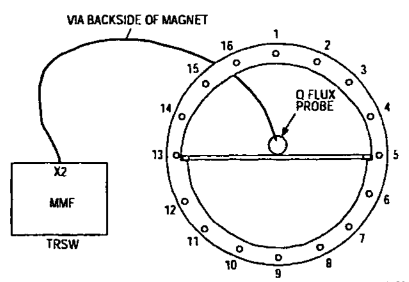


Figure 1 Measurement set-up (Front view)

18. Determine the values for the frequency offset and the Q-factor.

$$Offset \leq \pm\ 10\ kHz$$
$$Q \geq 190$$

19. If the measured values are within specification, restore the original situation and continue with paragraph "Determination of body coil parameters".

    If the measured offset or Q-value are not within specification, adjustment of this mode must be done, according the next paragraph "Hybrid box + QBC resonance frequency adjustment".

1.3. HYBRID BOX + QBC RESONANCE FREQUENCY ADJUSTMENT

If one of the values in the previous paragraph were not within specification adjustment of the hybrid box must be done as follows:

Trimmers C31 and C32 in the hybrid box are used to adjust the two modes of the QBC and hybrid box. Remove the hybrid box top plate to reach the trimmers. See Figure 2 .

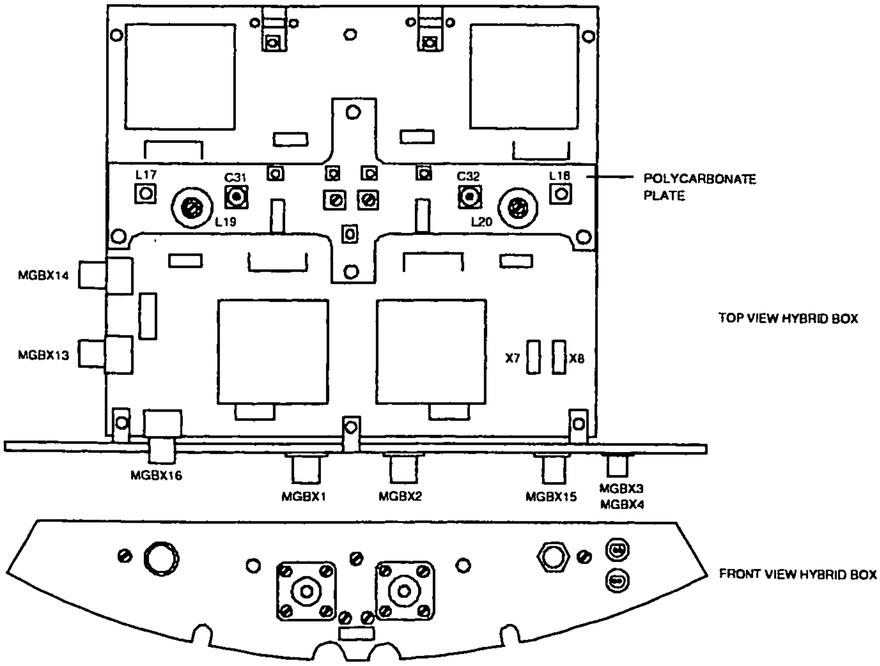


Figure 2 25 kW hybrid box with trimmers C31 and C32

1. Position the **linear flux probe** with use of the rotation knob in the QBC flux probe holder in the center of the QBC. Connect the cable of the flux probe to tune A at connector MMFX2. The cable should leave the magnet via the rear side to prevent coupling with the hybrid box. See Figure 3.

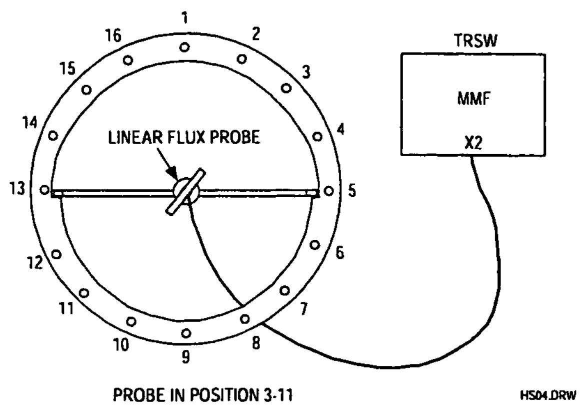


Figure 3 Frequency adjustment mode 3 - 11 (front view)

Procedure:

2. Position the flux probe in position 3-11 (Figure 3).

3. Login : GYROSCAN.
4. Select. SCAN CONTROL.
5. Select: SCAN UTILITIES.
6. Select: ENTER SERVICE MODE.
7. Select: SYSTEM TUNING
8. Select SYSTEM TUNING TOOLS.
9. Select: INSTALLATION PROCEDURES.
10. Select: BODY COIL PROCEDURES
11. Select: ADJUST QBC RECEIVE

12. Adjust the frequency with trimmer C31 of the hybrid box as good as possible to the resonance frequency. The offset can be checked on the monitor.

Specification: *F(1) < 10 kHz*

13. Position the flux probe in position 7-15 (Figure 4). The cable must leave the bore via the rear side of the magnet.

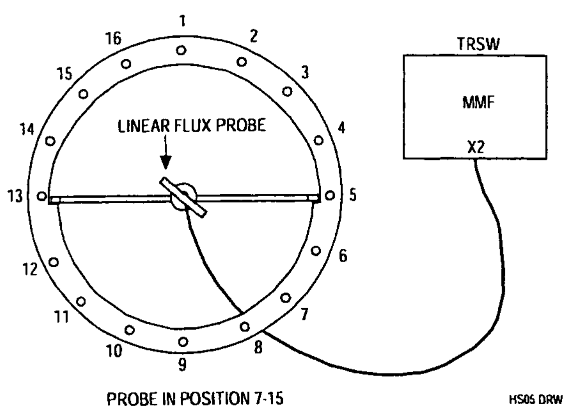


Figure 4 Frequency adjustment mode 7 - 15

14. Adjust the frequency with trimmer C32 of the hybrid box as good as possible to the resonance frequency. The offset can be checked on the monitor.

Specification: *F(2) < 10 kHz*

15. Mount the **Quadrature** flux probe in the QBC flux probe holder and position the holder in the center of the Body coil **with the connector of the Q-flux probe pointed to the back side of the magnet.** The cable must be lead via the back side of the magnet.
16. Connect a 50 ohm coaxial cable between tune A at connector MMFX2 and the quadrature flux probe See Figure 5.

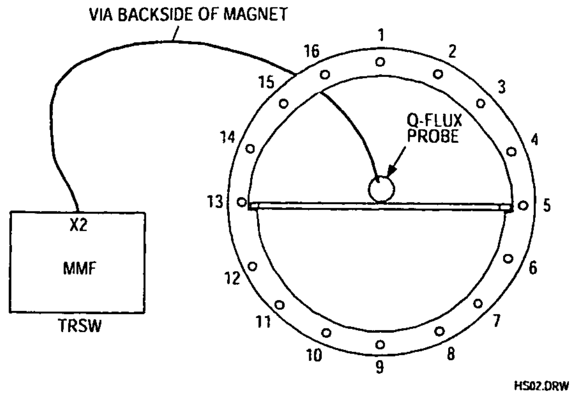


Figure 5 Connection of the Quadrature flux probe (Front View)

17. Select : ADJUST QBC RECEIVE
18. Determine the values for the frequency offset and the Q-factor.

Offset ≤ ± 10 kHz
Q ≥ 190

19. If the specification can not be reached, adjustment of the complete body coil is necessary. See appendix A "Readjustment procedure Quad Body Coil" at the end of this section.

1.4. DETERMINATION OF BODY COIL PARAMETERS

To perform this measurement the cone-covers and the bridge section must be mounted.

1. Mount the **quadrature** flux probe in the QBC flux probe holder and position the holder in the center of the body coil **with the connector of the Q-flux probe pointed to the backside of the magnet**. Also the cable must be lead via the backside of the magnet.
2. Connect a cable between tune A at connector MMFX2 and the quadrature flux probe. See Figure 5.

   The program uses 63.870 Mhz as reference frequency.

3. Login : GYROSCAN.
4. Select: SCAN CONTROL.
5. Select: SCAN UTILITIES.
6. Select: ENTER SERVICE MODE.
7. Select: SYSTEM TUNING
8. Select: SYSTEM TUNING TOOLS.
9. Select: INSTALLATION PROCEDURES.
10. Select: BODY COIL PROCEDURES
11. Select: DETERMINE BODY COIL PARAMETERS
12. Now the measurement can be started with <Proceed>. No errors are allowed.
13. Write the actual body coil frequency and Q-factor in the MRL as a reference.

2. ADJUSTMENT OF THE QUAD HEAD COIL 1.5T

2.1. INTRODUCTION

The Quad Head Coil (QHC) for the Intera1.5T Release 5 system is provided with an electronic load
The quad head coil is already adjusted in the factory Adjustment during installation is only necessary if the cable has to be installed at the other side of the coil or if the Image Quality measurements for the head coil are not within specifications.

2.2. ADAPTATION CABLE TO THE RIGHT SIDE

The procedure to adapt the cable to the other side of the coil, is written in Appendix B at the end of this section. After the adaptation of the cable, readjustment/check of the head coil is necessary.

2.3. QUAD HEAD COIL FREQUENCY AND ORTHOGONALITY ADJUSTMENT

NOTE

The following procedure is only necessary when the cable is adapted to the right side of the coil, or when the QHC adjustment is suspected e.g. when the Image Quality measurements are not within specification.

The QHC is checked on frequency, Q-factor and orthogonality. The adjustment frequency for the QHC is 64.050 Mhz. Frequency-offset with respect to this frequency and orthogonality can be adjusted if not within specification. Q-factor can not be adjusted but is optimized by adjusting frequency and orthogonality.

The quad head coil can be seen as two coils working in different modes: there is a phase shift of 90° between both coils. Each coil has a pre-amplifier. By disconnecting one coil (removal of a jumper on the quad pre-amp board), the resonance frequency of the other coil can be adjusted.
The adjustment for orthogonality must be done to ensure an exact 90°phase difference between the two coils.

Preparation:

1. Read the previous Note if not already done.
2. Remove the left magnet top cover if not already done.
3. Remove both tune coil cables MMFX2 and MMFX3 from the TRSW board (PFEI).
4. Connect the head coil to the surface coil connector 1 (SC1).
5. Move the slideable part of the head coil to the front (scanning) position. This is necessary to reduce the influence of the quad driver board on the coil.

6. Unscrew the rear cover of the coil and mount it back to the head coil using tape in stead of screws. For the adjustment you need to take the cover off but the measurements are only valid when the rear cover is closed, because the values are influenced by this cover.
7. Place the head coil on the table in such a way that the rear of the head coil aligns with the end of the patient table, closest to the magnet (reference point).
8. Use the light visor and the marker on the head coil to position the head coil in the iso-center of the magnet via the "travel-to-scanplane".
9. Disconnect both tune cables MMFX2 and MMFX3 from the TRSW board and connect the **linear flux probe** cable to MMFX2. (DO NOT USE THE QUADRATURE FLUX PROBE)
10. Position the linear flux probe in position L8 - L12 (see Figure 6). The flux probe holder must be positioned against the front ring of the head coil The flux probe holder must be used without the 360°rotation knob.

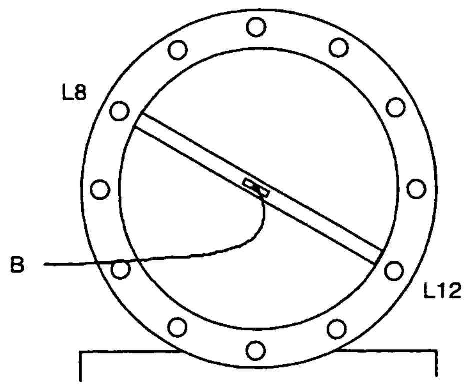


Figure 6 Flux probe position L8 - L12, seen from patient support end.

NOTE

On the front ring of the head coil, marker lines indicate rod L5, L8, L11 and L12. Make sure the flux probe cable is apart from the head coil cable and that it leaves the bore as straight as possible.

Start the adjustment loop as follows:

1. Login : GYROSCAN.
2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES.
4. Select: ENTER SERVICE MODE.
5. Select: SYSTEM TUNING
6. Select: SYSTEM TUNING TOOLS.
7. Select: INSTALLATION PROCEDURES.
8. Select: HEAD COIL PROCEDURES
9. Enter : <PROCEED> (2x) to start the measurement process
10. Ignore the message: "Difference with designed frequency is more then 100kHz.
11. Select: AUTOSCALE
12. When the optional LCD is present, select the magnet display with the button in the display window.

The center frequency used by the program is 64.050 MHz.

ADJUSTMENT QUAD HEADCOIL EL-LOAD ACS-NT

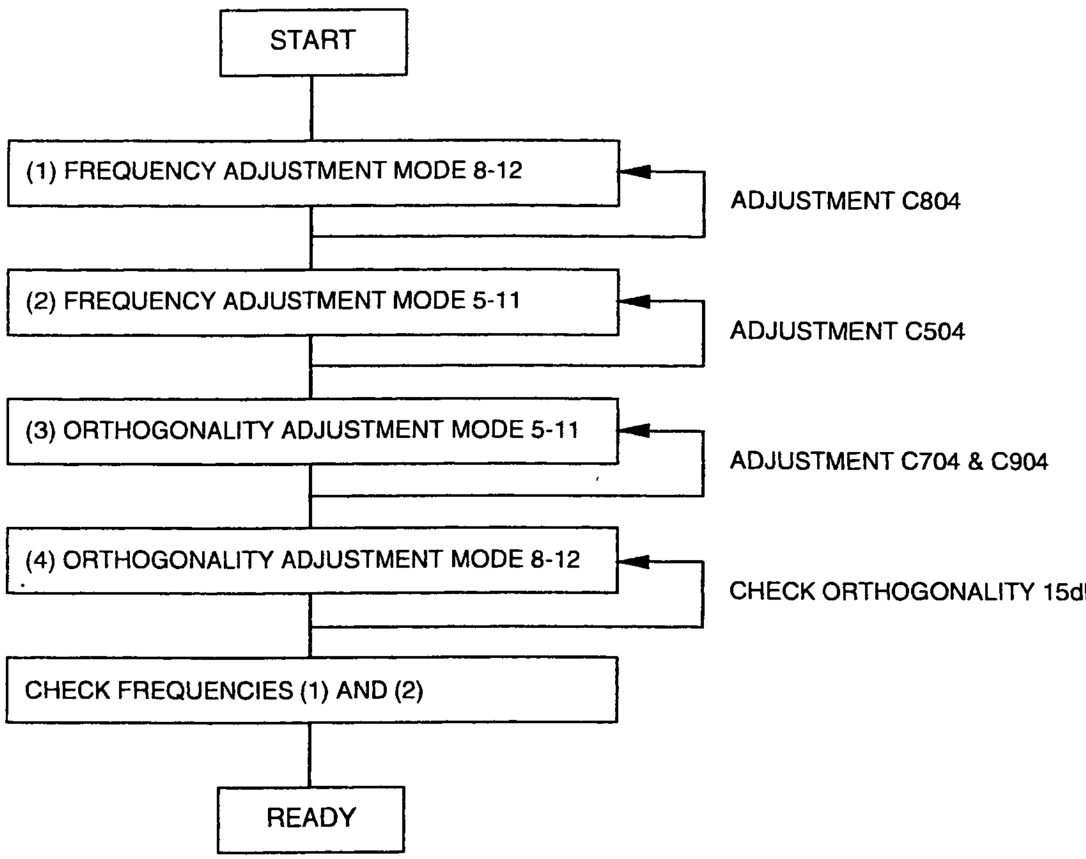


Figure 7 Flow diagram of adjustment Quad head coil 1.5T

NOTE

The measured values are only valid if the rear cover of the head coil is closed during the measurement.

STEP 1 : Frequency check/adjustment first mode

1. Place the flux probe in position L8-L12 if not already done
2. Remove rear cover.
   Jumper W1: Position 2
   Jumper W2: Removed
   See Figure 8 for jumper location. Refer also to Drawings manual Z3-7.19 and Z3-7.20.

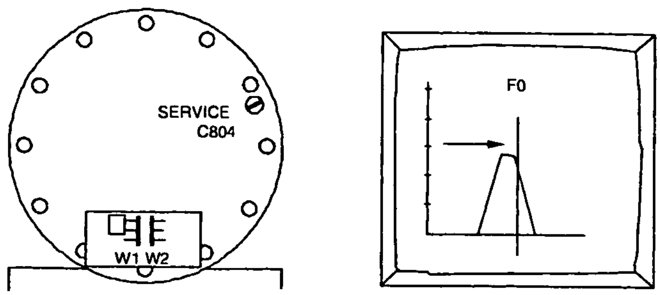


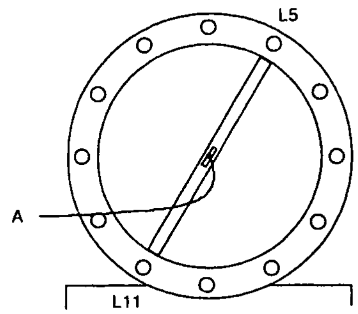
Figure 8 Frequency adjustment mode L8 - L12

3. Place back the cover. Read the following values and write them down in table 1 (at the end of this paragraph):
   - Q(1) = Coil Q-factor.
   - U_1 = Voltage at Tune Top (maximum voltage)
   - F(1) = offset frequency

   Specifications:
   - Q(1): > 90
   - F(1): < ± 25 kHz
4. If F(1) does not meet the specifications, adjust the frequency with trimmer C804 (See Figure 8)
5. If the frequency is within specification, write down the Q-value (Q(1)), Voltage at Tune Top U1 and the Offset frequency (F(1)).

STEP 2 : Frequency check/adjustment second mode

1. Install the flux probe in position L5-L11.
2. Remove rear cover. Remove jumper W1 and install jumper W2 in position 2. See Figure 10 for jumper location. Refer also to Drawings manual Z3-7 19 and Z3-7.20.



3. Replace rear cover, read and write down:

Q(2)	= Coil Q-factor.
U_2	= Voltage at Tune Top (maximum voltage)
F(2)	= offset frequency
Specifications:	
Q(2)	> 90
F(2)	≤ ± 25 kHz and frequency difference F(1) and F(2) ≤ 25 kHz

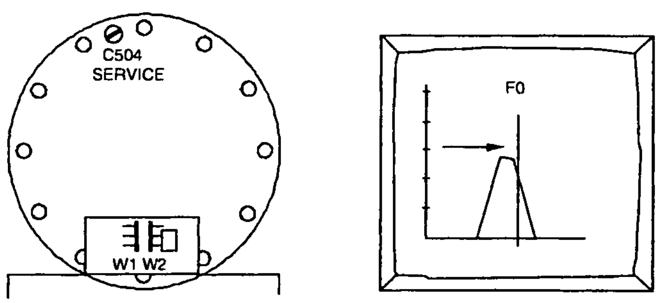
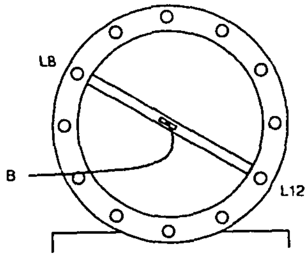


Figure 10 Frequency adjustment L5 - L11

4. If F(2) does not meet the specifications, adjust the frequency with trimmer C504 (see Figure 10).
5. If the frequency is within specification, write down the Q value (Q(2)), Voltage at Tune Top (U2) and the Offset frequency (F(2)).

STEP 3: Orthogonality check/adjustment (1)

1 Install flux probe to position L8 - L12 The flux probe must be positioned against the front ring of the head coil.



2. Read and write down:
   - U_3 = **Voltage at Center Frequency**

   Calculate the attenuation with next formula:

$$Att(1) = 20\log\left(\frac{U_2}{U_3}\right)$$

The typical value for the orthogonality is: Att(1) > 15 dB

If this value is reached, continue with **STEP 4**.

If this value is not reached, adjust the Tune Voltage at the marker position to a value as low as possible using the next procedure:
(See Drawings manual Z3-7.19 for exact trimmer location)

2.3.1. ADJUST ORTHOGONALITY

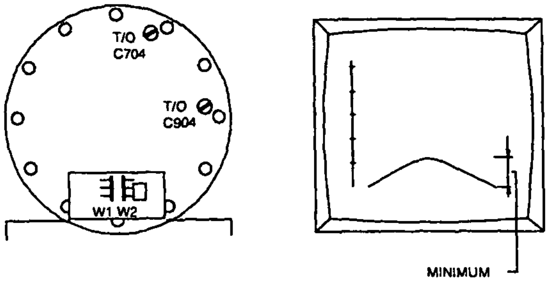


Figure 12 Adjustment of the orthogonality

1. Probe position between L8/L12 (As the previous adjustment).
2. Jumper position: W1 -> removed W2 -> position. See Figure 12.

3. Turn C704 a little clockwise, see if response decreases. If not turn C704 counter clockwise. Turn C904 the same amount in the other direction(*).

   (*) Other direction means. If you have set for more capacitance with C704, turn C904 in the other direction until you have the same amount capacitance less.

4. Minimize the response by turning both trimmers successively, as described above, until a minimum is reached.
   Write down the measured voltage U_3.

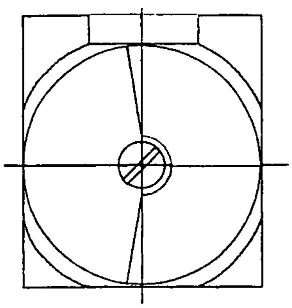


Figure 13 Trimmer in middle position 1/2 capacitance

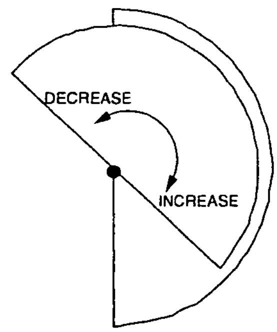


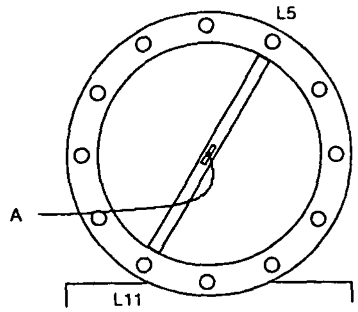
Figure 14 Increasing / decreasing capacitance

NOTE

The orthogonality measurement is not accurate due to the mechanical construction of the coil. Adjust the orthogonality as good as possible. The correct functioning of the coil is checked during the Image Quality performance measurements.

STEP 4: Orthogonality check/adjustment (2)

1. Install flux probe to position L5 - L11. The flux probe must be positioned against the front ring of the head coil and put the jumpers in the next position:
   W1 position 2
   W2 removed (See Figure 16)



2. Read and write down:
   - U_4 = **Voltage at Center Frequency**

   Calculate the attenuation with next formula:

$$Att(2) = 20\log(\frac{U_3}{U_4})$$

The typical value for the orthogonality is: Att(1) > 15 dB

If this value is reached, continue with paragraph **"Final actions"**

If this value is not reached, adjust the Tune Voltage at the marker position to a value as low as possible using the next procedure:
(See Drawings manual Z3-7.19 for exact trimmer location)

2.3.2. ADJUST ORTHOGONALITY

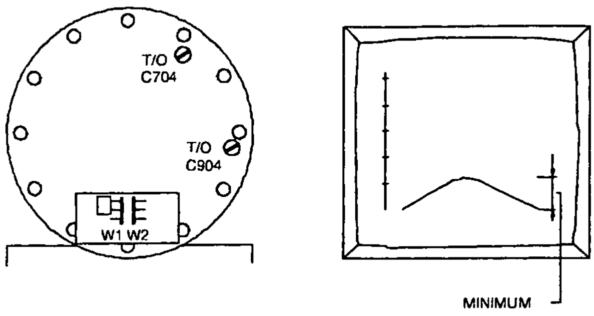


Figure 16 Adjustment of the orthogonality

3. Probe position between L5/L11 (As the previous adjustment). Put the jumpers in the following position:
   W1 -> position 2
   W2 -> removed (see Figure 16)
4. Turn C704 a little clockwise, see if response decreases. If not turn C704 counter clockwise **and** turn C904 the same amount in the other direction(*).

   (*) Other direction means: If you have set for more capacitance with C704 turn C904 in the other direction until you have the same amount capacitance less. Minimize the response by turning both trimmers successively, as described above, until a minimum is reached.

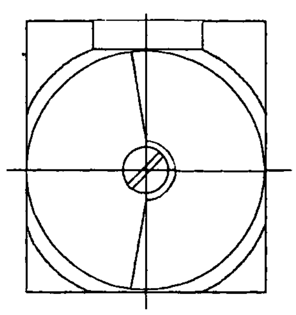


Figure 17 Trimmer in middle position: 1/2 capacitance

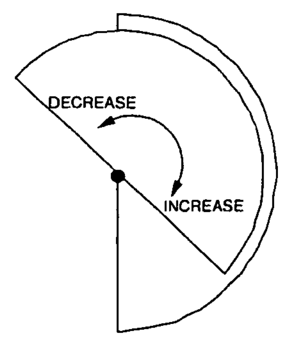


Figure 18 Increasing / decreasing capacitance

NOTE

The orthogonality measurement is not accurate due to the mechanical construction of the coil. Adjust the orthogonality as good as possible. The correct functioning of the coil is checked during the Image Quality performance measurements.

2.4. FINAL ACTIONS

1. Repeat step 1 - 4 until <u>no</u> adjustments are necessary. Every adjustment can influence other parameters.

2. If you have followed the procedure and no adjustments were necessary, write down the measured values, in the MRL according Table 1.(next page)
3. Install both jumpers in position 2 and mount the rear cover.
4. Remove the flux-probe holder out of the head coil and re-connect tune coil cables MMFX2 and MMFX3 to the TRSW board.
5. Read the following values from the screen and write them down as a reference in MRL (no specification):

F(head) = Offset frequency
Q(head) = Coil Q-factor

6. Stop the measurement
7. Restore normal situation.

Probe position		Parameters with specification		Measured values
Step 1) L8 - L12	Jumper position: W1: Position 2 W2: Removed	Q(1)	> 90	
		Voltage U_1		
		F(1)	< ± 25 kHz	
Step 2) L5 - L11	Jumper position: W1. Removed W2. Position 2	Q(2)	> 90	
		Voltage U_2		
		F(2)	< ± 25 kHz F(1) - F(2) ≤ 25 kHz	
Step 3) L8 - L12	Jumper position: W1: Removed W2: Position 2	Voltage U_3		
		Att(1)	typical value > 15 dB	
Step 4) L5 - L11	Jumper position: W1: Position 2 W2: Removed	Voltage U_4		
		Att(2)	typical value > 15 dB	
Both jumpers positioned (reference values)		F(head)	n.a.	
		Q(head)	n.a.	

Table 1 - Measurement results

3. APPENDIX A: READJUSTMENT PROCEDURE QUAD BODY COIL

Only if the hybrid box + quad body coil could not be adjusted within the required specifications, (re)adjustment of the separated QBC should be done. This adjustment is also necessary after the QBC has been replaced.

The re-adjustment is done in three steps:

1. Horizontal and vertical mode are adjusted to the resonance frequency.
2. The diagonal modes are set to the resonance frequency
3. The complete QBC is set to the correct frequency.

3.1. TOOLS

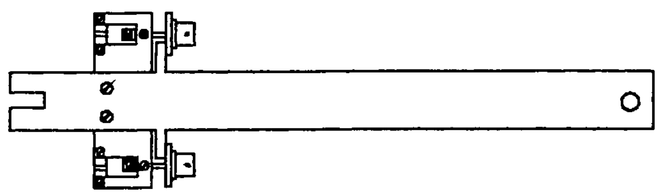
The following tools are necessary during the measurement and can be found in the measurement accessories.

- Flux probe holder
- Rotation knob for flux probe
- QBC adjustment tool 4522 131 35001
- 2 Extension cables 4522 131 32531 (belonging to QBC adjustment tool)
- 50 ohm coaxial cable 15 m
- Quadrature flux probe 64 Mhz. 4522 117 70051
- Two special trimmer adjustment tools.

3.2. MEASUREMENT SET-UP

1. Switch off the gradient system.
2. Lower the patient table.
3. Switch off the FEPSU in the system filter box.
4. Remove the bridge section, the PICU cover and the front cone cover. The hybrid box and QBC are visible now.

4 Disconnect all cables from the hybrid box.
5 Open the hybrid box by taking off the top-plate and loosen the screws of the connectors for the two rigid coax connections of the QBC (two screws, per coax connection!) Refer to drawing Z3-7.15
6. Remove the hybrid box.
7. Remove the inner top-cover of the QBC by carefully pulling out the covers by hand. start on one side of the cover, when one side is loose, the cover can be taken out. Mark the front and rear side of the top-cover of the QBC to be able to put it back in the same way. It has an influence on the shimming of the magnet.
8 Remove the inner lower-cover: start on one side.
9. Mount the special QBC adjustment tool at the QBC rigid coax connection. Do not use force to fasten the screws.



4522 131 3500 SERVICE TOOL QBC ADJUST T10-NT / ACS-NT

Figure 19 QBC adjustment tool

10 The QBC adjustment tool (with the two special extension cables (0.35 meter) must be connected to the QBC connections in stead of the hybrid box. (see Figure 22). The extension cables must be left open on one end. **(not terminated).**

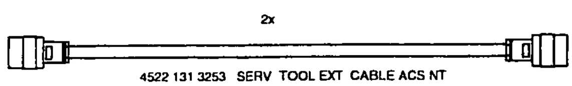


Figure 20

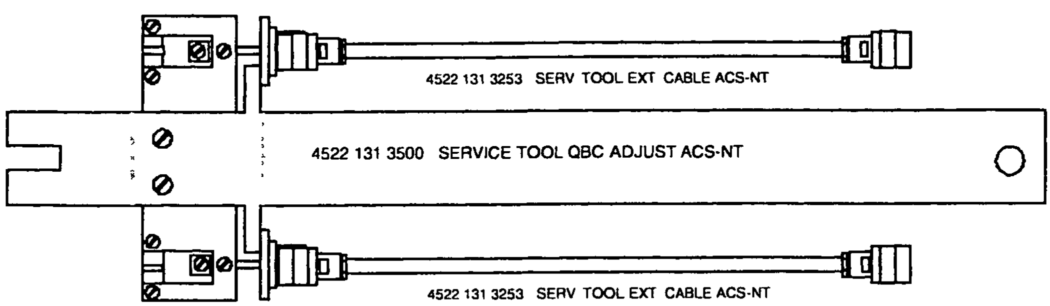


Figure 21 QBC measurement tool set-up

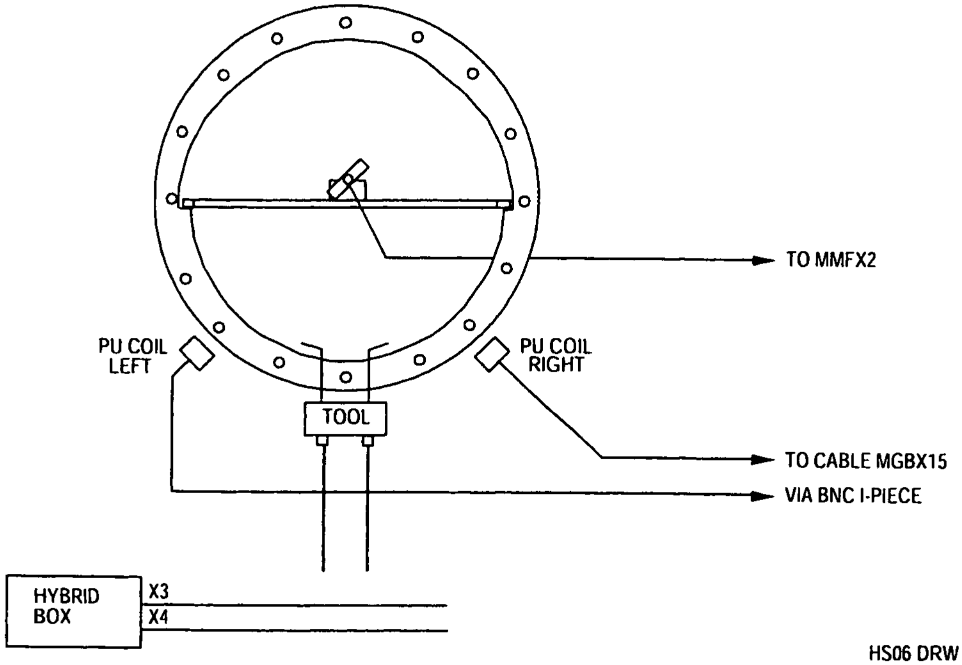


Figure 22 Measurement set-up for adjusting the QBC without hybrid box (front view)

11. Mount the top plate to the hybrid box with a few screws.
12. Position the hybrid box just outside the magnet bore in such a way that the cables MGBX3 and MGBX4 can be connected to the hybrid box. This will prevent that you will get error messages during the Manual coil tuning program. Make sure that the bottom side of the hybrid box does not make contact with the gradient coil. The top plate of the hybrid box must be present to avoid touching the 300 Volts.
13. Connect the left pick-up coil (looking from the front of the magnet) to the MR response cable MGBX15 by using a BNC I-piece. The pick-up coils are used as receive path.
14. Mark or measure the present position of the Pick-up coils and shift both Pick-up coils as close as possible to the QBC to get the best signal.

15. On the front side and the back side of the QBC you can find the adjustment screws for the trimmers. Loosen the screws that lock the trimmers for all QBC rods (see Figure 23 and Figure 24).

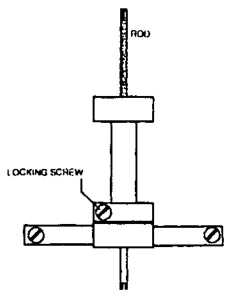


Figure 23 QBC trimmer in detail

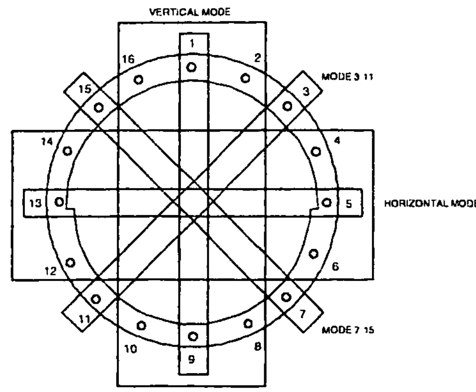


Figure 24 QBC frequency modes

You can see that two types of rods are used: thick rods (with thick trimmers) and thin rods (with thin trimmers). For each type of trimmer a separate trimmer tool must be used.

16. Mount the QBC top-inner cover (check the correct orientation). Position one side of the cover in its place and use your hands to push the other side in position. Use your fist if necessary to make the cover "click" in position.
    During the measurement the top-inner cover must be mounted because it introduces a frequency shift. The lower cover can be left out.
17. Remove the tune cable MMFX2 and MMFX3 from the TRSW board (PFEI)
18. Position the QBC flux probe holder in the center of the QBC and mount the linear flux probe via the rotation knob in the flux probe holder. Connect the flux probe to tune A at connector MMFX2. It is advised to route the cable via the rear side of the magnet to the tune box. The flux probe is used for transmit.
19. Switch on the FEPSU.

3.3. ADJUSTMENT QBC

1. Login : GYROSCAN.
   Select: SCAN CONTROL.
   Select: SCAN UTILITIES.
   Select: ENTER SERVICE MODE.
   Select: SYSTEM TUNING
   Select. SYSTEM TUNING TOOLS.
   Select: INSTALLATION PROCEDURES.
   Select: BODY COIL PROCEDURES

2. Select Parameter set ADJUST FREQUENCY QBC and start the measurement.

3.4. FINE ADJUSTMENT SEPARATE MODES

1. Position the flux probe to position 7 - 15 (see Figure 25).

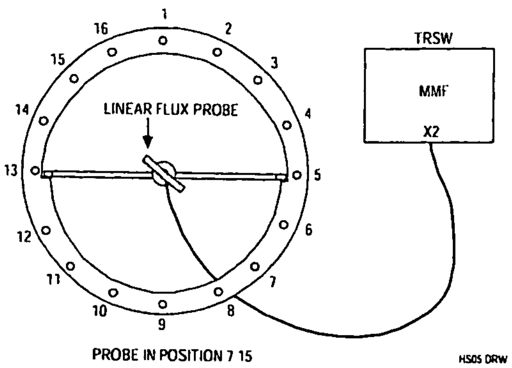


Figure 25 Flux probe position 7 - 15

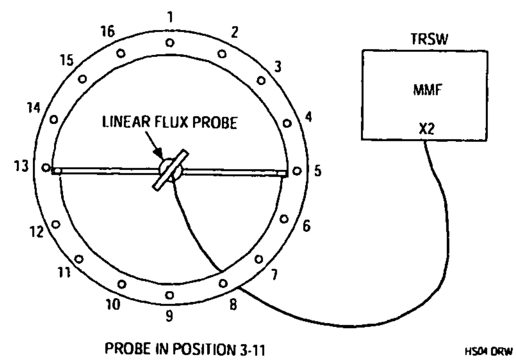


Figure 26 Flux probe position 3 - 11

2. Measure the frequency offset and write down this value in the table below.
3. Position the flux probe in position 3 - 11 (see Figure 26) and connect the right pick-up coil in stead of the left pick-up coil to the receive cable MGBX15.
4. Measure the offset and write down the value in the table below.
5. Connect the left pick-up coil and position the flux probe in horizontal position and measure the offset. Fill in the table.
6. Put the flux probe in vertical position and measure the offset. Fill in the table.

Linear Flux probe	Specification	Measured
Mode 7 - 15	F < ± 10 kHz	
Mode 3 - 11	F < ± 10 kHz	
Mode horizontal	F < ± 10 kHz	
Mode vertical	F < ± 10 kHz	

To adjust the QBC, use the following trimmers according the measured offset: (See Figure 24)

Mode 7 - 15:	trimmers 7 and 15
Mode 3 - 11:	trimmers 3 and 11
Horizontal mode:	For offset > 20 kHz. use trimmers 4, 5, 6 and 12, 13, 14 For offset < 20 kHz. use trimmers 5 and 13
Vertical mode:	For offset > 20 kHz. use trimmers 16, 1, 2 and 8, 9, 10 For offset < 20 kHz. use trimmers 1 and 9

NOTE

Turn all trimmers in the same direction (inwards (clockwise) or outwards (counter-clock wise)). For each rod the trimmers on the front and backside have to be turned with the same amount in the same direction.

7 Start with the mode with the largest offset and adjust all modes to the center frequency (± 10 kHz.)
8. Use the left pickup coil for modes 7 - 15, horizontal mode and vertical mode, and the right pickup coil for the mode 3 - 11.
9. Use small frequency steps (e.g. 1/4 turn at a time).
10 Check the modes until no adjustments are necessary any more.
11. Dismount the QBC inner covers and lock all trimmers

NOTE

Locking the trimmers may give a frequency shift.
Therefore the specification in the next table for the adjusted frequencies is ñ 20 kHz.

---

12. Re-install the QBC inner covers.
13. Check all modes and fill in the following table.

Linear Flux probe	Specification	Measured
Mode 7 - 15	F < ± 20 kHz	
Mode 3 - 11	F < ± 20 kHz	
Mode horizontal	F < ± 20 kHz	
Mode vertical	F < ± 20 kHz	

NOTE

If you loose count during adjustment, you should start with the initial position.
The procedure to adjust the QBC from the start position is described in DSS.

3.5. PREPARATION FOR MEASURING QBC RECEIVE MODE

1. Switch off the FEPSU.
2. Remove the flux probe and flux probe holder from the QBC.
3. Disconnect cables MGBX3 and MGBX4 from the hybrid box.
4. Remove the QBC adjustment tool from the QBC. Be careful with the coaxial connections of the QBC when disconnecting.
5. Mount the hybrid box in front of the QBC and connect all the cables to it. Position the pick-up coils back to their original position. (Check the marks or position to 6 mm from the QBC front ring)
   Make sure that the "computer cable" is lead in the middle underneath the hybrid box.
6. Switch on the FEPSU

3.6. HYBRID BOX + QBC RESONANCE FREQUENCY ADJUSTMENT

Trimmers C31 and C32 in the hybrid box are used to adjust the two modes of the QBC and hybrid box. Remove the top plate of the hybrid box to reach the trimmers.

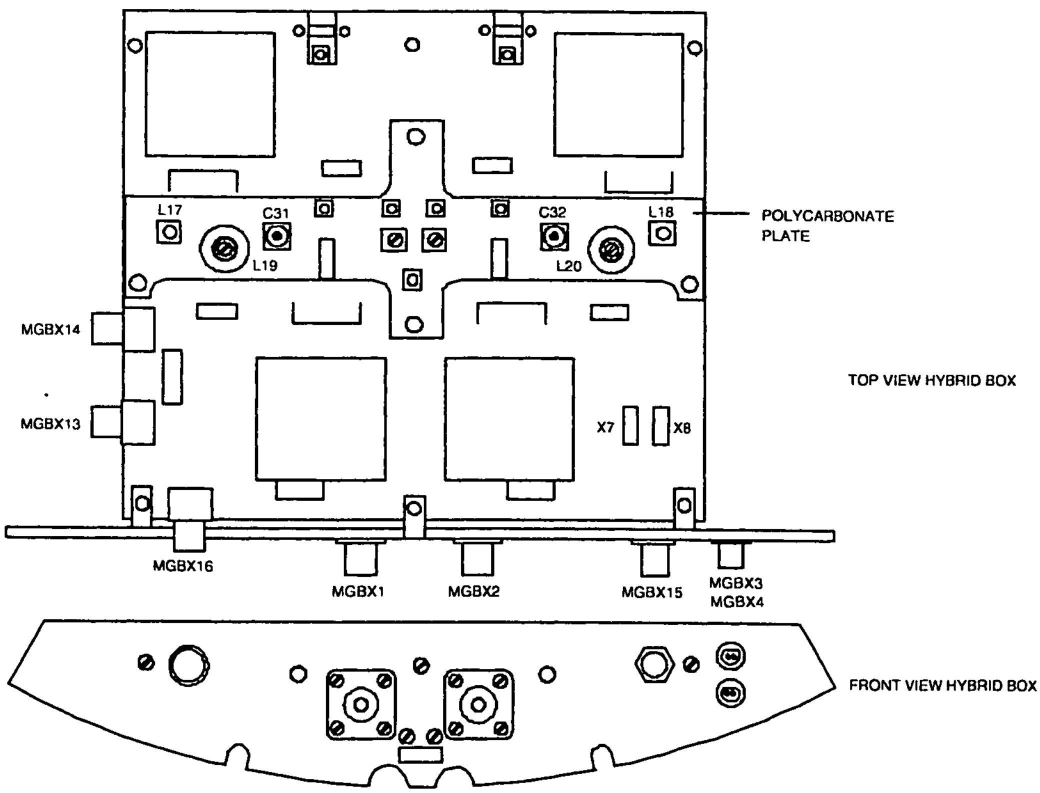
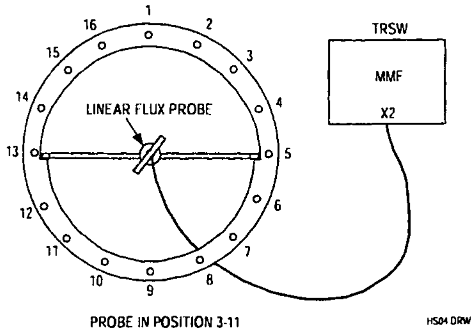


Figure 27 25 kW hybrid box with trimmers C31 and C32

1. Position the linear flux probe with use of the rotation knob in the QBC flux probe holder in the center of the QBC. Connect the cable of the flux probe to tune A signal at connector MMFX2 of the TRSW board. See Figure 28.



To prevent coupling problems with the hybrid box, the cable must leave the bore via the rear side of the magnet.

Figure 28 Frequency adjustment mode 3 - 11

Procedure:

2. Position the flux probe in position 3-11 (Figure 28).

3. Login : GYROSCAN.
4 Select: SCAN CONTROL.
5. Select: SCAN UTILITIES.
6. Select: ENTER SERVICE MODE.
7. Select: SYSTEM TUNING
8. Select: SYSTEM TUNING TOOLS.
9. Select: INSTALLATION PROCEDURES.
10. Select: BODY COIL PROCEDURES
11 Select: ADJUST QBC RECEIVE
12. Enter: Proceed to start the measurement
13 Adjust the frequency with trimmer C31 to the resonance frequency. The offset can be checked on the monitor.
Specification: *F(1) < 10 kHz*
4. Position the flux probe in position 7-15 (Figure 29)

mind to lead the flux probe cable via the rear
of the magnet.

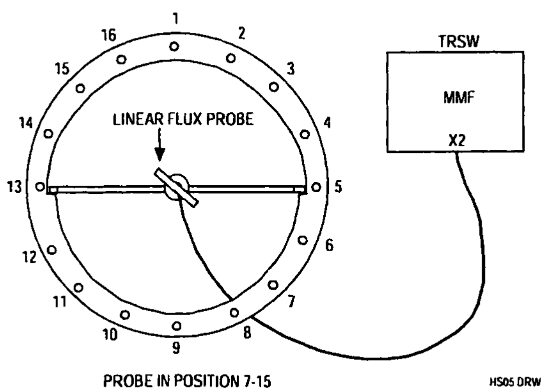


Figure 29 Frequency adjustment mode 7 - 15

15. Adjust the frequency with trimmer C32 to the resonance frequency. The offset can be checked on the monitor.
    Specification: *F(2) < 10 kHz*
16. Mount the quadrature flux probe in the QBC flux probe holder and position the holder in the center of the body coil with the connector of the Q-flux probe pointed to the backside of the magnet Also the cable must be lead via the backside of the magnet.
17. Connect a cable between tune A at connector MMFX2 and the quadrature flux probe.
    See Figure 30.

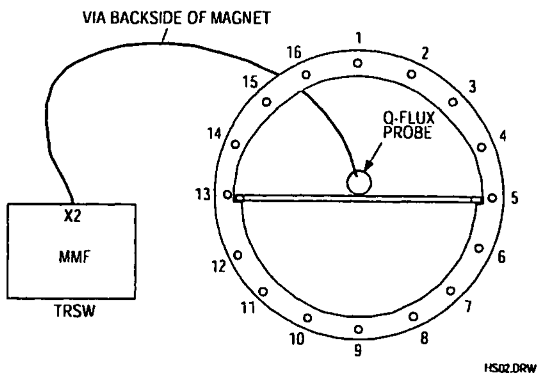


Figure 30 Connection of the Quadrature flux probe (Front View)

18. The same program is used : ADJUST QBC RECEIVE
19. Determine the values for the frequency offset and the Q-factor.

Offset $\leq \pm$ 10 kHz
$Q \geq 190$

3.7. RESTORING NORMAL CONDITION

1. Switch off the FEPSU.
2. Connect all cables to their original position.
3. Mount the front cone cover and the bridge section.
4. Switch on the FEPSU.

3.8. DETERMINATION OF BODY COIL PARAMETERS

To perform this measurement the cone-covers and the bridge section must be mounted.

1. Mount the **quadrature** flux probe in the QBC flux probe holder and position the holder in the center of the body coil **with the connector of the Q-flux probe pointed to the backside of the magnet.** Also the cable must be lead via the backside of the magnet.
2. Disconnect both Tune cable MMFX2 and MMFX3 and connect a cable between tune A (MMFX2) and the quadrature flux probe. See Figure 30.
   The program uses 63.870 Mhz as reference frequency.

3. Login : GYROSCAN.
4. Select: SCAN CONTROL.
5. Select: SCAN UTILITIES.
6. Select: ENTER SERVICE MODE.
7. Select: SYSTEM TUNING
8. Select: SYSTEM TUNING TOOLS.
9. Select: INSTALLATION PROCEDURES.
10. Select: BODY COIL PROCEDURES
11. Select: DETERMINE BODY COIL PARAMETERS
12. Now the measurement can be started with <Proceed>.
    No errors are allowed.

11\. Write the actual body coil frequency and Q-factor in the MRL as a reference.

4. APPENDIX B: ADAPTATION QHC CABLE TO RIGHT SIDE

4.1. INTRODUCTION

The quad head coil is standard delivered with the cable on the left side of the coil.
This procedure can be used for changing the cable to the right side.

4.2. PROCEDURE DESCRIPTION

1. Put the head coil on a table and remove the cover on the bottom side that covers the Driver PCB.
2. Disconnect the cables to the PCB and dismount the driver. On the other side of the coil, the bottom cover is already prepared to mount the driver board. In the PCB two extra holes are already present to mount the board on the right side.
3. Remove the bracket from the right side that covers the hole for the cable feed through.
4. Remove the cable from the left side. The hole can be covered with the bracket.
5. Dismount the rear cover of the Head coil and dismount the cable bracket from the cover.
6. Turn it over 180 degrees so that the cable points to the right side of the coil
7. Mount the bracket again and mount the cover back to the coil.
8. Mount the cables on the right side of the coil and connect them to the driver PCB.
   The connection cables on each side of the driver PCB should be separated from each other as good as possible to reduce the coupling. Use the already present cable support to secure the cables to the bottom cover.
9. Mount the cover for the PCB.

4.3. ADJUSTMENT QHC

After the cable has been moved to the right side the coil has to be readjusted.
For procedure, see the relevant paragraphs in this section (15).

Section 15 RF Coils

Contents

1. QUAD BODY COIL (QBC) CHECK/ADJUSTMENT

The quad body coil is already adjusted in the factory Only a test has to be done to check the QBC and to determine the values for the MRL as a reference.
Verify that the QBC did not move during transportation by checking all excenter screws. When the excenters were loose, the QBC has changed position and re-adjustment of the QBC is necessary (See appendix A).

1.1. TOOLS

- Quadrature flux probe (4522 117 70051)
- QBC flux probe holder
- 15 meter 50 ohm coaxial cable
- Linear flux probe
- Rotation knob

1.2. HYBRID BOX + QBC RESONANCE FREQUENCY

1. Switch-off the gradient system
2. Lower the patient table.
3. Remove the bridge section, the PICU cover and the front cone cover.
   The QBC and the hybrid box in front of the QBC are visible now.
4. Mount the **quadrature** flux probe in the QBC flux probe holder from the backside of the magnet and position the holder horizontal in the center of the body coil with the connector of the Q-flux probe pointed to the backside of the magnet. The cable must be routed via the backside of the magnet.
5. Remove the left magnet top cover and disconnect tune cables MMFX2 and MMFX3 from the TRSW board (PFEI).
6. Connect a 50 ohm coaxial cable between tune A signal at connector MMFX2 and the quadrature flux probe. See Figure 1.
7. Login : GYROSCAN.
8. Select: SCAN CONTROL.
9. Select: SCAN UTILITIES.
10. Select: ENTER SERVICE MODE.
11. Select: SYSTEM TUNING
12. Select: SYSTEM TUNING TOOLS.
13. Select: INSTALLATION PROCEDURES.
14. Select: BODY COIL PROCEDURES
15. Select: ADJUST QBC RECEIVE
16. Enter: Proceed (2x) to start the measurement
17. Select: AUTOSCALE
18. Determine the values for the frequency offset and the Q-factor.

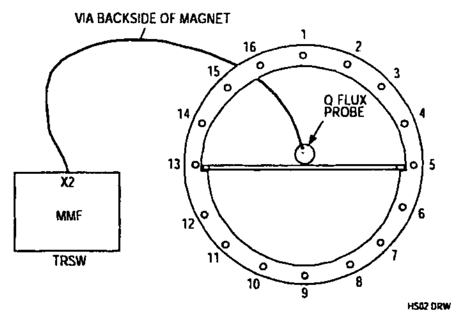


Figure 1 Measurement set-up (Front view)

Offset $\leq \pm$ 20 kHz
Q: Typical value $\geq$ 170

19. If the measured values are within specification, restore the original situation and continue with paragraph "Determination of body coil parameters".

    If the offset or Q-value are not within specification, adjustment of this mode must be done, according next paragraph "Hybrid box + QBC resonance frequency adjustment.

1.3. HYBRID BOX + QBC RESONANCE FREQUENCY ADJUSTMENT

If one of the values in previous paragraph were not within specification, adjustment of the hybrid box must be done as follows:

Trimmers C31 and C32 in the hybrid box are used to adjust the two modes of the QBC and hybrid box. Remove the hybrid box top plate to reach the trimmers. See Figure 2.

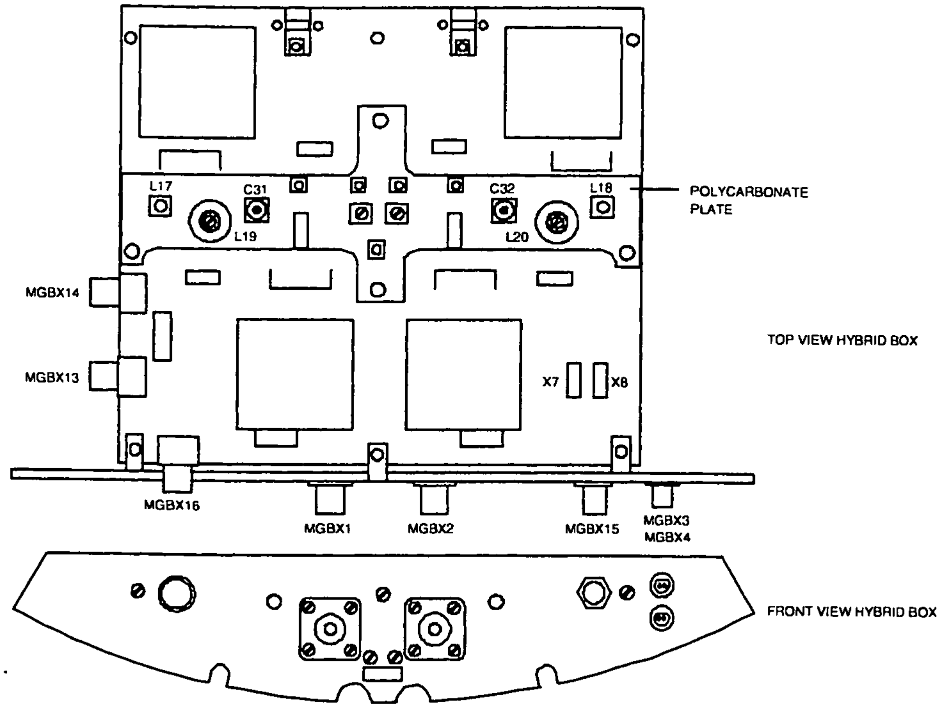


Figure 2 Hybrid box with trimmer location

Position the **linear flux probe** with use of the rotation knob in the QBC flux probe holder horizontal in the center of the QBC. Connect the cable of the flux probe to tune A signal at connector MMFX2. The cable should leave the magnet via the backside to prevent coupling with the hybrid box. See Figure 3.

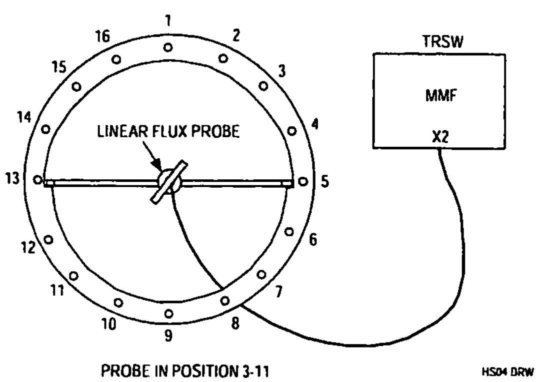


Figure 3 Frequency adjustment mode 3 - 11 (front view)

Procedure:

2. Position the flux probe in position 3-11 (Figure 3).
3. Login : GYROSCAN.
4. Select: SCAN CONTROL
5. Select: SCAN UTILITIES.
6. Select: ENTER SERVICE MODE.
7. Select: SYSTEM TUNING
8. Select: SYSTEM TUNING TOOLS.
9. Select: INSTALLATION PROCEDURES.
10. Select. BODY COIL PROCEDURES
11. Select: ADJUST QBC RECEIVE

12. Adjust the frequency with trimmer C31 of the hybrid box as good as possible to the resonance frequency. The offset can be checked on the monitor.

    Specification: *F(1) < 3 kHz*

13. Position the flux probe in position 7-15 (Figure 4). The cable must leave the bore via the backside of the magnet.

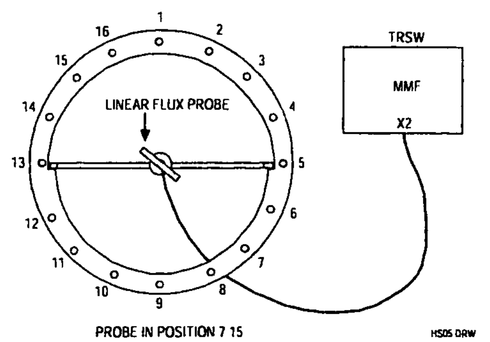


Figure 4 Frequency adjustment mode 7 - 15

14. Adjust the frequency with trimmer C32 of the hybrid box as good as possible to the resonance frequency. The offset can be checked on the monitor.

    Specification: *F(2) < 3 kHz*

15. Mount the **quadrature** flux probe in the QBC flux probe holder and position the holder in the center of the body coil **with the connector of the Q-flux probe pointed to the backside of the magnet**. Also the cable must be lead via the backside of the magnet.
16. Connect a 50 ohm coaxial cable between tune A at connector MMFX2 and the quadrature flux probe. See Figure 5.

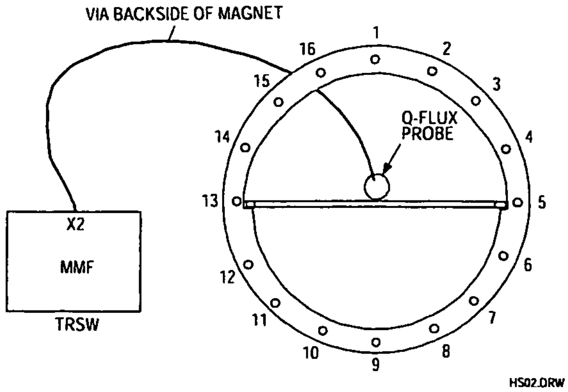


Figure 5 Connection of the Quadrature flux probe (Front View)

17. Select: ADJUST QBC RECEIVE
18. Determine the values for the frequency offset and the Q-factor.

Offset ≤ ± 20 kHz
Typical value Q: ≥ 170

19. When the Q factor is far below typical value and the performance measurements are not within specification, adjustment of the complete body coil is necessary. See appendix A at the end of this section.

1.4. DETERMINATION OF BODY COIL PARAMETERS

To perform this measurement the bridge section and both cone-covers must be mounted.

1. Mount the **quadrature** flux probe in the QBC flux probe holder and position the holder in the center of the body coil **with the connector of the Q-flux probe pointed to the backside of the magnet**. Also the cable must be lead via the backside of the magnet.
2. Connect a cable between tune A at connector MMFX2 and the quadrature flux probe. See Figure 5.

The program uses 63.870 MHz as reference frequency.

1. Login : GYROSCAN.
2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES.
4. Select: ENTER SERVICE MODE.
5. Select: SYSTEM TUNING
6. Select: SYSTEM TUNING TOOLS.
7. Select: INSTALLATION PROCEDURES.
8. Select: BODY COIL PROCEDURES
9. Select: DETERMINE BODY COIL PARAMETERS
10. Now the measurement can be started with <Proceed>.
    No errors are allowed.
11. Write the actual body coil frequency and Q-factor in the MRL (Section 22) as a reference.

2. DETERMINATION QUAD HEAD COIL INTERA 1.5T PARAMETERS

2.1. INTRODUCTION

The Quad Head Coil (QHC) for the Gyroscan Intera 1.5T system is provided with an electronic load.
The quad head coil is already adjusted in the factory. Adjustment during installation is only necessary if the cable has to be installed at the other side of the coil or if the Image Quality measurements for the head coil are not within specifications.

2.2. ADAPTATION CABLE TO THE RIGHT SIDE

The procedure to adapt the cable to the other side of the coil, is written in Appendix B at the end of this section. After the adaptation of the cable, readjustment/check of the head coil is necessary.

2.3. QUAD HEAD COIL FREQUENCY AND ORTHOGONALITY ADJUSTMENT

NOTE

The following procedure is only necessary when the cable is adapted to the right side of the coil, or when the QHC adjustment is suspected e.g. when the Image Quality measurements are not within specification.

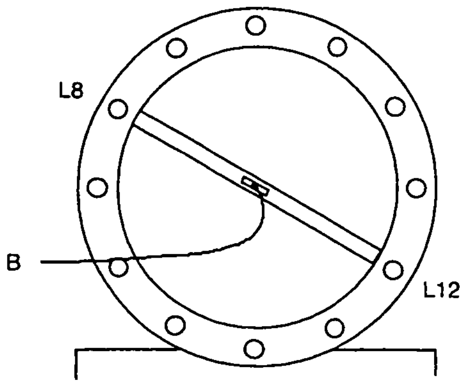
The QHC is checked on frequency, Q-factor and orthogonality. The adjustment frequency for the QHC is 64.050 Mhz Frequency-offset with respect to this frequency and orthogonality can be adjusted if not within specification. Q-factor can not be adjusted but is optimized by adjusting frequency and orthogonality.

The quad head coil can be seen as two coils working in different modes: there is a phase shift of 90° between both coils. Each coil has a pre-amplifier. By disconnecting one coil (removal of a jumper on the quad pre-amp board), the resonance frequency of the other coil can be adjusted
The adjustment for orthogonality must be done to ensure an exact 90° phase difference between the two coils.

Preparation:

1. Read the previous Note if not already done.
2. Remove the left magnet top cover.
3. Remove both tune coil cables MMFX2 and MMFX3 from the TRSW board (PFEI).
4. Connect the head coil to the Surface Coil connector 1 (SC1).
5. Move the slideable part of the head coil to the front (scanning) position. This is necessary to reduce the influence of the quad driver board on the coil.

6. Unscrew the rear cover of the coil and mount it back to the head coil using tape in stead of screws For the adjustment you need to take the cover off but the measurements are only valid when the rear cover is closed, because the values are influenced by this cover.
7. Place the head coil on the table in such a way that the rear of the head coil aligns with the end of the patient table, closest to the magnet (reference point).
8. Use the light visor and the marker on the head coil to position the head coil in the iso-center of the magnet via the "travel-to-scan-plane".
9. Connect the **linear flux probe** cable to MMFX2. (DO NOT USE THE QUADRATURE flux probe)
10. Position the linear flux probe in position L8 - L12 (see Figure 6). The flux probe holder must be positioned against the front ring of the head coil. The flux probe holder must be used without the 360° rotation knob



NOTE

On the front ring of the head coil, marker lines indicate rod L5, L8, L11 and L12.
Make sure the flux probe cable is apart from the head coil cable and that it leaves the bore as straight as possible.

Start the adjustment loop as follows:

1. Login : GYROSCAN.
2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES.
4. Select: ENTER SERVICE MODE.
5. Select: SYSTEM TUNING
6. Select: SYSTEM TUNING TOOLS.
7. Select: INSTALLATION PROCEDURES.
8. Select: HEAD COIL PROCEDURES
9. Enter : <PROCEED> to start the measurement process
10. Ignore the message: "Difference with the designed frequency is more then 100kHz"
11. Select: AUTOSCALE

The center frequency used by the program is 64.050 MHz.

ADJUSTMENT QUAD HEADCOIL EL-LOAD ACS-NT

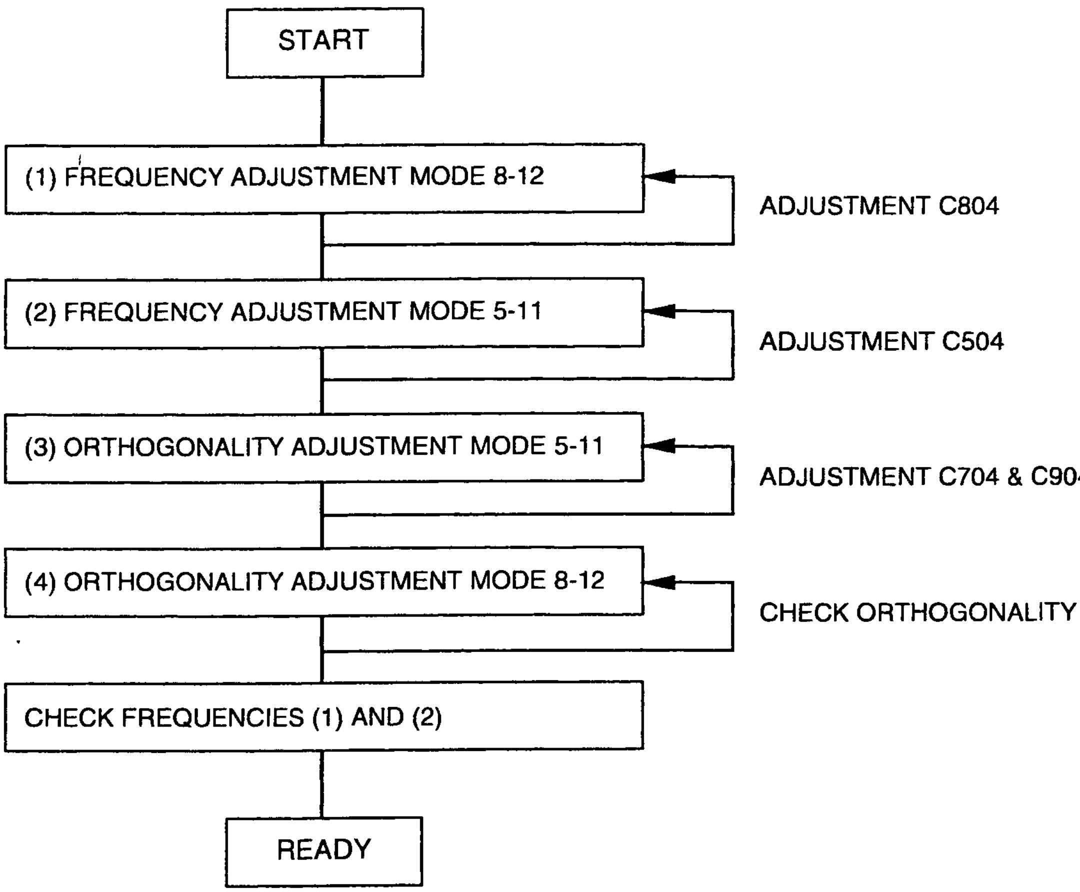


Figure 7 Flow diagram of adjusting the Quad head coil Intera 1.5T

NOTE

The measured values are only valid in case the rear cover of the head coil is closed during the measurement.

STEP 1 : Frequency check/adjustment first mode

1. Place the flux probe in position L8-L12 if not already done
2. Remove rear cover.
   Jumper W1: Position 2
   Jumper W2: Removed
   See Figure 8 for jumper location. Refer also to Drawings manual Z3-7.19 and Z3-7.20.

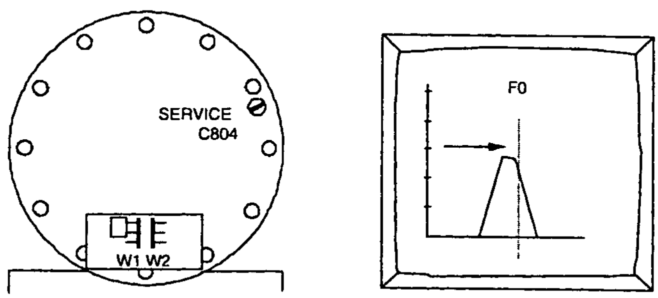
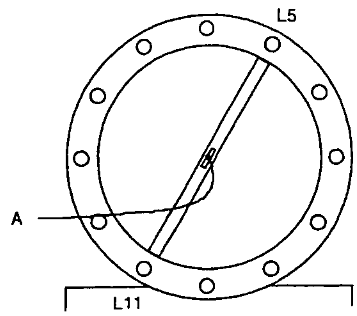


Figure 8 Frequency adjustment mode L8 - L12

3. Place back the cover. Read the following values and write them down in table 1 (at the end of this paragraph):
   Q(1) = Coil Q-factor.
   U_1 = Voltage at Tune Top (maximum voltage)
   F(1) = offset frequency
   Specifications:
   Q(1): > 90
   F(1): < ± 25 kHz
4. If F(1) does not meet the specifications, adjust the frequency with trimmer C804 (See Figure 8)
5. If the frequency is within specification, write down the Q-value (Q(1)), Voltage at Tune Top U_1 and the Offset frequency (F(1)).

STEP 2: Frequency check/adjustment second mode

1. Install the flux probe in position L5-L11.
2. Remove rear cover. Remove jumper W1 and install jumper W2 in position 2. See Figure 10 for jumper location Refer also to Drawings manual Z3-7.19 and Z3-7.20



3. Replace rear cover, read and write down:

Q(2)	= Coil Q-factor.
U_2	= Voltage at Tune Top (maximum voltage)
F(2)	= offset frequency
Specifications:	
Q(2)	> 90
F(2)	≤ ± 25 kHz and frequency difference F(1) and F(2) ≤ 25 kHz

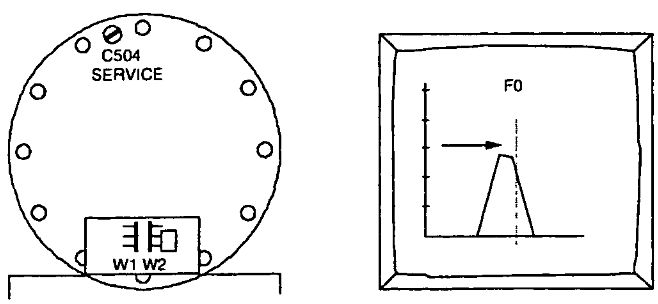
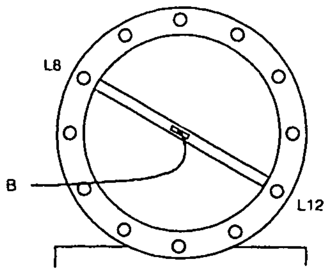


Figure 10 Frequency adjustment L5 - L11

4. If F(2) does not meet the specifications, adjust the frequency with trimmer C504 (see Figure 10).
5. If the frequency is within specification, write down the Q value (Q(2)), Voltage at Tune Top (U_2) and the Offset frequency (F(2))

STEP 3 : Orthogonality check/adjustment (1)

1. Install flux probe to position L8 - L12. The flux probe must be positioned against the front ring of the head coil.



2. Read and write down:
   - U_3 = **Voltage at Center Frequency**

   Calculate the attenuation with next formula:

$$Att(1) = 20\log\left(\frac{U_2}{U_3}\right)$$

The typical value for the orthogonality is: Att(1) > 15 dB

If this value is reached, continue with **STEP 4**.

If this value is not reached, adjust the Tune Voltage at the marker position to a value as low as possible using the next procedure:
(See Drawings manual Z3-7.20 for exact trimmer location)

2.3.1. ADJUST ORTHOGONALITY

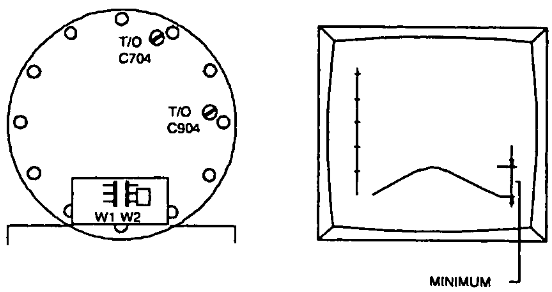


Figure 12 Adjustment of the orthogonality

1. Probe position between L8/L12 (As the previous adjustment).
2. Jumper position: W1 -> removed W2 -> position. See Figure 12.

3. Turn C704 a little clockwise, see if response decreases. If not turn C704 counter clockwise. Turn C904 the same amount in the other direction(*).

   (*) Other direction means: If you have set for more capacitance with C704, turn C904 in the other direction until you have the same amount capacitance less

4. Minimize the response by turning both trimmers successively, as described above, until a minimum is reached
   Write down the measured voltage U_3.

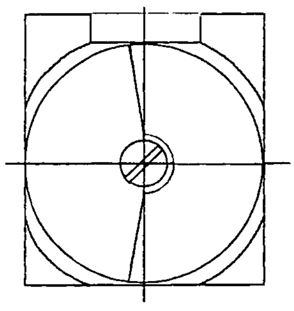


Figure 13 Trimmer in middle position 1/2 capacitance

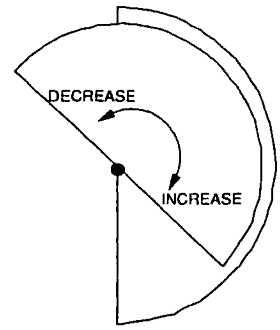


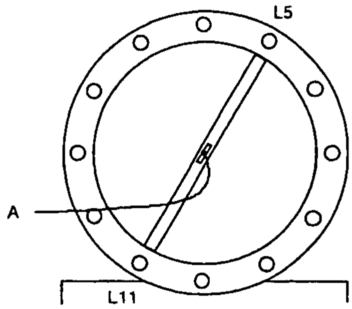
Figure 14 Increasing / decreasing capacitance

NOTE

The orthogonality measurement is not accurate due to the mechanical construction of the coil. Adjust the orthogonality as good as possible. The correct functioning of the coil is checked during the Image Quality performance measurements.

STEP 4 : Orthogonality check/adjustment (2)

1. Install flux probe to position L5 - L11. The flux probe must be positioned against the front ring of the head coil and put the jumpers in the next position:
   W1 position 2
   W2 removed (See Figure 16)



2. Read and write down:
   - U_4 = **Voltage at Center Frequency**

   Calculate the attenuation with next formula:

$$Att(2) = 20\log(\frac{U_1}{U_4})$$

The typical value for the orthogonality is: Att(1) > 15 dB

If this value is reached, continue with paragraph "**Final actions**"

If this value is not reached, adjust the Tune Voltage at the marker position to a value as low as possible using the next procedure:
(See Drawings manual Z3-7.19 for exact trimmer location)

2.3.2. ADJUST ORTHOGONALITY

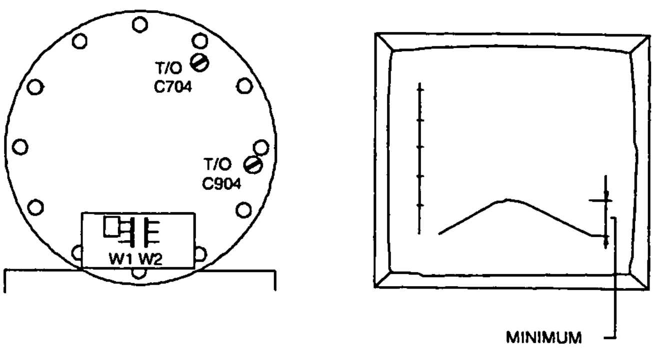


Figure 16 Adjustment of the orthogonality

3 Probe position between L5/L11 (As the previous adjustment). Put the jumpers in the following position·
W1 -> position 2
W2 -> removed (see Figure 16)

4. Turn C704 a little clockwise, see if response decreases. If not turn C704 counter clockwise **and** turn C904 the same amount in the other direction(*).

(*) Other direction means: If you have set for more capacitance with C704 turn C904 in the other direction until you have the same amount capacitance less. Minimize the response by turning both trimmers successively, as described above, until a minimum is reached.

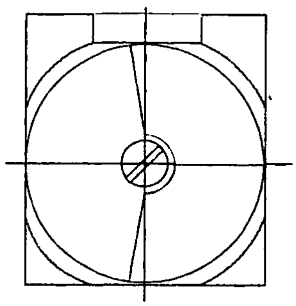


Figure 17 Trimmer in middle position1/2 capacitance

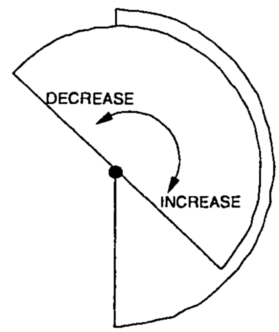


Figure 18 Increasing / decreasing capacitance

NOTE

The orthogonality measurement is not accurate due to the mechanical construction of the coil.
Adjust the orthogonality as good as possible.
The correct functioning of the coil is checked during the Image Quality performance measurements.

2.4. FINAL ACTIONS

1. Repeat step 1 - 4 until no adjustments are necessary. Every adjustment can influence other parameters.
2. If you have followed the procedure and no adjustments were necessary, write down the measured values, in the MRL according Table 1.(next page)
3. Install both jumpers in position 2 and mount the rear cover.
4. Remove the flux-probe holder out of the head coil and re-connect tune coil cables MMFX2 and MMFX3 to the TRSW board..
5. Read the following values from the screen and write them down as a reference in MRL (no specification):

F(head) = Offset frequency
Q(head) = Coil Q-factor

6. Stop the measurement
7 Restore normal situation.
8. Write down the final values in the table in the MRL (Section 22) as a reference.

Probe position	Parameters with specification		Measured values
Step 1) Jumper position: W1: Position 2 W2: Removed L8 - L12	Q(1)	> 90	
	Voltage U_1		
	F(1)	< ± 25 kHz	
Step 2) Jumper position: W1: Removed W2: Position 2 L5 - L11	Q(2)	> 90	
	Voltage U_2		
	F(2)	< ± 25 kHz F(1) - F(2) ≤ 25 kHz	
Step 3) Jumper position: W1: Removed W2: Position 2 L8 - L12	Voltage U_3		
	Att(1)	typical value > 15 dB	
Step 4) Jumper position: W1: Position 2 W2: Removed L5 - L11	Voltage U_4		
	Att(2)	typical value > 15 dB	
Both jumpers positioned (reference values)	F(head)	n.a.	
	Q(head)	n.a.	

Table 1 - Measurement results

3. APPENDIX A: READJUSTMENT PROCEDURE QUAD BODY COIL

Only if the hybrid box + quad body coil could not be adjusted within the required specifications, (re)adjustment of the separated QBC should be done. Adjustment is also necessary when the QBC is replaced

The re-adjustment is done in three steps:

1. Horizontal and vertical mode are adjusted to the resonance frequency.
2. The diagonal modes are set to the resonance frequency
3. The complete QBC is set to the correct frequency.

3.1. TOOLS

- Flux probe holder
- Rotation knob for flux probe
- QBC adjustment tool 4522 131 35001
- 2 Extension cables 4522 131 32531 (belonging to QBC adjustment tool)
- 50 ohm coaxial cable 15 m
- Quadrature flux probe 64 Mhz. 4522 117 70051
- Two special trimmer adjustment tools.

3.2. MEASUREMENT SET-UP

1. Switch off the gradient system.
2. Lower the patient table.
3. Switch off the FEPSU (system filter box).
4. Remove the bridge section, the PICU cover and the front cone cover (if not already done).
   The hybrid box and QBC are visible now.
5. Disconnect the cables from the hybrid box.
6. Open the hybrid box by taking off the top-plate and loosen the screws of the connectors for the two rigid coax connections of the QBC (two screws, per coax connection!) Refer to drawing Z3-7.15
7. Remove the hybrid box.
8. Remove the inner top-cover of the QBC by carefully pulling out the covers by hand: start on one side of the cover, when one side is loose, the cover can be taken out. Mark the front and rear side of the Top-cover of the QBC to be able to put it back in the same way. It has an influence on the shimming of the magnet.
9. Remove the inner lower-cover: start on one side.

10. Mount the special QBC adjustment tool at the QBC rigid coax connection. Do not use force to fasten the screws.

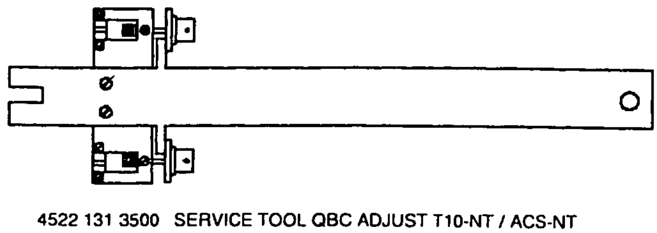


Figure 19 QBC adjustment tool

10. The QBC adjustment tool (with the two special extension cables (0.35 meter) must be connected to the QBC connections in stead of the hybrid box. (see Figure 22). The extension cables must be left open on one end. **(not terminated)**. See Figure 21.

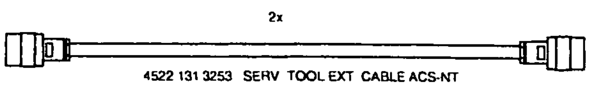


Figure 20 Extension cable

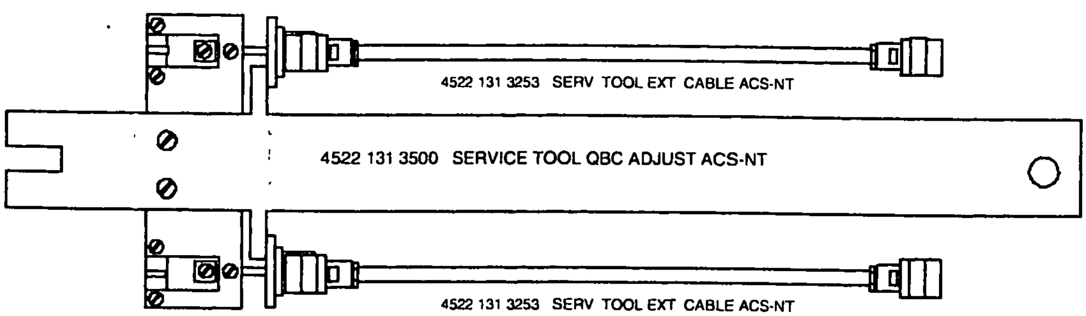


Figure 21 Measurement set-up QBC adjustment tool

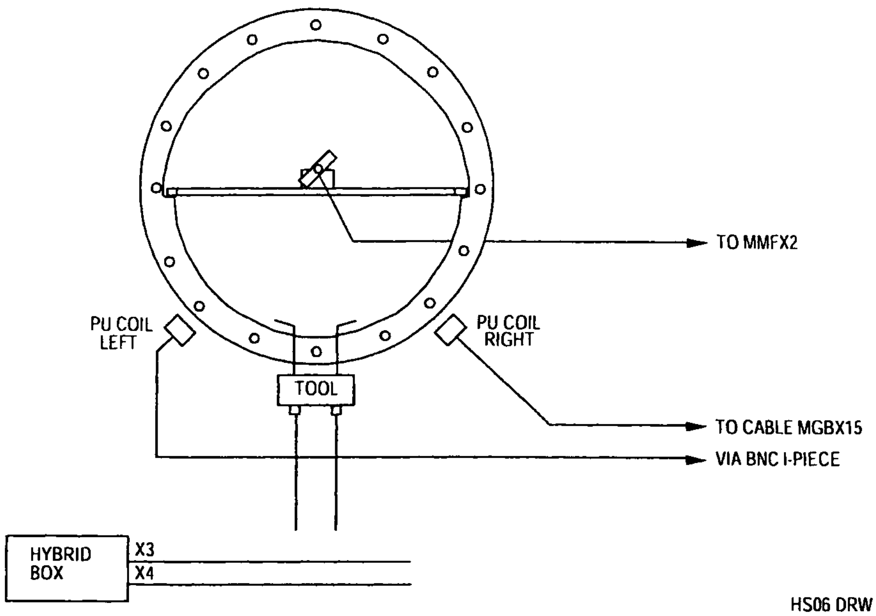


Figure 22 Measurement set-up for adjusting the QBC without hybrid box (front view)

11. Mount the top plate to the hybrid box with a few screws.
12. Position the hybrid box just outside the magnet bore in such a way that the cables MGBX3 and MGBX4 can be connected to the hybrid box. This will prevent that you will get error messages during the Manual coil tuning program. Make sure that the bottom side of the hybrid box does not make contact with the gradient coil. The top plate of the hybrid box must be present to avoid touching the 300 Volts.
13. Connect the left pick-up coil (looking from the front of the magnet) to the MR response cable MGBX15 by using a BNC I-piece. The pick-up coils are used for receive.
14. Mark or measure the present position of the pick-up coils and shift both pick-up coils as close as possible to the QBC to get the best signal.

15. On the front side and the back side of the QBC you can find the adjustment screws for the trimmers. Loosen the screws that lock the trimmers for all QBC rods. (see Figure 23 and Figure 24).

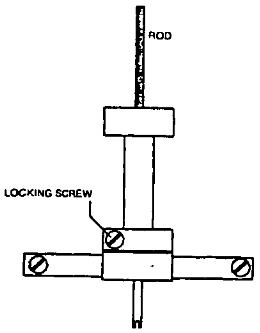


Figure 23 QBC trimmer

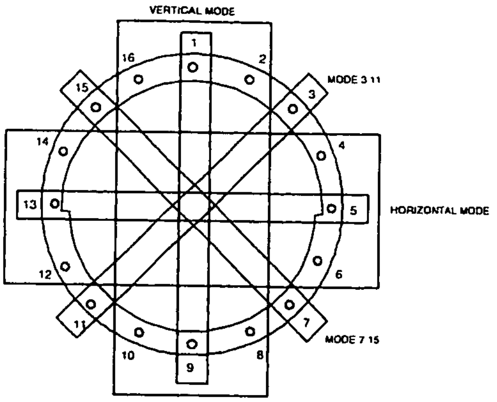


Figure 24 QBC frequency modes

You can see that two types of rods are used. thick rods (with thick trimmers) and thin rods (with thin trimmers). For each type of trimmer a separate trimmer tool must be used.

16. Mount the QBC top-inner cover (check the correct orientation). Position one side of the cover in its place and use your hands to push the other side in position. Use you fist if necessary to make the cover "click" in position.
    During the measurement the top-inner cover must be mounted because it introduces a frequency shift. The bottom cover can be left out.
17. Position the QBC flux probe holder horizontal in the center of the QBC and mount the linear flux probe via the rotation knob in the flux probe holder.
18. Remove borh tune cable MMFX2 and MMFX3 from the TRSW board (PFEI) and connect the flux probe to tune A at connector MMFX2. The cable must be routed via the backside of the magnet to the tune box. The flux probe is used for transmit.
19. Switch on the FEPSU.

3.3. ADJUSTMENT QBC

1. Login : GYROSCAN.
   Select: SCAN CONTROL.
   Select: SCAN UTILITIES.
   Select: ENTER SERVICE MODE.
   Select: SYSTEM TUNING
   Select: SYSTEM TUNING TOOLS.
   Select: INSTALLATION PROCEDURES.
   Select: BODY COIL PROCEDURES

2. Select Parameter set ADJUST FREQUENCY QBC and start the measurement.

3.4. FINE ADJUSTMENT SEPARATE MODES

1. Position the flux probe to position 7 - 15 (see Figure 25).

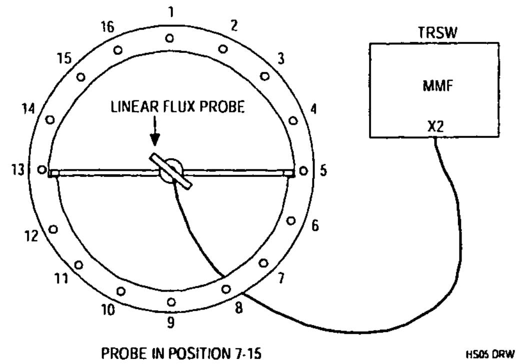


Figure 25 Flux probe position 7 - 15

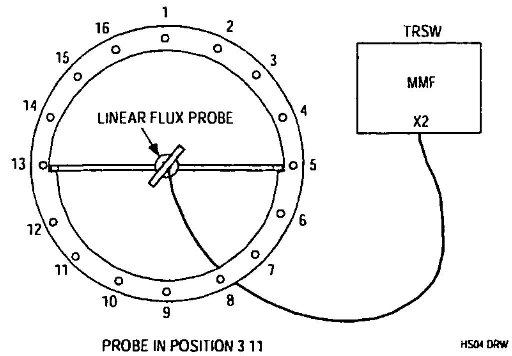


Figure 26 Flux probe position 3 - 11

2. Measure the frequency offset and write down this value in the table below.
3. Position the flux probe in position 3 - 11 (see Figure 26) and connect the right pick-up coil in stead of the left pick-up coil to the receive cable MGBX15.
4. Measure the offset and write down the value in the table below.
5. Connect the left pick-up coil and position the flux probe in horizontal position and measure the offset. Fill in the table.
6. Put the flux probe in vertical position and measure the offset. Fill in the table.

Linear Flux probe	Specification	Measured
Mode 7 - 15	F < ± 20 kHz	
Mode 3 - 11	F < ± 20 kHz	
Mode horizontal	F < ± 20 kHz	
Mode vertical	F < ± 20 kHz	

To adjust the QBC, use the following trimmers according the measured offset: (See Figure 24)

Mode 7 - 15:	trimmers 7 and 15
Mode 3 - 11:	trimmers 3 and 11
Horizontal mode:	For offset > 40 kHz. use trimmers 4, 5, 6 and 12, 13, 14
	For offset < 40 kHz. use trimmers 5 and 13
Vertical mode:	For offset > 40 kHz. use trimmers 16, 1, 2 and 8, 9, 10
	For offset < 40 kHz. use trimmers 1 and 9

NOTE

Turn all trimmers in the same direction (inwards (clockwise) or outwards (counter-clock wise)). For each rod the trimmers on the front and backside have to be turned with the same amount in the same direction.

7. Start with the mode with the largest offset and adjust all modes to the center frequency (± 10 kHz.)
8. Use the left pickup coil for modes 7 - 15, horizontal mode and vertical mode, and the right pickup coil for the mode 3 - 11.
9. Use small frequency steps (e.g. 1/4 turn at a time).
10 Check the modes until no adjustments are necessary any more.
11. Dismount the QBC inner covers and lock all trimmers.

NOTE

Locking the trimmers may give a frequency shift.
Therefore the specification in the next table for the adjusted frequencies is ± 20 kHz.

12. Re-install the QBC inner covers.
13. Check all modes and fill in the following table.

Linear Flux probe	Specification	Measured
Mode 7 – 15	F < ± 20 kHz	
Mode 3 – 11	F < ± 20 kHz	
Mode horizontal	F < ± 20 kHz	
Mode vertical	F < ± 20 kHz	

NOTE

If you loose count during adjustment, you should start with the initial position. The procedure to adjust the QBC from the start position is described in DSS.

3.5. PREPARATION FOR MEASURING QBC RECEIVE MODE

1. Switch off the FEPSU.
2. Remove the flux probe and flux probe holder from the QBC.
3. Disconnect cables MGBX3 and MGBX4 from the hybrid box.
4. Remove the QBC adjustment tool from the QBC. Be careful with the coaxial connections of the QBC when disconnecting.
5. Position the pick-up coils back to their original position. (Check the marks or position them to 8 mm from the QBC front ring)
   Make sure that the "computer cable" is lead in the middle underneath the hybrid box.
6. Switch on the FEPSU

3.6. HYBRID BOX + QBC RESONANCE FREQUENCY ADJUSTMENT

Trimmers C31 and C32 in the hybrid box are used to adjust the two modes of the QBC and hybrid box. Remove the top plate of the hybrid box to reach the trimmers.

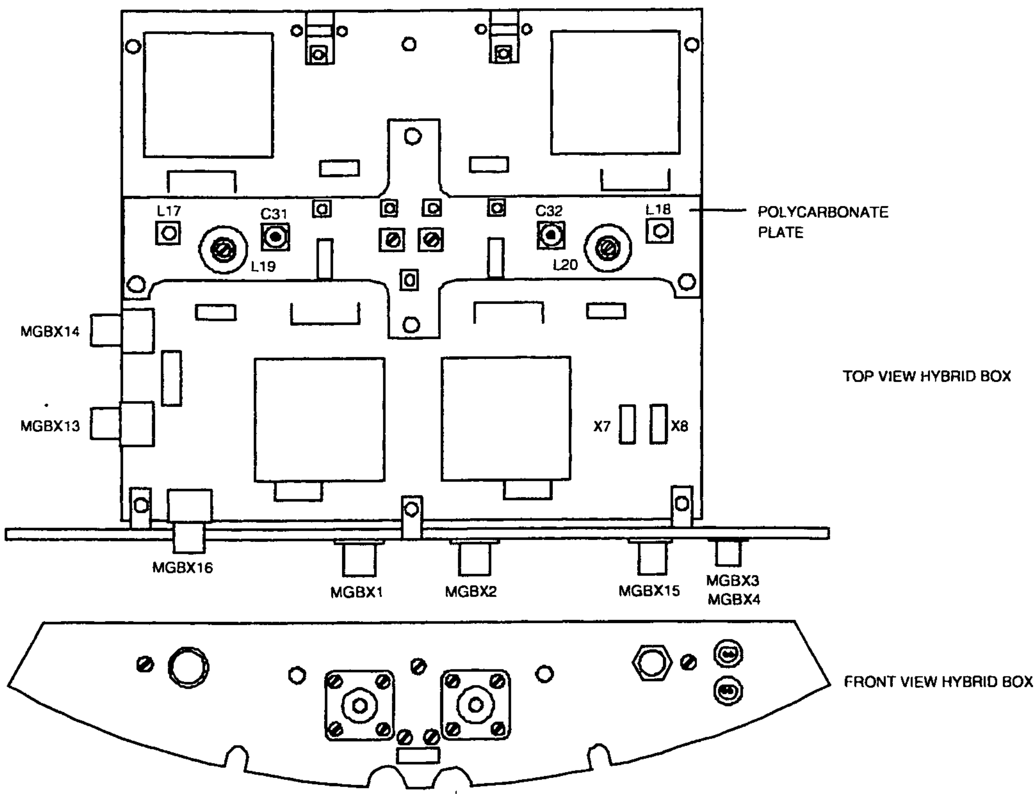
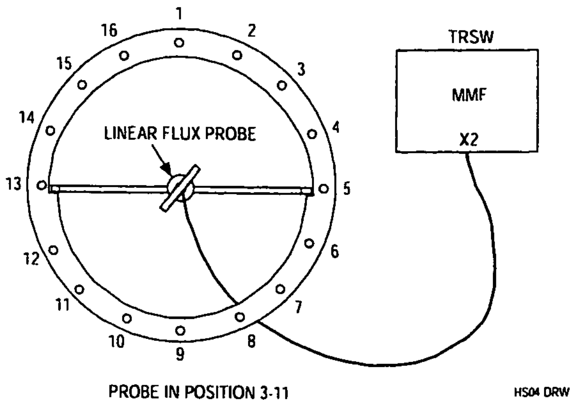


Figure 27 Hybrid box with trimmer location

1. Position the linear flux probe with use of the rotation knob in the QBC flux probe holder in the center of the QBC. Connect the cable of the flux probe to tune A signal at connector MMFX2. See Figure 28.



To prevent coupling problems with the Hybrid box, the cable must leave the bore via the backside of the magnet.

Figure 28 Frequency adjustment mode 3 - 11

Procedure:

2. Position the flux probe in position 3-11 (Figure 28).
3. Login . GYROSCAN.
4. Select: SCAN CONTROL.
5. Select: SCAN UTILITIES.
6. Select: ENTER SERVICE MODE.
7. Select. SYSTEM TUNING
8. Select: SYSTEM TUNING TOOLS.
9. Select: · INSTALLATION PROCEDURES.
10. Select: BODY COIL PROCEDURES
11. Select: ADJUST QBC RECEIVE
12. Enter: Proceed to start the measurement
13. Adjust the frequency with trimmer C31 to the resonance frequency. The offset can be checked on the monitor.
    Specification: *F(1) < 3 kHz*
14. Position the flux probe in position 7-15 (Figure 29).

Remind to lead the flux probe cable via the backside of the magnet.

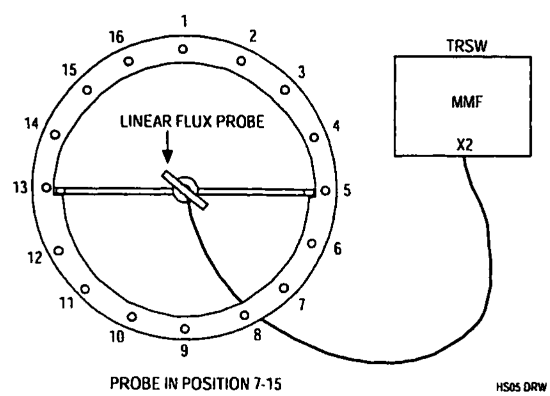


Figure 29 Frequency adjustment mode 7 - 15

15. Adjust the frequency with trimmer C32 to the resonance frequency. The offset can be checked on the monitor.

    Specification: *F(2) < 3 kHz*

16. Mount the quadrature flux probe in the QBC flux probe holder and position the holder in the center of the body coil with the connector of the Q-flux probe pointed to the backside of the magnet. Also the cable must be lead via the backside of the magnet.
17. Connect a cable between tune A at connector MMFX2 and the quadrature flux probe. See Figure 30.

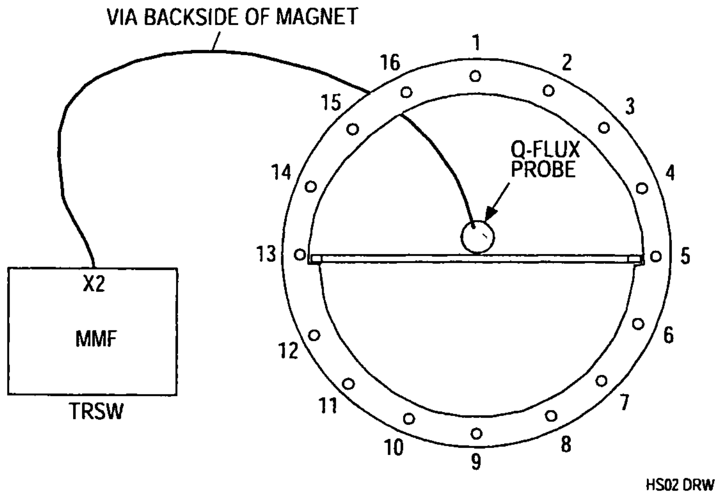


Figure 30 Connection of the Quadrature flux probe (Front View)

18. The same program is used : ADJUST QBC RECEIVE
19. Determine the values for the frequency offset and the Q-factor.

Offset ≤ ± 20 kHz
Typical value Q: ≥ 170

3.7. RESTORING NORMAL CONDITION

1. Switch off the FEPSU.
2. Connect all cables to their original position.
3. Mount the front cone cover and the bridge section.
4. Switch on the FEPSU.

3.8. DETERMINATION OF BODY COIL PARAMETERS

To perform this measurement the bridge section and both cone-covers must be mounted.

1. Mount the **quadrature** flux probe in the QBC flux probe holder and position the holder in the center of the body coil **with the connector of the Q-flux probe pointed to the backside of the magnet**. Also the cable must be lead via the backside of the magnet.
2. Remove both tune cable MMFx2 and MMFX3 and connect a cable between MMFX2 and the quadrature flux probe. See Figure 30.

   The program uses 63.870 Mhz as reference frequency.

3. Login : GYROSCAN.
4. Select: SCAN CONTROL.
5. Select: SCAN UTILITIES.
6. Select: ENTER SERVICE MODE.
7. Select: SYSTEM TUNING
8. Select: SYSTEM TUNING TOOLS.
9. Select: INSTALLATION PROCEDURES.
10. Select: BODY COIL PROCEDURES
11. Select: DETERMINE BODY COIL PARAMETERS
12. Now the measurement can be started with <Proceed>.
    No errors are allowed.
13. Write the actual body coil frequency and Q-factor in the MRL as a reference.

4. APPENDIX B: ADAPTATION QHC CABLE TO RIGHT SIDE

4.1. INTRODUCTION

The quad head coil is standard delivered with the cable on the left side of the coil.
This procedure can be used for changing the cable to the right side.

4.2. PROCEDURE DESCRIPTION

1. Put the head coil on a table and remove the cover on the bottomside that covers the driver PCB.
2. Disconnect the cables to the PCB and dismount the driver. On the other side of the coil, the bottom cover is already prepared to mount the driver board. In the PCB two extra holes are already present to mount the board on the right side.
3. Remove the bracket from the right side that covers the hole for the cable feed through.
4. Remove the cable from the left side. The hole can be covered with the bracket.
5. Dismount the rear cover of the head coil and dismount the cable bracket from the cover.
6. Turn it over 180 degrees so that the cable points to the right side of the coil.
7. Mount the bracket again and mount the cover back to the coil
8. Mount the cables on the right side of the coil and connect them to the driver PCB
   The connection cables on each side of the driver PCB should be seperated from each other as good as possible to reduce the coupling. Use the already present cable support to secure the cables to the bottom cover.
9. Mount the cover for the PCB.

4.3. ADJUSTMENT QHC

After the cable has been moved to the right side the coil has to be readjusted.
For procedure, see the relevant paragraphs in this section (15).